Ideology in Government: Evidence from the Office of Indian Affairs and the Assimilation Era*

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Current Draft: October 23, 2026
First Draft: August 27, 2025

Abstract

This paper studies bureaucratic ideology and coercive policy by examining the Office of Indian Affairs, an institution that held authority over land, education, and legal governance of Indigenous populations in the United States. We digitize bureaucratic reports and use computational tools to measure support for assimilationist policies during the late 19th and early 20th centuries. Our measure distinguishes normative endorsement of assimilation from merely describing or implementing federal policy. We validate it using human evaluations and separate archival records of agents' policy advocacy. Across reports, assimilationist commitment was widespread but varied over time and across personnel regimes: support was especially strong among religious nominees and moderated as the appointment system changed, even as federal policy remained firmly assimilationist. Finally, we study the Dawes Act, a pivotal policy that dismantled collective tribal landholding, and find that agencies led by more assimilationist agents carried out more land redistribution immediately after its passage.

JEL Codes: D73, M5, J15, N41.

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1 Introduction

Governments throughout history have implemented coercive policies targeting racial or ethnic groups under their jurisdiction. In the United States, Jim Crow laws enforced racial segregation and disenfranchisement for decades (Woodward, 1955). Nazi Germany used racial ideology to organize law and administration through the Nuremberg statutes, registries, and confiscations—well before the escalation to mass violence (Koonz, 2003). Apartheid-era pass laws in South Africa constrained Black people's movement and residency (Posel, 1991). Together, these cases illustrate a common repertoire of policy instruments—detention, segregation, and surveillance—through which states pursue repressive projects of social engineering.

A fundamental question for the exercise of coercive state power is how much these policies depend on the ideology of the government officials charged with their administration. Bureaucrats need not be neutral functionaries: a large qualitative literature on everyday bureaucracy documents how officials' beliefs, values, and attitudes influence their decisions in routine administrative settings (e.g., Lipsky, 1980; Maynard-Moody and Musheno, 2003). Related theories of bureaucracy emphasize that public officials may also be intrinsically motivated by substantive policy goals, with important consequences for how they use discretion and implement policy (e.g., Besley and Ghatak, 2005; Gailmard and Patty, 2007). Yet, we know much less about whether and how bureaucratic ideology matters in coercive policy settings, where researchers rarely observe both officials' beliefs and direct measures of their policy implementation.

This paper provides a novel study of bureaucratic ideology—and whether it shapes the implementation of coercive policy—by examining a formative institution in U.S. history: the Office of Indian Affairs (OIA). For more than a century, the Office exercised extensive control over the land, education, and legal governance of Indigenous populations. We focus on the dominant ideology that emerged and structured the agency's work in the late 19th and early 20th centuries: the aggressive promotion of Western norms over traditional cultural practices. During this period, federal policy toward Native Americans underwent major reforms, including the dismantling of collective tribal landholding and the expansion of off-reservation schools intended to "civilize" Native children.

The foundation of our analysis is an original, large-scale digitization of decades of archival records from the OIA. The data allow us to measure bureaucrats' ideological commitments and trace how the ideological composition of the bureaucracy evolved amid major shifts in appointment rules and federal assimilation policy. Moreover, we examine

whether variation in ideological commitments predicts the implementation of a pivotal assimilation policy. Key to our analysis is that Indian agents, the bureaucrats tasked with enforcing federal policy on reservations, were required to submit annual assessments describing their operations and activities. We use these reports to recover support for assimilation among OIA personnel from the tone and framing of their accounts, distinguishing normative endorsement of assimilation from merely describing or carrying out federal policy. To connect these beliefs to policy implementation, we digitize additional archival materials on early land allotment under the Dawes Act—a defining assimilation policy whose rollout required sustained local effort amid tribal resistance and ultimately contributed to large-scale dispossession.

To recover ideological commitment at scale, we extract systematic information on the intensity of each bureaucrat's support for assimilationist goals and policies using large language models (LLMs). Crucially, we calibrate our classification prompt using a benchmark sample of manually evaluated reports, coding each author's stance toward assimilation as strong support versus neutrality or opposition. This approach is similar to other recent efforts to use LLM-based classification methods to extract attitudes and sentiment from narrative documents (Lagakos, Michalopoulos and Voth, 2025) and builds on a broader literature using text to infer ideology (e.g., Laver, Benoit and Garry, 2003; Gentzkow, Kelly and Taddy, 2019; Kistner et al., 2025). To validate our classifier, we assess stability across repeated model runs, compare its classifications with blinded evaluations by history educators, and test whether the resulting ideology measure predicts an independent expression of policy support. Specifically, we examine whether assimilation scores from annual reports predict whether the same agents endorsed proposals for a federally funded Indian police force—a policy intended to weaken tribal authority and advance assimilation.

To understand our approach, consider two reports that discuss similar assimilation-related activity but differ sharply in their evaluative stance. William Daugherty's 1878 report from the Crow Creek Agency includes standard administrative updates—recounting the number of acres planted and the condition of agency buildings. But Daugherty also describes traditional beliefs as deeply entrenched "native vices," and warns that military force may be necessary to overcome the influence of local leaders opposed to assimilation. His report therefore expresses clear assimilationist support. By contrast, James Roberts's 1873 report from Camp Apache reflects a more neutral stance toward assimilation. His account focuses on logistical challenges such as insufficient supplies and the need for farm-

ing equipment and medical aid. Although Roberts notes that some Apache leaders wear Western clothing and that the community has embraced agriculture, he does not celebrate these changes as signs of moral or social advancement. Our measure is designed to capture this distinction: whether officials merely describe or implement assimilation-related policies, or normatively endorse assimilation as a policy objective.

Applying our measure to the full set of reports reveals that assimilationist commitment was widespread but changed markedly over time. Overall, 73 percent of reports written by Indian agents express strong support for assimilationist policies or goals. Support rose sharply in the 1870s, peaked in the 1880s, and began to moderate after 1890, even as federal assimilation policies, such as efforts to break up tribal land and expand Indian boarding schools, continued or even intensified. This moderation does not appear to reflect a broader retreat from assimilation policy by high-level leadership: Commissioner reports remained strongly supportive of assimilation throughout this period. Instead, the ideological shifts closely track changes in the institutions used to select OIA personnel. The rise in support coincided with the influx of religious nominees—who expressed especially strong assimilationist commitments—under Grant's Peace Policy, while the later moderation followed the end of denominational appointments and the growing presence of military officers and civil-service appointees, both of whom were less supportive of assimilation.

Finally, we ask whether the ideological commitments we recover from officials' reports predict consequential differences in the subsequent implementation of coercive policy. We focus on the early implementation of the Dawes Act of 1887, a defining assimilation policy of the era that aimed to break up tribal lands by assigning tracts of land (allotments) to individual Native Americans. Although the Dawes Act was a federal law, implementation proceeded gradually across reservations and depended on local agents to facilitate allotment in the face of tribal opposition. This setting allows us to relate assimilationist commitments measured from agents' reports written before Dawes to their agencies' allotment activity after the law's passage.

Our main finding is that local agencies led by agents with stronger prior assimilation support were substantially more likely to initiate allotment—and, once initiated, allotted more land—immediately following the law's enactment: a one standard deviation increase in assimilation support corresponds to an 11.4 percentage point increase in the probability of having any allotment and an increase of about 5,100 acres in the total land allotted during the first year of Dawes. These relationships remain broadly unchanged

after accounting for baseline agency characteristics and the personal attributes of agents. Moreover, balance and placebo exercises provide little evidence that our findings simply reflect the selective assignment of agents with stronger assimilation commitments to agencies predisposed to engage in early allotment activity. Thus, the ideology recovered from officials' writings predicts meaningful differences in subsequent policy implementation, providing behavioral evidence that our measure of ideology captures more than mere rhetoric in administrative reports.

This paper contributes to research studying major government campaigns that systematically repressed and coerced marginalized communities in the United States. This work has focused on the consequences of policies ranging from Jim Crow laws (Althoff and Reichardt, 2024; Olson, 2025), to immigrant language bans (Fouka, 2019, 2020), to the incarceration of Japanese Americans during World War II (Komisarchik, Sen and Velez, 2022).[1] Our work directly relates to studies of federal initiatives that disrupted Native communities. For example, Feir, Gillezeau and Jones (2024) show that the near-extinction of the bison—a state-enabled ecological shock—had lasting effects on Indigenous development and political organization. Related work studies flagship assimilation policies, including off-reservation boarding schools (e.g., Gregg, 2018; Maruthiah, 2024) and land allotment programs (e.g., Akee, 2020; Dippel, Frye and Leonard, 2024). Whereas this literature principally examines consequences for Native communities, we shift focus to those administering the policy—federal Indian agents—and introduce new data to recover direct measures of their ideology and trace its impacts on bureaucratic performance. Our large-scale analysis also builds on historical scholarship documenting the goals and ideological foundations of the OIA during the assimilation era (e.g., Priest, 1942; Hoxie, 1984; Prucha, 1984), as well as political science research on federal–tribal relations and Indigenous politics (Spirling, 2012; Evans, 2014).

Our work also contributes to a broad literature in political economy on state institutions and bureaucrats (Suryanarayan, 2024). Prior work highlights both that bureaucrats exercise meaningful discretion in policy implementation and that personnel institutions can shape the preferences and composition of the bureaucracy (e.g., Brehm and Gates, 1997; Bonica, Chen and Johnson, 2015). Within this literature, a theoretical strand emphasizes that bureaucrats may themselves value the substantive policy objectives they administer, and

[1] These studies form part of a broader literature on state repression and political control (e.g., Davenport, 2007; Hassan, Mattingly and Nugent, 2022), as well as research on assimilation and nation-building as strategies that states use to transform or incorporate minority populations (e.g., Mylonas, 2012; Fouka, 2024).

that such policy motivation can shape their incentives and behavior (e.g., Besley and Ghatak, 2005; Gailmard and Patty, 2007). We provide new empirical evidence on this form of policy motivation by directly measuring officials' normative commitment to a specific policy objective: cultural assimilation. This approach differs from related empirical work that has largely measured bureaucratic ideology through general partisan or ideological alignment (e.g., Clinton et al., 2012; Spenkuch, Teso and Xu, 2023), or has studied mission motivation in the delivery of public services rather than in coercive policy (e.g., Khan, 2025).[2] We then trace how personnel institutions were associated with variation in these commitments across the bureaucracy and show that stronger prior commitment predicted more aggressive subsequent implementation of the Dawes Act. These implementation results speak directly to theories of coercive state action, which emphasize that formal directives depend on the compliance of state bureaucrats (Tyson, 2018). Our evidence links this implementation margin to officials' substantive commitment to the state's objective.

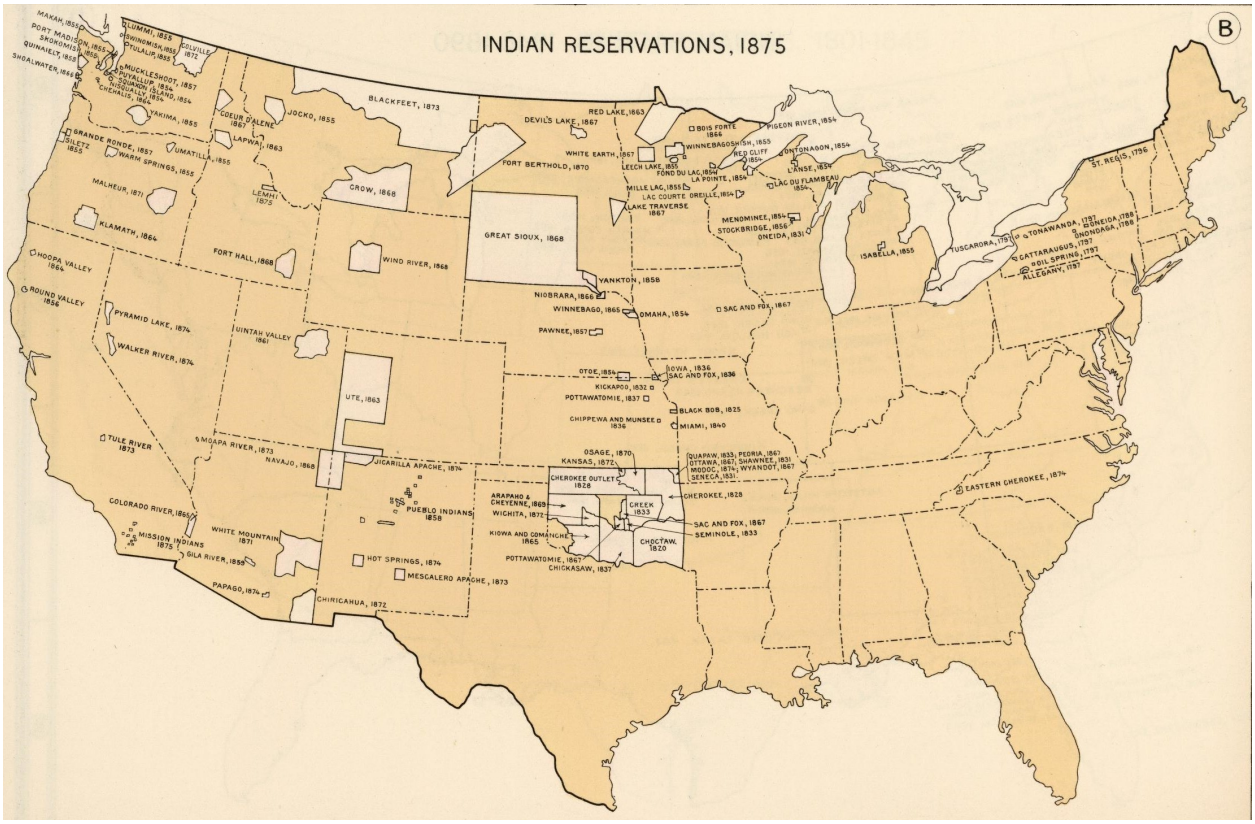
2 A History of Native American Reservations and the Office of Indian Affairs

The 19th century saw a profound transformation of Native American life as the United States displaced tribal nations through war, treaty-making, and forced removal to facilitate continental expansion. The Indian Removal Act of 1830 formalized a strategy of relocation, leading to the forced migration of tens of thousands of Native people from the southeastern United States to Indian Territory. In the decades that followed, U.S. expansion continued into the Great Plains and the Southwest, where many Indigenous communities were confined to reservations. Figure 1 provides a map of reservations in 1875 and highlights the system's wide territorial spread across much of the continental U.S. By the end of the century, this process had systematically undermined tribal sovereignty, leaving more than 200,000 Native people living under federal oversight on reservations across the United States.

The Office of Indian Affairs (OIA) was the principal federal authority governing reservation life. Established in 1824 within the War Department, the agency initially managed relations between the federal government and tribal nations during removal and treaty enforcement. In 1849, it was transferred to the newly created Department of the Interior, marking a shift from military control to civilian administration. Over the second half of the 19th century, as the number of reservations grew, the OIA expanded rapidly in both scale

[2]Our work also differs from and complements research examining the party affiliations of street-level bureaucrats such as police officers (Ba et al., 2025; Goncalves and Tuttle, 2025).

Figure 1: Map of U.S. Indian Reservations, 1875



Notes: Map shows Indian reservations as of 1875 (shaded light tan). Adapted from Plate B, "*Indian Reservations, 1875*," from Charles O. Paullin and John K. Wright, *Atlas of the Historical Geography of the United States* (Carnegie Institution of Washington, 1932). Digital scan courtesy of the Digital Scholarship Lab, University of Richmond. In the public domain.

and scope, with its workforce growing from fewer than 1,000 employees in the early 1870s to roughly 4,000 by the 1890s (Appendix Figure A1). The OIA's responsibilities ranged from distributing rations and managing land to running schools with teachers, courts with judges, and reservation police forces with their own officers and doctors. This unusually broad scope meant that the OIA often functioned as a de facto local government on many reservations.

By the late 19th century, the OIA had become the administrative center of an increasingly explicit federal campaign to assimilate Native Americans. Political leaders and a coalition of religious organizations, social reformers, and political elites advanced the view that Indigenous cultures should be dismantled and replaced with Western norms

(Priest, 1942).[3] In this framework, Native people were to become private landowners engaged in traditional Western farming and adopt Euro-American cultural practices (including converting to Christianity)—a vision that required deep intervention in family life, governance, and spiritual practice. This assimilationist agenda gave rise to a suite of coercive federal policies. Off-reservation boarding schools separated children from their families in order to suppress Native languages, religion, and identity (Adams, 1995; Lomawaima, 1995; Trennert, 1988; Vuckovic, 2024). The Dawes Act of 1887 authorized the division of tribal lands into individual allotments to erode collective land ownership and promote market agriculture (McDonnell, 1991). Allotments under Dawes also set in motion the reduction of Native landholdings by defining "surplus" acreage—land beyond what was needed for individual allotments—that would be opened for sale to non-Native settlers (Otis, 1973).[4] Additional policies banned ceremonial practices and imposed Anglo-American legal and religious systems. Together, these policies made the OIA bureaucracy the operational arm of a nationwide effort to transform Native social, cultural, and political life.

2.1 The OIA Bureaucracy, Discretion, and Personnel Selection

At the center of the OIA bureaucracy were Indian agents, the federal government's principal representatives on reservations who were appointed directly by the President, typically with Senate confirmation (Stuart, 1979). These agents exercised sweeping administrative, fiscal, and quasi-judicial authority over reservation affairs, and had substantial discretion to carry out federal policy. Until the expansion of civil service regulations in the late 19th century, Indian agents generally had the authority to appoint many of their subordinate employees, allowing them to shape the local administrative apparatus according to their own priorities and beliefs.[5] Staff included physicians, teachers, farmers,

[3] In the aftermath of the Civil War, a range of political and advocacy organizations promoted assimilationist reforms through lobbying, public campaigns, and missionary activity. These efforts included the creation of the Board of Indian Commissioners in 1869, which institutionalized Protestant and philanthropic oversight of Indian policy, and culminated in the annual Lake Mohonk Conferences, where federal officials, reformers, and religious leaders coordinated support for policies such as land allotment, off-reservation boarding schools, and the suppression of tribal governance and religious practices (Prucha, 1984).

[4] Contemporary officials frequently framed land allotment as an instrument of assimilation. Soon after the Act, Superintendent of Indian Schools Daniel Dorchester wrote in his annual report: "The breaking up of the more offensive features of the tribal relation is destined to follow the severalty allotments, and with that also must come, in time, habits of industry, individuality, and self-reliance, all of which will be conserving forces to strengthen the Indian youth upon whom the Government confers culture" (p. 335, U.S. Department of the Interior, 1889).

[5] While some subordinate agency employees were appointed directly through the OIA in Washington in response to concerns about corruption, Indian agents still shaped their staffs by recommending candidates

and clerks—many of whom were based on or near the reservations they served. Together, agents and their staffs administered routine duties and federal policies. The latter efforts often provoked resistance from tribal leaders and members who opposed the major assimilation policies of the era, such as boarding schools and land allotment (Adams, 1995; Otis, 1973).

The importance of individual agents in the implementation of federal policy was widely recognized by both OIA leadership and contemporary critics. As Commissioner Hiram Price observed in 1883, "The civilization and elevation of the Indian depends more upon the agents who have their immediate care and management than upon any and all other instrumentalities combined" (p. VIII, U.S. Department of the Interior, 1883). Yet, this same discretionary power raised deep concerns among reformers and Native advocates. Testifying to this, Native rights activist Thomas Tibbles wrote: "An Indian agent under our system is an absolute monarch. He can practice any cruelty and there is no appeal so long as he is in favor" (Unrau, 1972). Such concerns were reinforced by recurring allegations of corruption and abuse among agents during the 1870s and 1880s, including embezzlement of annuity funds, manipulation of census rolls, and the illicit sale of government rations (Prucha, 1984). Together, the breadth of agents' authority and persistent concerns over how that authority was exercised made the selection and oversight of Indian agents a recurring target of federal reform.

The process for selecting Indian agents became the subject of sustained debate and reform throughout the late 19th century. Prior to the election of President Ulysses S. Grant, Indian agents had primarily been appointed through a system of political patronage and selected based on partisan loyalties rather than competence or the ability to promote moral reform. In 1869, Grant implemented the so-called "Peace Policy," which initially attempted to promote non-political appointments by selecting army officers as agents. Under pressure from Congress, this approach was soon replaced by a system that delegated the power to nominate Indian agents to Christian religious denominations.[6] The system again reverted to political control by the 1880s as some members of Congress criticized the policy of religious favoritism, while others resented the loss of political patronage. Additional key reforms followed the Wounded Knee Massacre in 1890. Congress authorized military

and, at times, petitioning for the removal of unwanted appointees (Stuart, 1979).

[6] According to the account by Indian Agent Lawrie Tatum, the policy of delegating Indian agent appointments was initially promoted by the Society of Friends. Tatum wrote that Grant responded to the proposal by saying: "Gentlemen, your advice is good. I accept it. Now give me the names of some Friends for Indian agents and I will appoint them. If you can make Quakers out of the Indians, it will take the fight out of them." (p. 246, Sim, 2008).

officers to be detailed as Indian agents to fill vacancies, and a subsequent act enabled the Commissioner of Indian Affairs to abolish agent positions and elevate the superintendents of reservation schools to oversee agencies (United States, 1893). These superintendent positions fell under the classified Civil Service system, established earlier by the Pendleton Act, and therefore required candidates to pass competitive examinations. Together, these changes produced four distinct channels through which Indian agents entered office over our study period: political patronage, religious nomination, military detail, and civil-service promotion.

These shifting appointment policies, along with broader social and political changes, resulted in a heterogeneous OIA bureaucracy whose personnel arrived in office with different training and life experiences. Civilian appointees often had little or no prior experience working with Native populations (Stuart, 1979; Keller, 1983).[7] Military officers had a background in discipline and command, while many religious appointees were committed to the goal of moral reform and cultural transformation. Political appointees could arrive with quite different motivations, and contemporary accounts highlighted cases in which agents exploited their positions for private gain through fraudulent contracts and illicit sales. Later, civil-service superintendents generally rose through the OIA and brought prior administrative experience. Even among those appointed by reform-minded religious groups, motivations and levels of commitment were not uniform. John Armstrong was nominated for an agency position by his friends and initially declined his appointment to serve in New Mexico. After changing his mind and accepting the job, he ultimately resigned amid sustained accusations of misconduct (Bender, 1988). In contrast, Lawrie Tatum was given an unsolicited appointment but embraced the role as a moral calling and worked diligently to promote assimilationist goals such as agricultural reform (Cutler, 1971). The changing appointment system therefore altered not only how agents were selected, but also the mix of professional experiences, institutional ties, and motivations represented among officials who exercised broad authority over reservation life and implemented federal assimilation policy.

3 Data

Our analysis draws on a large corpus of annual reports authored by personnel of the OIA, which we link to an original collection of data on individual characteristics, agency

[7]For example, James M. Haworth was appointed as the agent to the Kiowa and Comanche despite lacking experience with Native populations and having no background in agency affairs (Hiatt, 1958).

characteristics, and policy outcomes. Using the combined data, we construct two main samples. The first is a report-level sample that treats each annual report produced by the OIA workforce as the unit of analysis. While this corpus includes reports from a range of OIA personnel, our principal analyses focus on reports written by Indian agents and other officials leading local agencies. The second is an agency-level sample, which links the prior ideological commitment of agencies' leadership with records of subsequent land allotment at reservations under the jurisdiction of these agencies. This section describes the primary data sources and provides an overview of the construction of each sample. Further details are provided in Appendix B.

Annual Reports: Our core data source consists of annual reports filed by Indian agents and other local OIA personnel stationed on or near reservations across the country. OIA regulations required that Indian agents submit a written report every year to the Commissioner of Indian Affairs, summarizing activities, progress, and notable developments during the previous year.[8,9] In addition to agents, annual or periodic reports were also filed by other local officials such as reservation school teachers, physicians, and farmers, often for inclusion in the agent's own annual report. We use the annual reports produced from 1868 to 1906, the period when the Annual Report of the Commissioner of Indian Affairs (ARCIA) systematically published reports authored by local OIA personnel alongside the Commissioner's own report. We begin in 1868, before President Grant's administration introduced reforms that altered the system for selecting Indian agents. We extracted these reports from each volume of the ARCIA and recorded metadata on the reporting agency, its location, the author's name as it appeared in the report signature, and the author's title (e.g., "Indian Agent" or "Physician in Charge").

Characteristics of Personnel: We rely on a range of sources to obtain key background characteristics of the authors of the annual reports in our corpus. A major limitation of the reports is that they contain no biographical details: in a large share of records (41 percent), the signature line includes only an initial for the first name alongside a full last

[8] Appendix Figure A2 reproduces the official annual reporting directions from the 1884 Regulations of the Indian Department.

[9] Before institutional reforms in the 1870s, many Indian agents reported to superintendents of Indian Affairs who oversaw multiple agencies and relayed information, including annual reports, to the Commissioner. These superintendent positions were gradually eliminated by congressional acts, with the last one abolished in 1878 (Prucha, 1984). We stress that the superintendents of Indian Affairs in this earlier system served as an intermediate administrative authority and should not be confused with the homonymous but distinct position of "superintendent" at local reservation schools.

name. To recover additional personnel characteristics, we draw on the federal biennial register, a personnel directory listing federal employees and information such as place of birth, salary, and, in some cases, complete first names. We supplement these records with automated and manual searches of state historical archives and genealogical sources to recover full names, birth years, and places of birth. All matches are manually reviewed and retained only when the external source directly documents the individual's employment or involvement with the OIA. Further details on this matching procedure are provided in Appendix B.

For authors identified as leading a local OIA agency, there are two types of formal positions: Indian Agents and reservation school superintendents (i.e., career civil servants) promoted to oversee an agency. As noted above, the promotion of superintendents to a broader leadership role became possible after Congressional reforms in 1893 (United States, 1893). Our analysis focuses on authors holding agency head positions, which we group in a binary indicator that codes both Indian agents and promoted superintendents as a single category. Consistent with this coding, we refer to all agency heads as "agents" throughout the paper. We measure appointment types and identify promoted superintendents using schedules in the ARCIA, which listed each agency and the personnel in charge during a given year. Among Indian agents, a key distinction is whether their appointment was through a military detail or a nomination by a religious denomination. Military appointments are identified in the schedule by the inclusion of a formal military rank (e.g., "Captain, U.S.A.") next to the agent's name, while promotions of school superintendents are indicated by the title "school superintendent" appearing alongside the individual's name. Religious appointments are inferred by matching a non-military agent's first year of service with the period during which their agency was officially assigned to a religious denomination, based on a separate ARCIA table that reported these assignments. Such religious appointees are temporally clustered in the 1870s and 1880s, reflecting the fact that the denomination-based appointment system was discontinued after 1881.

Dawes Allotment Records: As the measure for our analysis of the relationship between bureaucratic ideology and policy implementation, we use land allotment statistics reported in the ARCIA. The Dawes Act passed in 1887, and we focus on its initial implementation using statistics on the number of allotments and total acres allotted from the 1888 ARCIA. The original records are presented for individual tribes. Since agents were assigned to agencies that often encompassed multiple tribes, we aggregate the underlying

records to the agency level to align with our ideology measures. As robustness checks, we also use comparable allotment statistics from the 1889 and 1890 ARCIA.

Characteristics of Agencies: We digitize measures of agency-level characteristics from several benchmark years to include as controls in our empirical exercises. For our analysis of variation in assimilationist commitments across personnel and agencies, we digitize information on total acres under an agency's jurisdiction, total Native population, the fraction of the Native population reported wearing Western dress (as a proxy for cultural change), and livestock holdings (as a measure of economic conditions). We draw these measures from three benchmark "anchor" years (1875, 1885, and 1895), selected to provide broad coverage and comparability across the study period. To align the annual reports with these benchmarks, we assign each report to the closest anchor year within its decade (e.g., reports from 1868–1879 are linked to 1875; 1880–1889 to 1885; and 1890–1906 to 1895), with the goal of assigning each report a corresponding set of agency controls and avoiding missing observations. As detailed in Appendix B, we construct harmonized crosswalks for each period to link the agency listed in an annual report to its counterpart in the relevant benchmark year. This procedure accounts for the fact that agencies frequently opened, closed, merged, or were renamed during the period, meaning that report-year agency titles do not always map cleanly onto a single benchmark-year agency.[10] In addition, for our analysis of the relationship between personnel ideology and the implementation of the Dawes Act, we digitize the same four agency-level characteristics from the 1887 ARCIA.

3.1 Samples for Main Analysis

Our main analysis uses two samples. The first is a report-level sample constructed to study the ideology of the OIA bureaucracy and its evolution over our nearly four-decade study period. The full corpus from 1868 to 1906 contains 4,114 annual reports. We exclude reports with fewer than 500 words, as these contain limited information for inferring author ideology and beliefs. The resulting sample includes 3,554 reports written by 1,301 distinct authors. Authors typically appear in the corpus for only a small number of years, contributing fewer than three reports on average, consistent with substantial turnover among OIA personnel. A total of 2,384 reports were written while the author

[10]For example, the Mescalero and Jicarilla agencies operated separately in 1881 but were merged in 1882 into a single agency (U.S. House of Representatives, 1882). Accordingly, all reports from the separate Mescalero and Jicarilla agencies in 1881 are mapped to the combined agency's statistics reported in the 1885 ARCIA.

was serving as a local agency head, while the remaining 1,170 reports were written by other OIA personnel. Our principal analyses focus on the agent reports, while we use the broader corpus to characterize ideology across the OIA and to assess the properties of our text-based measure. Additional descriptive characteristics of report authors are available in Appendix B.5.[11] The final data is augmented by including agency-level characteristics from several benchmark years as discussed above.[12]

The second sample is defined at the agency level and is used to examine whether the ideological commitments of agency leadership predict the implementation of the Dawes Act. As detailed in Section 5, the Act was passed in 1887, midway through our observation period, and stands as a natural test of whether personnel with stronger assimilationist commitments acted more aggressively when tasked with executing a high-stakes, assimilation-oriented policy. Because we observe agents' expressed views on assimilation in reports written before 1887, we can treat these ideological positions as predetermined with respect to subsequent allotment activity.

The agency-level sample includes 55 agencies that were in operation in 1887 and whose leadership at the passage of the Dawes Act had sufficient reporting history to measure the strength of their prior commitment to assimilation.[13] For all agencies in our sample, we construct pre-1887 measures of ideology using all available annual reports written by the agent in charge. We augment these data with digitized records describing each agency's social and economic context in 1887, including the total acres under supervision, the total Native population, the fraction of the Native population reported wearing Western dress, and the size of livestock holdings.

[11] Notably, Appendix Table A1 shows that authors typically appear in only a small number of annual reports: the average author contributes 2.7 reports, while authors who ever served as agency heads contribute about 3.1. The appendix also documents a notable geographic contrast: although most reports come from agencies in the West and Midwest, most linked authors were born in the Northeast or Midwest rather than the West.

[12] A small number of non-agent reports were written from non-reservation institutions such as military outposts (e.g., Fort Apache). There are no agency-level measures—such as the number of acres overseen—to link to these reports. We retain such reports in relevant analyses, setting missing values to zero and including indicators for missing data.

[13] We exclude four agencies that experienced leadership changes in the year immediately preceding the Dawes Act. In addition, we exclude the Union Agency, which had responsibility for the "Five Civilized Tribes"—the Cherokee, Chickasaw, Choctaw, Creek, and Seminole in what is now eastern Oklahoma—who were exempted from the Dawes Act of 1887.

4 Methodology for Identifying Assimilation Ideology

This section describes our main methodology for identifying ideological commitment to assimilation policies within the Indian Affairs bureaucracy. We begin with an overview of our LLM classification approach. To better understand our measure, we discuss in detail one annual report and the resulting LLM-generated assimilation support score and justification. Next, we discuss the classification results across the full corpus, focusing in particular on whether differences in measured assimilation support reflect differences in the topics personnel discuss or instead differences in their evaluative framing of those topics. To validate the measure, we rely on comparisons with human coding and an independent measure of assimilation policy support from a separate survey of Indian agents by the Board of Indian Commissioners. Finally, we conduct complementary textual analyses of the reports and assess the stability of our classifications across repeated model runs and alternative prompt specifications.

4.1 Overview of LLM Classification Approach

To measure the ideological commitments of bureaucrats, we developed a classification approach using a large language model (LLM) to identify whether annual reports express normative support for assimilation.[14] Our structured prompt was developed using OpenAI's GPT-4o and instructed the model to rate each report's attitude toward assimilation objectives and policies such as boarding schools and land allotment.[15] We use a binary classification in which "1" indicates clear support and "0" indicates neutrality or opposition. Importantly, a neutral description or discussion of the implementation of federal policy is not sufficient evidence of ideological support: the prompt assigns a supportive classification only when the text contains clear normative endorsement or approval of assimilation objectives.[16] In addition to the score, the model returned a detailed justification and three separate pieces of evidence, each tied to a quote from the report. This approach broadly follows the LLM approaches in Lagakos, Michalopoulos and Voth (2025) and Fang, Li and Lu (2025). The supporting quotations make the basis for each classification

[14] Related work has used LLMs to construct measures of executive-agency ideology on a general liberal–conservative dimension (Napolio, 2025); our approach instead uses officials' own reports to recover their normative support for a specific substantive policy objective.

[15] Specifically, our main classifications were produced on June 10, 2025 using the `gpt-4o-2024-08-06` release.

[16] Note that we do not separate paternalistic expressions from those that express the view that Native culture should be eliminated on moral grounds. This is because both lead to support for the OIA's assimilation policies.

directly auditable, and we verify that quoted material appears in the underlying source report.[17]

Further details on the prompt and our development process are reported in Appendix C. To summarize, we developed the prompt through a calibration exercise using 30 randomly selected annual reports that we manually coded based on our definition of assimilation support. We then developed an initial candidate prompt and compared its classifications in the sample of 30 reports with our benchmark scores. We iteratively refined the prompt by editing its decision rules, treatment of edge cases, and the examples contained in the prompt. The final prompt agreed with our manual classifications in 28 of the 30 reports (93 percent).[18]

4.2 Illustrative Example

To demonstrate how our classification approach operates in practice, Figure 2 presents selected excerpts from one annual report alongside the corresponding output from our LLM prompt. Specifically, we return to William E. Daugherty, the Indian agent we briefly described in Section 1. We now focus on his 1880 report from the Lower Brule Agency in South Dakota. As shown in Panel (a), the report includes a direct endorsement of federal efforts to restructure Native life through disarmament, compulsory education, and the abolition of tribal practices. Daugherty describes tribal sovereignty and customs as "superstitious practices" and calls for their elimination, later emphasizing that these changes can only be achieved through the use of physical force.

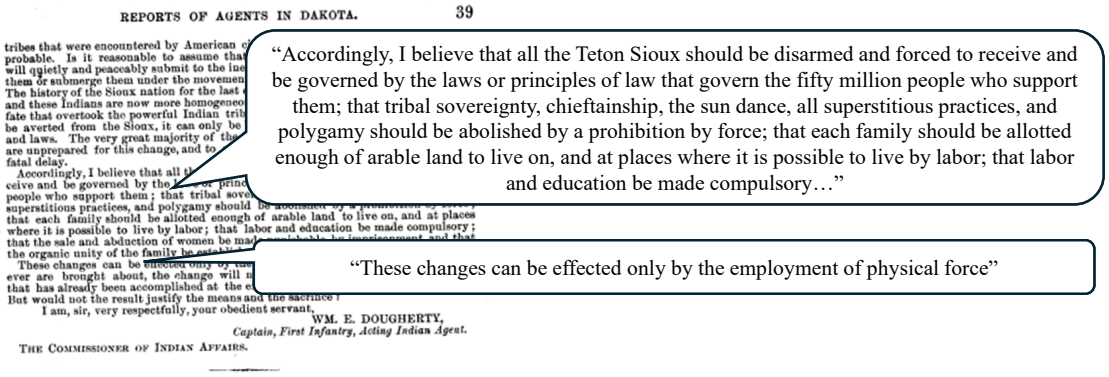
Panel (b) displays the model's structured output, including the support score (i.e., a value of "1" for strong assimilationist support) and the LLM-generated justification. As expected given our prompt, the justification output identifies the features of the original record highlighted above: the writer explicitly endorses specific federal assimilation efforts, including the disarmament of the Sioux and the imposition of external cultural norms. The model's reasoning clearly picks up the author's evaluative framing of Native traditions as obstacles to be overcome based on the normative language.

[17] The prompt also included detailed decision rules and concrete examples that clarify the distinction between mere compliance and ideological support. For instance, an agent might describe the transfer of students to a federal boarding school without endorsing the program; our approach codes only explicit normative approval or evaluative language as ideological support, not neutral accounts of compliance.

[18] Appendix Table A2 presents a subset of illustrative examples, showing the manual and GPT-4o scores together with our justification statements and representative quotes. Note that our use of explicit instructions and examples followed recommended best practices from the OpenAI (2024) prompt guidelines.

Figure 2: Annual Reports and Assimilation Score Examples

(a) 1880 Lower Brule, Wm. E. Daugherty, Excerpts


REPORTS OF AGENTS IN DAKOTA. 39

tribes that were encountered by American
probable. Is it reasonable to assume that
will quietly and peaceably submit to the in
them or submerge them under the movemen
The history of the Sioux nation for the last
and these Indians are now more homogeneo
fate that overtook the powerful Indian trib
be averted from the Sioux, it can only be
and laws. The very great majority of the
are unprepared for this change, and to
fatal delay.

Accordingly, I believe that all t
ceive and be governed by the
people who support them; that tribal sove
superstitious practices, and polygamy should
that each family should be allotted enough of arable land to live on, and at places
where it is possible to live by labor; that labor and education be made compulsory;
that the sale and abduction of women be mad
the organic unity of the family be est

These changes can be effected only by
ever are brought about, the change will
that has already been accomplished at the
But would not the result justify the means and the sacrifice?

I am, sir, very respectfully, your obedient servant,

WM. E. DOUGHERTY,
Captain, First Infantry, Acting Indian Agent.

THE COMMISSIONER OF INDIAN AFFAIRS.

"Accordingly, I believe that all the Teton Sioux should be disarmed and forced to receive and be governed by the laws or principles of law that govern the fifty million people who support them; that tribal sovereignty, chieftainship, the sun dance, all superstitious practices, and polygamy should be abolished by a prohibition by force; that each family should be allotted enough of arable land to live on, and at places where it is possible to live by labor; that labor and education be made compulsory…"

"These changes can be effected only by the employment of physical force"

(b) Score and Rationale for 1880 Lower Brule, Wm. E. Daugherty

Assimilation score	**Assimilation score justification**
1	The text expresses a clear endorsement of assimilation policies, advocating for the disarmament of the Sioux and the imposition of laws and principles of the dominant society. The writer supports the abolition of tribal sovereignty and cultural practices, viewing them as obstacles to progress. The language used indicates a belief that assimilation is necessary and morally justified.

Notes: Panel (a) is a selected excerpt of William Daugherty's 1880 Annual Report from the Lower Brule Agency. The text produced in the callout box highlights a key portion of text that informs our large language model (LLM) assimilation support scoring model. Panel (b) shows the resulting LLM output which includes the binary score along with the LLM's overall scoring justification. In addition to the overall justification, the full output from our scoring model includes three quotes from the original text which includes the highlighted portion.

4.3 Interpreting Classifications at Scale: Aggregate Results, Topics, and Justification Analysis

Once the prompt was finalized, we applied it to our main sample of 3,554 annual reports produced by Indian Affairs personnel between 1868 and 1906. Table 1 summarizes the resulting classifications and several characteristics of the reports. Column 1 describes the full corpus, while Columns 2 and 3 separately describe reports written by agents and other OIA personnel. Because our subsequent substantive analyses focus primarily on agency leadership, Columns 4 and 5 then divide agent reports according to whether they are classified as assimilation supporting or neutral/non-supporting.

As shown in Panel A of Table 1, the model identifies 70 percent of all reports as express-

ing strong assimilationist attitudes—defined by normative endorsement of assimilation-related policies such as boarding schools, allotment, and the dismantling of tribal institutions. This rate of ideological support provides early evidence that assimilationist beliefs were widespread in the OIA workforce. Assimilation support is particularly widespread among agents, with nearly three-quarters of their reports classified as supportive. Panel A also reports average report length and shows that assimilation-supporting agent reports tended to be somewhat longer on average than neutral reports.[19]

A natural question is how the assimilation scores are related to the underlying topics discussed in each report. This distinction is important because text-based measures of ideology may incidentally capture differences in the topics actors discuss rather than their underlying ideological positions (Grimmer and Stewart, 2013). We use a simple keyword-based classification approach to identify whether a report discusses (i) boarding schools, (ii) land or allotment policy, (iii) either of these two types of policies, or (iv) both. Boarding schools and land allotment were among the most prominent and contested components of federal assimilation policy during our sample period, and thus provide a reasonable test of how topics are associated with the model's ideological classifications. Using the flags for assimilation policy, we compare topic prevalence across assimilation-supporting and neutral/non-supporting reports. Panel B first reports the prevalence of these topics for agent and non-agent reports and then, in Columns 4 and 5, compares topic prevalence within the agent sample by assimilation classification. The latter comparison is particularly informative because it holds fixed the broad institutional role of the author.

Overall, this analysis reveals no systematic differences in the frequency of discussing major topics. Nearly all agent reports discuss land or allotment policy regardless of whether they are classified as supportive or neutral/non-supporting (99.2 versus 99.4 percent), and the corresponding shares discussing either major assimilation policy are 99.4 and 99.5 percent. Differences in discussion of boarding schools (35.5 versus 32.6 percent) and both policies (35.4 versus 32.4 percent) are also modest. Thus, among the officials who form the core of our later analysis, variation in measured assimilation ideology is not simply a consequence of whether they discuss major assimilation policies. Appendix Table A3 reports the corresponding support/neutral comparisons for the full corpus and for non-agent reports.

Finally, Panel C reports results from examining the justification statements returned

[19] In subsequent empirical exercises, we control for word count, recognizing that longer text may reveal more information from the authors in our corpus. Controlling for word count leaves our results virtually identical in magnitude and significance.

Table 1: Annual Reports, Assimilation Topics, and Ideological Classification

	(1)	(2)	(3)	(4)	(5)
	All Reports	Agents	Non-agents	Agent Reports	
				Support	Neutral
A. Reports					
Assimilation score	0.70	0.73	0.65	1.00	0.00
Word count (avg.)	1,917	2,175	1,393	2,226	2,040
N (reports)	3,554	2,384	1,170	1,733	651
B. Major Topics					
Topic: Boarding schools	0.386	0.347	0.467	0.355	0.326
Topic: Land/allotment	0.937	0.993	0.823	0.992	0.994
Topic: Either policy	0.968	0.994	0.915	0.994	0.995
Topic: Both policies	0.355	0.346	0.374	0.354	0.324
C. Justification Word Statistics (Agent Reports)					
superiority	–	–	–	0.259	0.000
beneficial	–	–	–	0.193	0.005
factual	–	–	–	0.000	0.708
descriptive	–	–	–	0.000	0.559
administrative	–	–	–	0.000	0.498

Notes: This table summarizes report characteristics, major assimilation-policy topics, and selected words from the LLM-generated justification statements. Column 1 uses all annual reports in the analysis sample. Columns 2 and 3 split reports according to whether the author was serving as an agent at the time the report was written; agents include Indian agents and, after 1893, reservation school superintendents promoted to agency leadership. Columns 4 and 5 restrict the sample to agent reports and separate reports classified as assimilation supporting from those classified as neutral/non-supporting. Panel A reports the mean assimilation score, average report word count, and number of reports. Panel B reports the share of reports discussing boarding schools, land or allotment policy, either policy, and both policies. Panel C shows the frequency of selected words used in the LLM's justification statements as a test of the differences in rhetorical style. The list of words in Panel C was identified by comparing word frequencies across assimilation-supportive and neutral reports. Words that appeared disproportionately in supportive or neutral reports were then used to construct the set of keywords we focus on.

by the model for each report. We conduct a simple textual analysis of the model's justifications, comparing common words across assimilation-supporting and neutral/non-supporting cases. Justifications for assimilation-supporting reports are substantially more likely to use terms such as "superiority," "beneficial," and "civilizing," whereas justifications for neutral reports are more likely to include language such as "factual," "descriptive," or "administrative." These patterns are consistent with the distinction built into our measure: supportive reports contain evaluative or normative endorsement of assimilation, whereas neutral reports may discuss the same policies without expressing such

endorsement. Because these justifications are produced by the same model that assigns classifications, we view this exercise as a basic diagnostic test that makes the basis of the classifications more transparent. We turn next to independent validation of the measure we create.

4.4 Validation of the Assimilation Support Measure

We next assess whether our text-based assimilation measure corresponds to assessments and policy preferences observed independently of the LLM classification procedure. We conduct two complementary validation exercises. First, we compare the LLM classifications with judgments from human evaluators who apply the same substantive definition of assimilation support to a randomly selected set of annual reports. Second, we link agents' annual-report classifications to their independently recorded responses to a contemporaneous policy proposal concerning the creation of federal Indian police forces. Together, these exercises test whether the measure corresponds both to human readings of the underlying texts and to policy preferences expressed outside the annual-report setting.

Human Audit Comparisons: We begin assessing validity by comparing our LLM classifications to those from a blinded audit with five US-born high school teachers who each hold undergraduate degrees in History. We hired these auditors online via CloudResearch Connect's existing participant-targeting questions. Our audit sample characteristics were chosen to ensure that auditors possessed relevant contextual knowledge of U.S. history needed to accurately read the annual reports. Each auditor received the substantive instructions from our final prompt and independently classified the same 10 annual reports, which were selected at random from our full sample. The auditors were paid $15 each for the task that we estimated would take 90 minutes. Auditors were informed that compensation would only be paid after we verified that they completed all scores and provided justifications.

While the literature on LLM performance is evolving (Gilardi, Alizadeh and Kubli, 2023; Bojić et al., 2025; Törnberg, 2023), we make progress on establishing useful practices for testing LLM classification performance using two exercises. First, we use a novel and intuitive hold-out exercise to assess the performance of the LLM against human evaluators. Using each human assessment, we consider whether the LLM achieves a higher or equal agreement rate with the remaining human evaluators. This test assesses whether

Table 2: Pairwise Comparison of LLM versus Human Raters for 10 Reports

	A. Hold out R1			**B. Hold out R2**			**C. Hold out R3**			**D. Hold out R4**			**E. Hold out R5**		
	Agree %			Agree %			Agree %			Agree %			Agree %		
Report	LLM	R1	≥	LLM	R2	≥	LLM	R3	≥	LLM	R4	≥	LLM	R5	≥
1	.75	.75	1	.75	.75	1	.75	.75	1	.75	.75	1	1	0	1
2	.75	.25	1	.5	.5	1	.5	.5	1	.75	.25	1	.5	.5	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	.75	.75	1	1	0	1	.75	.75	1	.75	.75	1	.75	.75	1
5	.5	.5	1	.5	.5	1	.75	.25	1	.75	.25	1	.5	.5	1
6	.75	.25	1	.5	.5	1	.5	.5	1	.5	.5	1	.75	.25	1
7	.5	.5	1	.75	.25	1	.5	.5	1	.5	.5	1	.75	.25	1
8	.25	.75	0	.25	.75	0	0	0	1	.25	.75	0	.25	.75	0
9	.5	.5	1	.25	.25	1	.5	.5	1	.5	.5	1	.25	.25	1
10	.75	.75	1	.75	.75	1	1	0	1	.75	.75	1	.75	.75	1
Test	$p < .05$			$p < .05$			$p < .01$			$p < .05$			$p < .05$		

Notes: This table compares LLM performance versus each of five human evaluators in a series of hold-out comparisons. Each row in the table is a report in our audit study. For each report, we compute the agreement rate of the LLM and a held-out human (indicated in a given panel) with the remaining four human evaluators. For instance, the first row and second column shows the LLM agrees with auditors 2,3, 4, and 5 on report "1" in 75% of cases. The final column in each panel (indicated with ≥) indicates whether the LLM's agreement rate is greater than or equal to the held out human rater. Across all five hold out exercises, the LLM matches or outperforms the human in either 9 or 10 of the 10 reports. We use a one-sided test to assess whether the LLM's performance is higher or greater than a given human rater across all 10 documents. For all the 5 auditors, we reject the hypothesis that the LLM performance is worse than a given human evaluator.

a given human evaluator would outperform the LLM. Table 2 reports the agreement rates associated with five hold-out exercises that we can construct. Across all exercises, the LLM matches or exceeds the agreement rate of the held-out human rater in either 9 or all 10 of the scored reports. Using a one-sided comparison test for matched pairs, we reject the null hypothesis that the LLM performs worse than the human rater at the 5 percent or lower level in all cases. This test demonstrates that LLM performance can be meaningfully validated even when using a limited sample of human evaluators and a small set of reports.

As an additional exercise, we compare LLM classification with human responses using

Fleiss' κ, a standard measure of inter-rater reliability that captures the extent to which multiple raters agree in their evaluations beyond what would be expected by chance.[20] We begin by computing κ using only the evaluations from the five history educators. We then recompute the same statistic after including the GPT-4o model as a sixth "rater." If the LLM's judgments were misaligned with the human raters, we would expect its inclusion to lower or leave unchanged the overall level of agreement.

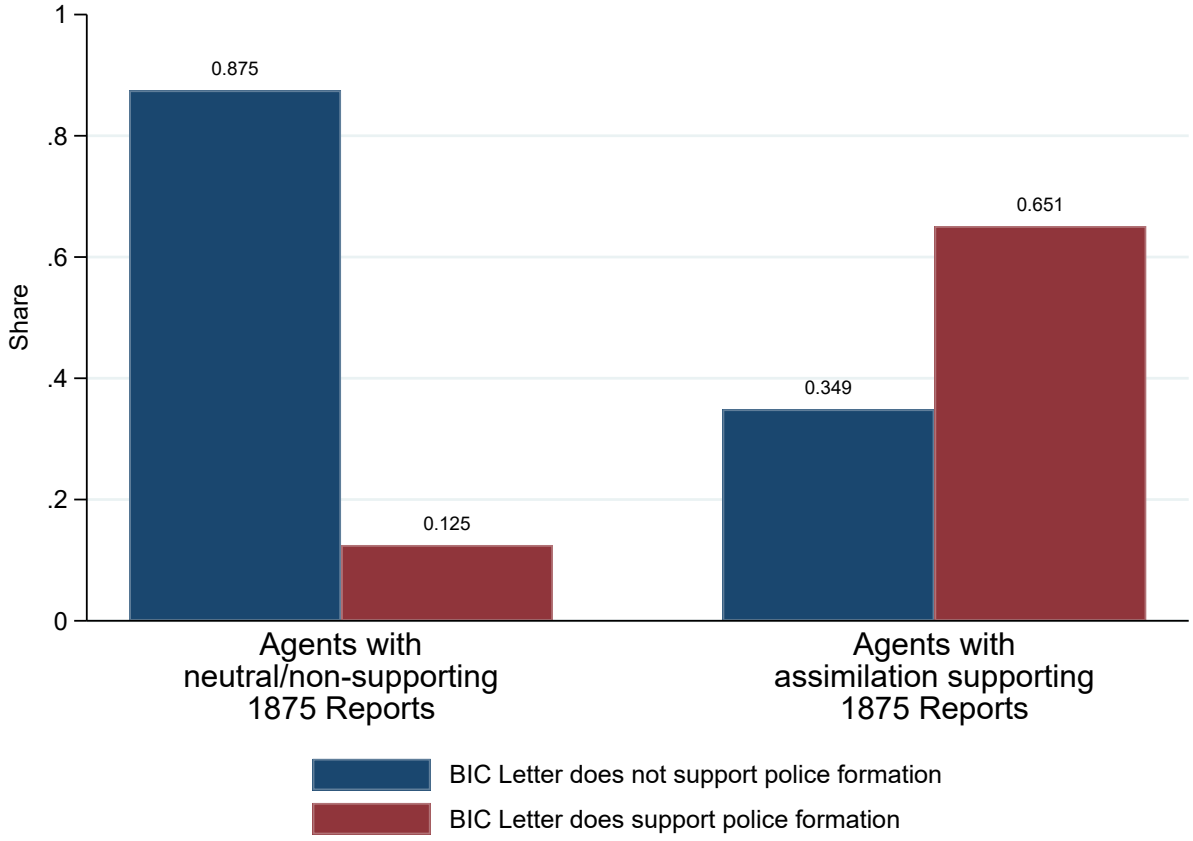
Our main finding is that the κ increases by approximately 20 percent. In line with our hold-out sample testing, the improvement in κ indicates that the LLM scores are more in line with the human raters, and, as a result, its inclusion brings the overall set of evaluations into closer consensus. To test whether this improvement could occur by chance, we perform a placebo-based randomization test, confirming that the improvement is unlikely to be driven by chance ($p < 0.05$).

Predictive Validity: Next, to assess the predictive validity of our ideology measure, we test whether it is correlated with support for a consequential reform closely tied to the federal assimilation agenda. In 1875, the Board of Indian Commissioners (BIC)—a quasi-official advisory body created by Congress—sent letters to every Indian agency, asking whether the agent would endorse the formation of a federally funded Indian police force. The development of a local police force was in part designed to weaken traditional tribal authority by establishing a state-controlled institution of law enforcement on reservations. As Commissioner of Indian Affairs Hiram Price later described, "[t]he police force is a perpetual educator. It is a power entirely independent of the chiefs. It weakens, and will finally destroy, the power of tribes and bands..." (p. XVII-XVIII, U.S. Department of the Interior, 1881). Because the responses to the BIC were written prior to the widespread adoption of police forces and come from separate archival correspondence, they offer a unique opportunity to assess how agent beliefs—captured in annual reports—translated into real-time policy preferences.

We manually coded the full set of 1875 BIC responses from responding local agents, classifying each letter as either supportive or non-supportive of creation of a police force at their agency. We then linked these letters to the same set of agents for whom we had assimilation policy support scores based on their 1875 annual report. In total, the resulting matched sample is a set of 55 agents who submitted both an annual report and a

[20] Fleiss' κ is calculated as $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$, where $\bar{P}$ is the observed agreement among raters and $\bar{P}_e$ is the expected agreement by chance, given the marginal distributions of responses. A value of 1 indicates perfect agreement, while a value of 0 reflects agreement no better than random guessing.

Figure 3: Predictive Validity of Assimilation Scores: Evidence from BIC Letters



Notes: This figure shows the share of Indian agents supporting or opposing a policy proposal to create a federally funded Indian police force at their local agency, broken out by their assimilation support scores. The sample is based on 55 agents who submitted both an annual report and a BIC response in 1875. Assimilation scores are based on each agent's 1875 annual reports using our LLM-based classification, while police support is a binary measure manually coded from responses to Board of Indian Commissioners letters sent that year.

BIC response in that year.[21] Figure 3 shows the share of agents who do not and do support the creation of a police force broken out by the 1875 assimilation scores assigned by our LLM approach.

In sum, the results show a strong association between model-classified ideology and policy support. Among the agents classified as strong assimilationist in 1875, 66 percent

[21]Responses to the BIC letters were published in the 1875 Annual Report of the Board of Indian Commissioners to the Secretary of the Interior (Board of Indian Commissioners, 1875). There are a total of 63 responses to the BIC inquiry regarding police forces, implying that we have a match rate of 87 percent (= 55/63).

supported the police reform. In contrast, only 13 percent of agents classified as neutral or non-supporting did so. This 53-percentage-point difference is statistically significant at the one percent level. The magnitude of the association suggests that our classifier captures more than linguistic framing: it recovers meaningful variation in bureaucratic beliefs that shaped real policy positions at the time.

4.5 Textual Analysis, Stability, and Sensitivity

We conclude our measurement analysis with three sets of complementary exercises that further characterize the assimilation classifications and assess the robustness of our scoring procedure. First, we use standard textual analysis methods to examine whether assimilation-supporting and neutral/non-supporting reports exhibit systematic differences in their underlying language and semantic content. Second, we assess the stability of the classifications by repeatedly scoring the same reports while holding the prompt and model settings fixed. Third, we examine sensitivity to alternative versions of the classification prompt, including changes to its framing, instructions, and examples. We summarize the main findings from these exercises below and report additional methodological details and results in Appendix Section D.

Standard Textual Analysis: We first use a standard dictionary-based approach to ask whether reports assigned different assimilation scores also differ in their evaluative tone. Specifically, we use the NRC Word-Emotion Association Lexicon, which maps words to positive and negative sentiment and to several emotional categories (Mohammad and Turney, 2013). Because some words have distinctive meanings in our historical administrative setting, we remove a set of domain-specific terms before applying the lexicon. The resulting differences are modest but systematic. Assimilation-supporting reports contain approximately nine more positive-sentiment terms per 1,000 words than neutral/non-supporting reports and make greater use of words associated with emotions such as joy, trust, and anticipation, while using fewer terms associated with fear, disgust, and sadness. The positive-sentiment difference remains when we additionally exclude terms such as "civilize," "influence," and "progress" that could mechanically flag assimilationist ideology, although some of the differences across individual emotion categories attenuate.

As a second exercise, we compare which words are disproportionately represented in assimilation-supporting and neutral/non-supporting reports. Rather than beginning with a predefined dictionary, this approach identifies terms that occur unusually often

in one group relative to the other while limiting the influence of very rare words (Monroe, Colaresi and Quinn, 2008). The resulting vocabulary differences are substantively interpretable. Terms most characteristic of assimilation-supporting reports include "civilize," "dance," and "progress," which are closely connected to cultural transformation and assimilation. In contrast, terms such as "guardian," "council," and "lease" are more characteristic of neutral/non-supporting reports and largely reflect administrative subject matter.

Our third and most comprehensive textual exercise asks whether the two sets of reports differ in their broader semantic content rather than the frequency of particular words. We use pretrained sentence-embedding models, which represent passages of text numerically such that passages with similar meanings receive similar representations even when they do not use the same vocabulary. For example, two passages advocating the adoption of Christianity or Western farming practices can be semantically similar despite sharing relatively few individual words, while both will be distant from a passage reporting financial accounts. We divide each annual report into shorter passages, generate an embedding for each passage, and average these representations to construct a single semantic representation of the report. We then ask how well these document-level representations can distinguish reports classified by the LLM as assimilation-supporting from those classified as neutral/non-supporting.

The semantic representations contain substantial information about the LLM classifications. Using our baseline embedding model, a classifier trained only on these representations achieves a balanced accuracy of 0.69 and an AUC of 0.76, compared with 0.50 for either measure under simple benchmark classifiers; average precision is 0.87 compared with a benchmark of 0.70. Performance improves further when we use higher-dimensional and instruction-based embedding models. These results show that reports assigned different assimilation scores differ systematically in their semantic content, not simply in a small number of individual words or phrases. Because the outcome being predicted remains our LLM-assigned classification, we interpret this exercise as a complementary textual diagnostic rather than an independent validation test.

Stability/Consistency: Large language models are not fully deterministic: even when provided with identical inputs, they can return slightly different outputs due to their probabilistic architecture and internal sampling behavior (Zhao et al., 2023; Atil et al., 2025). To reduce randomness and prioritize stability, we set the model temperature to 0 in

all API calls. This setting minimizes variation in the model and increases the likelihood of consistent output across runs. To assess the practical importance of the random variation inherent in LLMs such as GPT-4o (even with temperature set to 0), we randomly sample 200 reports and evaluate each five times using our finalized prompt and all settings fixed. We find that 93.5 percent of letters received the same score in all five iterations, and 98 percent received the same score in at least four out of five iterations (Appendix Table A4). Overall, we interpret this evidence as suggesting that our prompt generates highly consistent scores.

Sensitivity/Robustness: Finally, we examine whether our classifications are sensitive to changes in the wording and structure of the prompt. Our analysis is motivated by the fact that the computer science literature has observed that both substantive and subtle changes to an LLM prompt can affect resulting output (Lu et al., 2022; Liu et al., 2023; Sahoo et al., 2024; White et al., 2023). We examine the sensitivity of our classifications to a range of alternative prompt specifications using a randomly selected sample of 100 reports. These modifications vary the model's assigned persona, reasoning instructions, treatment of historical context and potential bias, and use of labeled examples. Across the alternative specifications, agreement with our baseline classifications ranges from 88 to 97 percent. Overall, these results indicate that the classifications are relatively insensitive to substantively different formulations of the prompt.

5 Main Results

With our measure of assimilation support in hand, we turn to the substantive question of how ideology varied within the Indian Affairs bureaucracy and whether that variation was consequential. We first characterize the ideological composition of the OIA, tracing how assimilation support evolved over time and how it differed across personnel serving under distinct appointment regimes. These patterns help establish the extent to which a bureaucracy charged with implementing a common federal policy nonetheless contained meaningful ideological heterogeneity. We then ask whether this heterogeneity predicted subsequent policy implementation, focusing on agencies' initial implementation of land allotment under the Dawes Act.

5.1 *Ideological Composition of the Indian Affairs Bureaucracy*

Evolution of Assimilation Support Over Time: We begin our main analysis by tracing how support for assimilation evolved over the nearly four decades covered by our corpus. Our focus is on Indian agents, the officials who headed local agencies and were directly responsible for implementing federal policy. Assimilation support among agents—the majority author in our corpus—changes substantially over this period: decade averages rise from 72 percent in the 1870s to 81 percent in the 1880s, before falling to 71 percent in the 1890s and 65 percent in the 1900s (Appendix Table A5).[22] This pattern indicates that strong assimilationist commitment was widespread throughout the period but was not constant within the bureaucracy.

Figure 4 examines this pattern more closely by plotting biennial averages of assimilation support among agents.[23,24] Two features are worth noting. First, we see a steady rise in support through the 1870s, a period that coincides with the institutional consolidation of assimilationist reform. Second, the decline in support begins around 1890 and persists through the turn of the century, with no clear reversal in the final years of the sample. Thus, the moderation visible in the decade averages reflects a sustained change in the ideological composition of agency leadership rather than a short-lived fluctuation. We next examine how this variation in assimilation support relates to the changing institutional systems through which Indian agents were selected.

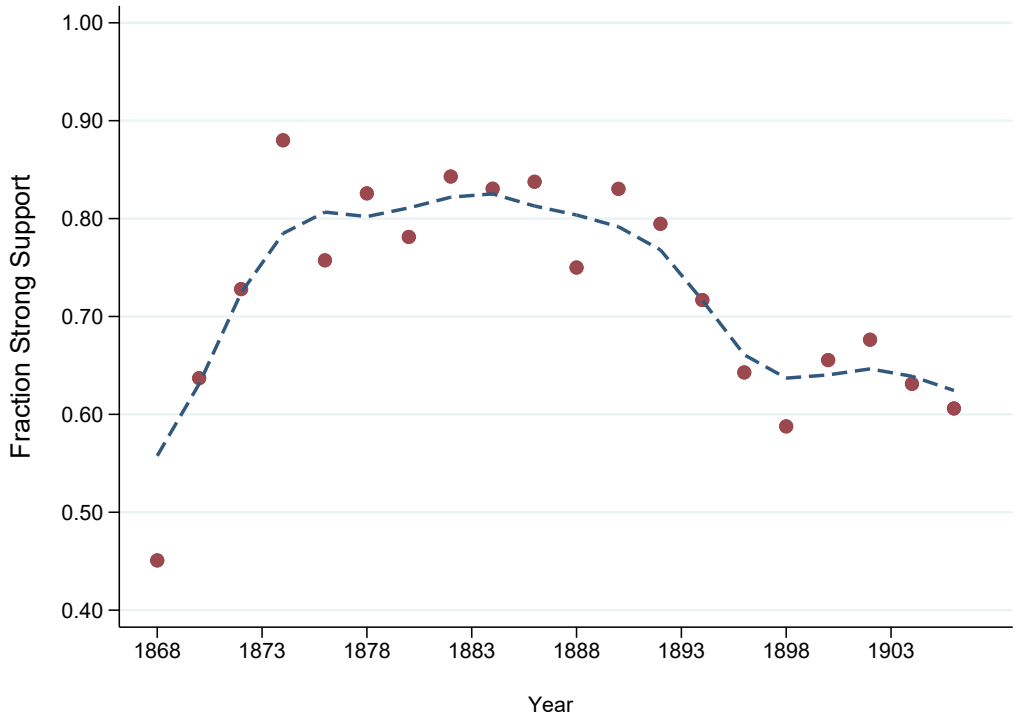
Assimilation Support Across Appointment Regimes: What might explain the rise and subsequent moderation in assimilation support within the bureaucracy of the OIA? A general explanation for ideological patterns within bureaucracies is that their internal character tends to track the dominant political or policy currents of the time. However, the timing of key federal initiatives does not align with the patterns we observe. The sharp rise in assimilation support among OIA personnel occurred during the 1870s, before the expansion of the off-reservation boarding school system in the 1880s and well in

[22] Appendix Table A5 also reports corresponding results for non-agent personnel, including teachers, physicians, farmers, and other supporting OIA staff. Across the full sample, assimilation support is higher among agents than non-agents (73 versus 65 percent), a difference that is statistically significant at the one percent level. The two groups nevertheless exhibit broadly similar changes in assimilation support over time.

[23] Appendix Figure A3 plots the biennial average for the pooled sample of all reports and non-agents alongside the agent averages. These series move closely in parallel.

[24] Table 1 shows that assimilation supporting reports are longer than neutral/non-supporting reports. Appendix Figure A4 shows that the evolution of assimilation support is substantively unchanged when we adjust for word count.

Figure 4: Assimilation Support Among Indian Agents, 1868-1906



Notes: This figure displays averages of a binary indicator for assimilation support, based on annual reports authored by Indian Affairs agents between 1868 and 1906. Each point represents the fraction of reports in a given two-year period that express strong support for assimilationist policies. The dashed line plots smoothed values from a locally weighted polynomial regression of the biennial series to visualize the trend.

advance of the Dawes Act of 1887. Conversely, the moderation in support that begins around 1890 does not coincide with a retreat from these initiatives: both off-reservation boarding schooling and land allotment policies intensified into the early 20th century. In addition, federal spending on the Indian Affairs bureaucracy rose steadily from the 1870s through the 1890s, and the number of staff on reservations grew in tandem (see Appendix Figure A1), reflecting the continued effort to support the federal government's broad assimilation agenda. Thus, the moderation in agent ideology occurred despite continued federal commitment to assimilation rather than alongside a retreat from it.

Another natural explanation for the shift in assimilation support is related to changes in the composition of the Indian Affairs workforce, associated with reforms to personnel appointment practices. At face value, key features of the institutional shifts reviewed in Section 2 align closely with the evolution of assimilation support. The rise in pro-

assimilation attitudes during the 1870s coincides with the influx of agents nominated by religious denominations who replaced the military officers that President Grant had initially appointed at the start of his administration. The subsequent moderation in ideological commitment emerges only after major changes to the appointment system took hold: the phasing out of religious nominations, the large-scale reappointment of military officers to agency roles following the Wounded Knee Massacre, and the eventual replacement of Indian agents with reservation school superintendents. The sharp timing of these shifts is particularly informative, as many alternative explanations—such as gradual cultural change within the bureaucracy or slow-moving regional economic trends—would be unlikely to generate such abrupt reversals in attitudes.

We estimate descriptive associations between assimilation support and the appointment type of the agency head who authored each report. We distinguish military officers, religious nominees, and civil-service superintendents from the omitted category of political/patronage appointees. This descriptive approach inevitably captures some of the same temporal patterns visible in the time series because appointment types are not evenly distributed across years. For example, religious nominees were exclusively appointed in the 1870s and early 1880s, and the typical agent served only about three years.[25] As a result, their observed attitudes may reflect both the appointment process and the broader ideological climate of that era. By contrast, army and political appointees appear throughout the sample period, allowing for comparisons that are less confounded by historical clustering.

Our analysis is based on the following specification:

$$A_{ir} = \pi + \gamma\, Army_i + \delta\, ReligiousNominee_i + \theta\, CivilServant_i + X'_{a(i)}\lambda + \varepsilon_{ir}, \tag{1}$$

where the dependent variable of interest is assimilation support A_{ir} for individual i in report r, and the independent variables include indicators for whether i was a military appointment ($Army_i$), a religious nomination ($ReligiousNominee_i$), or a career civil servant ($CivilServant_i$: a reservation school superintendent elevated to the position of being in charge of an agency). The omitted group in the specification is the set of reports by authors who were political appointees. A natural concern is that agents with different backgrounds may have been systematically assigned to reservations that had systematically different characteristics. With this in mind, the vector $X'_{a(i)}$ includes a set of controls

[25] Appendix Table A6 summarizes the distribution of appointment types and the timing of first appointments in the sample.

for key characteristics of the agency a that individual i is assigned to lead. These controls include the total acres under an agency's jurisdiction, total Native population, the fraction of the Native population reported as wearing Western dress, and total livestock holdings. As previewed in Section 3, we assign each report the characteristics from the nearest benchmark year (1875, 1885, 1895).[26] We further augment the vector $X'_{a(i)}$ to include controls for word count (given that assimilation supporting letters are slightly longer) and fixed effects for the region in which the agency is located.

Table 3 reports the descriptive results from equation 1. The results confirm that the types of appointees have importantly distinct ideological commitments. In Column 1, without any controls, the differences across appointment types are clear: reports by army officers or career civil servants are about 15 and 9 percentage points less likely to express strong assimilation support than those by political appointees, while religious nominees are roughly 6 percentage points more likely to do so. In Columns 2–4, we successively add word count, agency characteristics, and regional fixed effects controls. Notably, the estimated differences across appointment type change very little as these controls are added.[27,28] Taken together, these results show that the personnel regimes used by the OIA were associated with meaningfully different ideological compositions: religious nominees were more assimilation-supporting than political appointees, while military officers and civil-service personnel were less so.

Appendix Section E further examines two alternative explanations for the evolution of assimilation support. First, the moderation among agents could reflect changing ideological signals from higher-level OIA leadership (in this context, the Commissioner of Indian Affairs) rather than changes in the composition of agency personnel. We find that Commissioner reports remain overwhelmingly assimilation-supporting throughout the period, providing little evidence of a comparable moderation at the top of the bureaucracy. Second, assimilation support may have responded to the decline in violent conflict that

[26] The agency characteristics vary temporally across reports because each is assigned values from the nearest benchmark year (1875, 1885, 1895), but we omit an explicit time index to simplify notation.

[27] In the next section, we provide evidence that agent assimilation ideology is not systematically related to local reservation characteristics—consistent with minimal sorting of more assimilation-supporting agents into observably different posts.

[28] As an additional check on reporting-style differences across appointment types, Appendix Table A7 reports a parallel version of Table 3 that controls for the prevalence of moral language in each report. We construct this measure as the number of words from the moral evaluation category (MoralityGeneral) of the Moral Foundations Dictionary (Graham, Haidt and Nosek, 2009), scaled by total report length. The appointment-type estimates are substantively unchanged, suggesting that the results are not driven by religious nominees or other personnel groups simply using more moralized language.

Table 3: Assimilation Support and Agent Appointment Type

	(1)	(2)	(3)	(4)
	Dep. Variable: Assimilation Support			
Army	-0.148***	-0.147***	-0.134***	-0.138***
	(0.034)	(0.034)	(0.034)	(0.034)
Religious	0.061***	0.066***	0.044**	0.035*
	(0.020)	(0.020)	(0.020)	(0.020)
Civil Servant	-0.088***	-0.063*	-0.047	-0.070**
	(0.033)	(0.033)	(0.033)	(0.034)
Dep. Var Mean	0.702	0.702	0.702	0.702
Observations	2,384	2,384	2,384	2,384
Word Count Controls	No	Yes	Yes	Yes
Agency Controls	No	No	Yes	Yes
Region FE Controls	No	No	No	Yes

Notes: This table reports OLS estimates from equation (1) using the sample of annual reports written by authors holding an agent position (including, after 1893, reservation school superintendents promoted to agency leadership). The dependent variable is assimilation support in report r authored by agent i. The omitted category is political/patronage appointees. Columns 1–4 present results from models that vary the definition of the control variables, as indicated in the rows near the bottom of the table. The agency level controls include total acres, total Native population, the share of the Native population wearing Western dress, and total livestock holdings. Missing values for all variables are set to 0, and a full vector of indicators for missingness is included in the respective models. Region fixed effects are indicators for the four regions where agencies are located. Robust standard errors are reported in parentheses. Statistical significance is indicated by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

followed the especially turbulent 1870s and 1880s. We also construct an LLM-based measure of local violence from the annual reports and find that, although reported violence declines around the same time as assimilation support, reports mentioning violence do not exhibit meaningfully different assimilation attitudes from those that do not. These patterns provide little evidence that either changing senior-level ideology or declining local violence is the main explanation for the moderation we observe among agents.

5.2 Assimilation Support and Policy Implementation: Evidence from the Dawes Act

Do the ideological commitments of Indian Affairs personnel shape concrete policy outcomes? Linking support for assimilationist policy to implementation outcomes faces

several empirical challenges. Many key assimilation-related outcomes such as the adoption of agriculture and farming practices tend to evolve slowly over time, while many agents served for only a few years. More importantly, contemporaneous measures of bureaucratic beliefs may themselves respond to developments at the agencies where officials serve. For example, higher Native schooling attendance in a given year might encourage more confident or optimistic views of assimilation expressed in an agent's annual report written that same year. In this way, it becomes difficult to rule out the possibility that policy outcomes are shaping beliefs—a concern over simultaneity that complicates interpretation of ideology as an independent variable.

To empirically make progress on understanding ideology and policy implementation, we focus on initial land allotments under the Dawes Act of 1887. Critically, we measure assimilation support using the annual reports authored by agents before the law's passage. Moreover, allotment under Dawes proceeded gradually, with substantial variation across agencies in the timing and scale of initial land allocation.[29] The gradual implementation of the policy allows us to treat land allotment as a downstream outcome and test whether agencies led by bureaucrats who were stronger assimilation supporters moved more aggressively to break up tribal lands.

We focus on Indian agents due to their substantial autonomy over reservation affairs and their role in allotment implementation. Although the allotment process under Dawes was formally authorized by the President and often overseen by special agents, local agents often pressured the tribes under their authority to agree to allotment and coordinated necessary logistics to support the surveying work needed to divide tribal lands (Carlson, 1981). Their annual reports also describe direct involvement in these activities, indicating that even when they were not formally designated as special allotting agents, local agents could meaningfully influence the pace of implementation.[30,31]

[29] As Carlson (1981) notes, the pace of allotment was drawn out, and the number of acres allotted increased significantly after 1900.

[30] The process of allotment at the Devil's Lake agency demonstrates the close collaboration between Indian agents and special allotting agents. As recounted in the 1889 ARCIA, after Special Agent Malachi Krebs arrived to begin allotments, the Devil's Lake Sioux "most unexpectedly refused to make their selections." In response, the regular Indian agent, John Cramsie, collaborated with Krebs to convene two tribal councils aimed at defusing opposition and encouraging families to accept and select land. Cramsie's role was central to facilitating the allotment process.

[31] Dippel, Frye and Leonard (2024) also study a context in which Indian agent discretion mattered. Specifically, they focus on the period after the completion of allotment and show that agent fixed effects—i.e., which agent was stationed at a given location—mattered for whether Native families were given full property rights over their land allotment. Distinct from their analysis, we use new data and develop a methodology to measure assimilation policy support directly from agents' written reports, and test whether

As described in Section 3, our main analysis relies on allotment measures that we digitized for the initial year under Dawes from the 1888 ARCIA. We estimate the following specification:

$$y_a^{87-88} = \alpha + \beta \bar{A}_a^{pre} + X_a' \gamma + \epsilon_a, \tag{2}$$

where y_a^{87-88} is either an indicator for having any allotment, the number of allotments, or the total acres allotted at agency a during the period 1887-1888. The key independent variable is the strength of prior assimilation support expressed by the Indian agent assigned at agency a in the baseline year before the passage of the Dawes Act, denoted by $\bar{A}_a^{pre}$. The measure is defined as the average of the agent's assimilation support scores across all available annual reports written before Dawes, including any reports written while serving at another agency.[32] We standardize this measure to have mean zero and unit variance, so that coefficients are read per standard deviation. To improve precision and address interpretative concerns discussed further below, the term X_a' includes controls for baseline (1887) agency characteristics (the total number of acres, total Native population, the share of the Native population reported to be wearing Western dress, and total livestock) as well as characteristics of the agent in place in the baseline year: total prior experience as an Indian agent, annual salary, administrative capacity measured from pre-Dawes reports, and the average word count of those same reports (and its square).

Our main interest is the estimate of the coefficient β, which captures the degree to which allotment activity is associated with our measure of bureaucratic ideological support for assimilation. Two concerns are important for interpreting this estimate. First, agents with different ideological commitments may have been selectively assigned to different types of agencies. For example, more pro-assimilation agents could have been posted to agencies inherently more favorable to allotment. To shed light on this issue, Appendix Table A8 reports a balance analysis that examines the relationship between our set of baseline agency characteristics and our measure of ideology. We find no systematic relationships, suggesting that more (or less) assimilationist leadership was not simply concentrated in agencies with advantageous demographic, cultural, or economic characteristics.[33] This pattern reduces concerns that the relationship between ideology

this ideology predicts allotment activity.

[32] In practice, only 1 of the 55 agents in this sample authored any reports at a different agency than the one where they were stationed in the baseline year. On average, these agents wrote nearly 4 reports that contribute to our measure of assimilation support before Dawes.

[33] We also find no detectable association between the agency characteristics and an agent's total years of experience or with the text-based measure of administrative capacity.

and allotment is mechanically driven by observable differences across agencies. It is also consistent with features of the appointment process that generated substantial turnover and imperfect matching between personnel and posts. Abrupt resignations due to scandal or illness frequently resulted in temporary military appointments—introducing less assimilation-oriented officers into posts vacated by civilian agents—while the final cohort of denominational appointees, among the most assimilation-oriented, gradually left the service in the years leading up to Dawes.[34,35] Finally, the ideological views of prospective agents were likely only imperfectly observed by those making appointments, who may instead have prioritized considerations such as political ties or availability.

Second, even absent selective assignment of agents to reservations, our estimates of β could also reflect correlations between our measure of assimilation support and other agent attributes that independently affect implementation performance—such as greater experience in the Indian Service, higher salary, or stronger administrative capacity. To address this concern, we construct measures of agents' total prior experience, annual salary, and administrative capacity from their pre-Dawes reports and include them directly in our allotment specifications to assess whether the association between ideology and implementation persists after accounting for these attributes. We measure administrative capacity using a separate LLM classification that identifies explicit evidence of concrete, procedural, and repeatable practices that demonstrate administrative capacity. The prompt scores evidence across five administrative categories (planning/procedures; record-keeping; resource management; coordination/compliance; and problem-solving with follow-through) and assigns a binary indicator equal to one only when a report contains concrete evidence from at least one category. We then aggregate this pre-Dawes measure to the agent level (denoted $\bar{C}_a^{pre}$).[36] In our preferred specifications, we include experience, administrative capacity, and annual salary as controls in the vector X_a' and report these estimated coefficients separately to examine whether these agent characteristics predict allotment activity in their own right.

Our analysis shows that agencies led by Indian agents whose past reports revealed stronger support for assimilation policy engaged in more allotment activity in the first

[34] As highlighted in Section 2, fraud and mismanagement were prevalent among Indian agents, and in some cases, scandals led to their dismissal or resignation. For instance, allegations concerning agent Galen Eastman at the Navajo Agency included the creation of a nominal boarding school that existed only on paper; these accusations ultimately resulted in his resignation (McCluskey, 1980).

[35] In the sample for our Dawes analysis, 14 percent of agents are military or religious appointees.

[36] Appendix C.3 provides further details and reproduces the full administrative-capacity prompt. In our sample for the Dawes analysis, the mean of our capacity measure is 0.68.

Table 4: Analysis of Ideology and Land Allotment After Dawes (1887-1888)

	(1)	(2)	(3)	(4)	(5)	(6)
	Allotted		# of Allotments		Acres Allotted	
$\bar{A}_a^{pre}$	0.114***	0.134***	48.620**	66.582**	5,074.746**	6,758**
	(0.035)	(0.039)	(21.853)	(29.643)	(2,302.388)	(3,007.841)
Exp_a		-0.014**		-6.124		-654
		(0.006)		(3.944)		(397.915)
$\bar{C}_a^{pre}$		-0.117		-208		-21,018
		(0.160)		(167.120)		(16,144.780)
$Salary_a$		-0.000		0.008		1.047
		(0.000)		(0.014)		(1.369)
Dep. Var. Mean	0.145	0.145	60.891	60.891	6,348	6,348
Observations	55	55	55	55	55	55
Agency Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports OLS estimates from equation 2. The dependent variables are an indicator for having any allotment (columns 1-2); the number of allotments (columns 3-4); and total number of acres allotted (columns 5-6). All measures are based on the allotment activity during the first year after the passage of the Dawes Act reported in the 1888 ARCIA. All specifications include controls for word count and agency-level demographic, social and economic characteristics in the baseline (1887) year (i.e., total acres under the agency jurisdiction, total Native population, the fraction of the Native population that wears Western dress, and total livestock). Robust standard errors are reported in parentheses. Columns (2), (4), and (6) additionally control for the agent's total prior experience (Exp_a), pre-Dawes measured administrative capacity ($\bar{C}_a^{pre}$), and the agent's annual salary recorded in the Official Register of the United States. Note that the sample includes 55 agencies in operation in 1887 and whose leadership had sufficient reporting history (the Union Agency is excluded because it was exempt from the Dawes Act of 1887). Statistical significance is indicated by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

year of the Dawes Act. Table 4 reports estimates from equation 2 while varying whether the agent-specific controls for experience, salary, and administrative capacity are included. Column 1 shows that a one standard deviation increase in average assimilation support is associated with an 11.4 percentage point increase in the likelihood that an agency recorded any allotment—a large difference relative to the mean allotment rate of 14.5 percent. We similarly find large positive associations with the number of allotments (Column 3) and total acres allotted (Column 5). Consistent with the balance analysis, Appendix Table A9 shows that these estimates are very similar to those from specifications that do not include the 1887 agency characteristics, providing little indication that the relationship between agent ideology and allotment activity is driven by the observable differences

across agencies captured by these controls.

Table 4 also reports coefficients on the additional agent-specific characteristics. Reassuringly, the estimated relationship between assimilation support and allotment activity is essentially unchanged when we add controls for agent experience, annual salary, and administrative capacity (columns 2, 4, and 6). At the same time, we find some evidence that agent experience is negatively associated with early allotment: in column (2), experience is negatively associated with the probability of any allotment. While we interpret this pattern cautiously, one possibility is that longer-serving agents—having greater exposure to local conditions and Native communities—may have been more reluctant to advance allotment early on, even holding constant their expressed assimilation views. By contrast, we do not find clear statistical evidence that either administrative capacity or annual salary predicts allotment outcomes.[37,38]

To further address concerns about selective assignment of agents across agencies, we conduct a placebo analysis using the average ideology of predecessor agents who served at a given agency before the leadership in place when the Dawes Act was passed. Because these agents left their posts before Dawes was enacted, their ideology should not directly predict subsequent allotment activity through their own implementation decisions. Appendix Table A10 reports the results for our early allotment outcomes, showing no detectable effect of prior leadership ideology. For instance, a one standard deviation increase in the average assimilation support of the preceding agent is associated with a small and statistically insignificant 0.9 percentage-point *decrease* in the likelihood that an agency had any allotment ($p = 0.851$). The absence of any relationship is consistent with both our balance evidence and the institutional realities of the Indian Service, where high turnover, short tenures, and interim appointments often generated leadership changes that were not systematically related to ideology.

Finally, as a test of robustness, we examine whether the results persist when allotment activity is measured over a longer period after the passage of Dawes. Specifically, we use

[37]While measuring bureaucratic skill and linking it to performance has proven difficult in the literature, our null on administrative capacity is consistent with other findings from this period. For example, Moreira and Pérez (2024) show that Pendleton Act exams improved customs employees' professional backgrounds but not revenue collection performance.

[38]Salary levels can shape bureaucrat quality through both selection and effort. Higher base pay can attract higher-ability applicants (Dal Bó, Finan and Rossi, 2013), although lucrative financial incentives may also crowd out intrinsically motivated ones by signaling a less pro-social mission (Deserranno, 2019). For incumbents, however, large-scale experimental evidence has found limited effects of unconditional base-salary raises on performance (de Ree et al., 2018). We ultimately find that adding salary as a control has little effect on our estimates of ideology, and salary itself does not predict allotment outcomes.

ARCIA records to measure allotment activity through 1889 and 1890, respectively. These records do not report total acres allotted, so we focus on an indicator for any allotment and the number of allotments.[39] Appendix Table A11 reports these results, which continue to show a positive relationship between agents' prior assimilation support and subsequent allotment activity.

6 Conclusion

This paper examines the evolution of bureaucratic ideology and how it mattered for one of the most consequential policy campaigns in U.S. history: the federal government's effort to assimilate and socially transform Native populations. We study the Office of Indian Affairs (OIA), the federal agency that exercised sweeping authority over reservation life, effectively acting as a local government. Its personnel wielded extensive power over core aspects of reservation life, ranging from food and schooling to the governance of families and communities. During the late 19th century, the policies carried out by the OIA explicitly aimed to reshape Native culture and society. Implementing such a far-reaching project, often in the face of resistance, made officials' commitment to the government's assimilationist objectives potentially important for how policy was carried out. To recover these ideological commitments, we analyze how OIA personnel described their duties and work in their official reports. These reports allow us to study both how assimilationist commitment varied within the bureaucracy and whether that variation predicted differences in policy implementation.

Using thousands of reports spanning nearly four decades, we trace substantial changes in the ideological composition of the Indian Affairs bureaucracy. Support for assimilation rose sharply during the 1870s, coinciding with the delegation of personnel appointments to religious denominations under President Grant's Peace Policy. Strikingly, this rise in ideological intensity preceded major policies such as the spread of off-reservation boarding schools and the passage of the Dawes Act in 1887. Support subsequently moderated beginning around 1890, even as major federal assimilation policies continued or intensified. The timing strongly suggests that appointment policies—and the kinds of individuals they brought into the bureaucracy—were crucial in shaping the OIA's ideological commitments.

Our analysis further shows that differences in ideological commitment predict the pace

[39] During the period we study, allotments under Dawes typically assigned 160 acres to household heads and 80 acres to single adults or minors (United States, 1887).

of land allotment under one of the most far-reaching assimilation policies targeting Native Americans during the 19th and 20th centuries. Focusing on the early years of the Dawes Act, we find that agencies led by personnel with stronger assimilationist commitments were more likely to begin allotting land immediately after the law's passage. These results provide evidence that policy-specific ideological commitments were consequential for how a common federal directive was implemented across local agencies. We provide new insight into this formative period of the Dawes regime—the early phase of a policy that would ultimately lead to the loss of more than 90 million acres of Native land and a profound erosion of tribal sovereignty (McDonnell, 1991).

We conclude by noting that our substantive findings rest on, and in turn motivate, a methodological contribution: demonstrating how computational text analysis can illuminate the workings of bureaucracy and policymaking. Although these reports have long been available to historians, their scale has made it difficult to systematically study bureaucratic ideology over time. We address this challenge by developing and validating a classification approach using large language models (LLMs), designed to distinguish normative endorsement of policy objectives from merely describing or implementing them. The resulting classifications align closely with blinded human evaluations and predict agents' independently recorded support for a separate assimilation-related policy proposal. In doing so, we contribute to a growing literature that uses computational methods to recover beliefs and ideology from text (e.g., Jelveh, Kogut and Naidu, 2024; Card et al., 2022; Adukia et al., 2023; Adukia and Harrison, 2025), as well as to the economic history of Indigenous Peoples in the U.S. (Feir, 2025). More broadly, our findings provide new evidence that ideological heterogeneity inside government can predict meaningful differences in the implementation of state policy.

References

Adams, David Wallace. 1995. *Education for Extinction: American Indians and the Boarding School Experience, 1875–1928.* Lawrence: University Press of Kansas.

Adukia, Anjali, Alex Eble, Emileigh Harrison, Hakizumwami Birali Runesha, and Teodora Szasz. 2023. "What We Teach About Race and Gender: Representation in Images and Text of Children's Books." *The Quarterly Journal of Economics*, 138(4): 2225–2285.

Adukia, Anjali, and Emileigh Harrison. 2025. "Separation of Church and State Curricula? Examining Public and Religious Private School Textbooks." National Bureau of Economic Research.

Akee, Randall. 2020. "Land titles and dispossession: Allotment on American Indian reservations." *Journal of Economics, Race, and Policy*, 3(2): 123–143.

Althoff, Lukas, and Hugo Reichardt. 2024. "Jim Crow and Black economic progress after slavery." *The Quarterly Journal of Economics*, 139(4): 2279–2330.

Atil, Berk, Sarp Aykent, Alexa Chittams, Lisheng Fu, Rebecca J. Passonneau, Evan Radcliffe, Guru Rajan Rajagopal, Adam Sloan, Tomasz Tudrej, Ferhan Ture, Zhe Wu, Lixinyu Xu, and Breck Baldwin. 2025. "Non-Determinism of "Deterministic" LLM Settings."

Ba, Bocar A., Haosen Ge, Jacob Kaplan, Dean Knox, Mayya Komisarchik, Gregory Lanzalotto, Rei Mariman, Jonathan Mummolo, Roman Rivera, and Michelle Torres. 2025. "Political Diversity in U.S. Police Agencies." *American Journal of Political Science*. Forthcoming.

Bender, Norman J. 1988. "The Battle of Tierra Amarilla." *New Mexico Historical Review*, 63(3): 241–256.

Besley, Timothy, and Maitreesh Ghatak. 2005. "Competition and incentives with motivated agents." *American Economic Review*, 95(3): 616–636.

Board of Indian Commissioners. 1875. *Annual report of the Board of Indian Commissioners to the Secretary of the Interior.* US Government Printing Office.

Bojić, Ljubiša, Olga Zagovora, Asta Zelenkauskaite, Vuk Vuković, Milan Čabarkapa, Selma Veseljević Jerković, and Ana Jovančević. 2025. "Comparing large Language models and human annotators in latent content analysis of sentiment, political leaning, emotional intensity and sarcasm." *Scientific Reports*, 15(1): 11477.

Bonica, Adam, Jowei Chen, and Tim Johnson. 2015. "Senate gate-keeping, presidential staffing of "Inferior Offices," and the ideological composition of appointments to the public bureaucracy." *Quarterly Journal of Political Science*, 10(1): 5–40.

Brehm, John, and Scott Gates. 1997. *Working, Shirking, and Sabotage: Bureaucratic Response to a Democratic Public.* Ann Arbor: University of Michigan Press.

Brown, Dee. 1970. *Bury My Heart at Wounded Knee: An Indian History of the American West.* New York: Holt, Rinehart & Winston.

Card, Dallas, Serina Chang, Chris Becker, Julia Mendelsohn, Rob Voigt, Leah Boustan, Ran Abramitzky, and Dan Jurafsky. 2022. "Computational analysis of 140 years of US political speeches reveals more positive but increasingly polarized framing of immigration." *Proceedings of the National Academy of Sciences*, 119(31): e2120510119.

Carlson, Leonard A. 1981. *Indians, Bureaucrats, and Land: The Dawes Act and the Decline of Indian Farming.* Westport, CT: Greenwood Press.

Clinton, Joshua D, Anthony Bertelli, Christian R Grose, David E Lewis, and David C Nixon. 2012. "Separated powers in the United States: The ideology of agencies, presidents, and congress." *American Journal of Political Science*, 56(2): 341–354.

Cutler, Lee. 1971. "Lawrie Tatum and the Kiowa Agency: 1869-1873." *Arizona and the West*, 13(3): 221–244.

Dal Bó, Ernesto, Frederico Finan, and Martín A Rossi. 2013. "Strengthening state capabilities: The role of financial incentives in the call to public service." *The Quarterly Journal of Economics*, 128(3): 1169–1218.

Davenport, Christian. 2007. "State repression and political order." *Annu. Rev. Polit. Sci.*, 10(1): 1–23.

de Ree, Joppe, Karthik Muralidharan, Menno Pradhan, and Halsey Rogers. 2018. "Double for Nothing? Experimental Evidence on an Unconditional Teacher Salary Increase in Indonesia." *Quarterly Journal of Economics*, 133(2): 993–1039.

Deserranno, Erika. 2019. "Financial Incentives as Signals: Experimental Evidence from the Recruitment of Village Promoters in Uganda." *American Economic Journal: Applied Economics*, 11(1): 277–317.

Dippel, Christian, Dustin Frye, and Bryan Leonard. 2024. "Bureaucratic discretion in policy implementation: evidence from the Allotment Era." *Public Choice*, 199(3): 193–211.

Evans, Laura E. 2014. "Tribal-State Relations in the Anglosphere." *Annual Review of Political Science*, 17: 273–289.

Fang, Hanming, Ming Li, and Guangli Lu. 2025. "Decoding China's Industrial Policies." National Bureau of Economic Research.

Feir, Donn L. 2025. "Economic History and Indigenous Peoples in North America." *Annual Review of Economics*, 17(1): 479–509.

Feir, Donn L, Rob Gillezeau, and Maggie EC Jones. 2024. "The slaughter of the bison and reversal of fortunes on the Great Plains." *Review of Economic Studies*, 91(3): 1634–1670.

Fouka, Vasiliki. 2019. "How do immigrants respond to discrimination? The case of Germans in the US during World War I." *American Political Science Review*, 113(2): 405–422.

Fouka, Vasiliki. 2020. "Backlash: The unintended effects of language prohibition in US schools after World War I." *The Review of Economic Studies*, 87(1): 204–239.

Fouka, Vasiliki. 2024. "State policy and immigrant integration." *Annual Review of Political Science*, 27: 25–46.

Gailmard, Sean, and John W Patty. 2007. "Slackers and zealots: Civil service, policy discretion, and bureaucratic expertise." *American Journal of Political Science*, 51(4): 873–889.

Gentzkow, Matthew, Bryan Kelly, and Matt Taddy. 2019. "Text as data." *Journal of Economic Literature*, 57(3): 535–574.

Gilardi, Fabrizio, Meysam Alizadeh, and Maël Kubli. 2023. "ChatGPT outperforms crowd workers for text-annotation tasks." *Proceedings of the National Academy of Sciences*, 120(30): e2305016120.

Goncalves, Felipe, and Cody Tuttle. 2025. "Police Work and Political Identity." Working paper.

Graham, Jesse, Jonathan Haidt, and Brian A. Nosek. 2009. "Liberals and Conservatives Rely on Different Sets of Moral Foundations." *Journal of Personality and Social Psychology*, 96(5): 1029–1046.

Gregg, Matthew T. 2018. "The long-term effects of American Indian boarding schools." *Journal of Development Economics*, 130: 17–32.

Grimmer, Justin, and Brandon M Stewart. 2013. "Text as data: The promise and pitfalls of automatic content analysis methods for political texts." *Political analysis*, 21(3): 267–297.

Hassan, Mai, Daniel Mattingly, and Elizabeth R Nugent. 2022. "Political control." *Annual Review of Political Science*, 25: 155–174.

Hausladen, Carina I., Martin Fochmann, and Peter Mohr. 2024. "Predicting compliance: Leveraging chat data for supervised classification in experimental research." *Journal of Behavioral and Experimental Economics*, 109.

Hiatt, Burritt M. 1958. "James M. Haworth, Quaker Indian Agent." *Bulletin of Friends' Historical Association*, 47(2): 80–93.

Hoxie, Frederick E. 1984. *A Final Promise: The Campaign to Assimilate the Indians, 1880–1920.* Lincoln: University of Nebraska Press.

Hu, Tiancheng, and Nigel Collier. 2024. "Quantifying the Persona Effect in LLM Simulations." 10289–10307. Bangkok, Thailand: Association for Computational Linguistics.

Jelveh, Zubin, Bruce Kogut, and Suresh Naidu. 2024. "Political language in economics." *The Economic Journal*, 134(662): 2439–2469.

Jurafsky, Daniel, and James H. Martin. 2025. *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models.* . 3rd ed. Online manuscript released January 12, 2025.

Keller, Robert H. 1983. *American Protestantism and United States Indian Policy, 1869-82.* Lincoln: University of Nebraska Press.

Khan, Muhammad Yasir. 2025. "Mission Motivation and Public Sector Performance: Experimental Evidence from Pakistan." *American Economic Review*, 115(7): 2343–75.

Kistner, Michael R, Michael Heseltine, Robert Alvarez, Maya Fitch, Lucas Lothamer, and Elizabeth N Simas. 2025. "Measuring Partisanship and Representation in Online Congressional Communication." *Center for Effective Lawmaking Working Paper*, 02.

Kojima, Takeshi, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. "Large Language Models are Zero-Shot Reasoners." Vol. 35, 22199–22213. Curran Associates, Inc.

Komisarchik, Mayya, Maya Sen, and Yamil R Velez. 2022. "The political consequences of ethnically targeted incarceration: Evidence from Japanese American internment during World War II." *The Journal of Politics*, 84(3): 1497–1514.

Koonz, Claudia. 2003. *The nazi conscience.* Harvard University Press.

Lagakos, David, Stelios Michalopoulos, and Hans-Joachim Voth. 2025. "American Life Histories." National Bureau of Economic Research NBER Working Paper 33373, Cambridge, MA.

Laver, Michael, Kenneth Benoit, and John Garry. 2003. "Extracting policy positions from political texts using words as data." *American political science review*, 97(2): 311–331.

Li, Cheng, Jindong Wang, Yixuan Zhang, Kaijie Zhu, Wenxin Hou, Jianxun Lian, Fang Luo, Qiang Yang, and Xing Xie. 2023. "Large Language Models Understand and Can be Enhanced by Emotional Stimuli."

Lipsky, Michael. 1980. *Street-Level Bureaucracy: Dilemmas of the Individual in Public Services.* New York: Russell Sage Foundation.

Liu, Pengfei, Weizhe Yuan, Jinlan Fu, Zhengbao Jiang, Hiroaki Hayashi, and Graham Neubig. 2023. "Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing." *ACM Computing Surveys*, 55(9): 1–35.

Lomawaima, K. Tsianina. 1995. *They Called it Prairie Light: The Story of Chilocco Indian School.* Lincoln: University of Nebraska Press.

Lu, Yao, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. 2022. "Fantastically Ordered Prompts and Where to Find Them: Overcoming Few-Shot Prompt Order Sensitivity." 8086–8098. Dublin, Ireland: Association for Computational Linguistics.

Maruthiah, Christian. 2024. "Coercive assimilation policy and ethnic identification across generations." Technical report, Trinity College Dublin.

Maynard-Moody, Steven, and Michael Musheno. 2003. *Cops, Teachers, Counselors: Stories from the Front Lines of Public Service.* Ann Arbor: University of Michigan Press.

McCluskey, Stephen C. 1980. "Evangelists, educators, ethnographers, and the establishment of the Hopi reservation." *The Journal of Arizona History*, 21(4): 363–390.

McDonnell, Janet A. 1991. *The Dispossession of the American Indian, 1887–1934.* Bloomington: Indiana University Press.

Mohammad, Saif. 2023. "Best Practices in the Creation and Use of Emotion Lexicons." Dubrovnik, Croatia: Association for Computational Linguistics.

Mohammad, Saif M., and Peter D. Turney. 2013. "Crowdsourcing a Word-Emotion Association Lexicon." *Computational Intelligence*, 29(3): 436–465.

Monroe, Burt L., Michael P. Colaresi, and Kevin M. Quinn. 2008. "Fightin'words: Lexical feature selection and evaluation for identifying the content of political conflict." *Political Analysis*, 16(4): 372–403.

Moreira, Diana, and Santiago Pérez. 2024. "Civil Service Exams and Organizational Performance: Evidence from the Pendleton Act." *American Economic Journal: Applied Economics*, 16(3): 250–91.

Mylonas, Harris. 2012. *The Politics of Nation-Building: Making Co-Nationals, Refugees, and Minorities.* Cambridge University Press.

Napolio, Nicholas G. 2025. "Measuring executive agency ideology using large language models." *Journal of Political Institutions and Political Economy*, 6(2): 161–180.

Olson, Michael P. 2025. ""Restoration" and representation: Legislative consequences of Black disfranchisement in the American South, 1879–1916." *American Journal of Political Science*, 69(2): 387–405.

OpenAI. 2024. "Prompt Engineering Guide." https://platform.openai.com/docs/guides/prompt-engineering, Accessed: 2024-4-19.

Otis, Delos Sacket. 1973. *The Dawes Act and the Allotment of Indian Lands.* Norman, OK: University of Oklahoma Press.

Posel, Deborah. 1991. *The making of apartheid 1948–1961: Conflict and compromise.* Oxford University Press.

Priest, Loring Benson. 1942. *Uncle Sam's Stepchildren: The Reformation of United States Indian Policy, 1865–1887.* Rutgers University Press.

Prucha, Francis Paul. 1984. *The Great Father: The United States Government and the American Indians.* Lincoln: University of Nebraska Press.

Reimers, Nils, and Iryna Gurevych. 2019. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks."

Sahoo, Pranab, Ayush Kumar Singh, Sriparna Saha, Vinija Jain, Samrat Mondal, and Aman Chadha. 2024. "A Systematic Survey of Prompt Engineering in Large Language Models: Techniques and Applications."

Salewski, Leonard, Stephan Alaniz, Isabel Rio-Torto, Eric Schulz, and Zeynep Akata. 2023. "In-Context Impersonation Reveals Large Language Models' Strengths and Biases."

Silge, Julia, and David Robinson. 2016. "tidytext: Text Mining and Analysis Using Tidy Data Principles in R." *JOSS*, 1(3).

Sim, David. 2008. "The peace policy of Ulysses S. Grant." *American Nineteenth Century History*, 9(3): 241–268.

Spenkuch, Jörg L, Edoardo Teso, and Guo Xu. 2023. "Ideology and performance in public organizations." *Econometrica*, 91(4): 1171–1203.

Spirling, Arthur. 2012. "U.S. Treaty Making with American Indians: Institutional Change and Relative Power, 1784–1911." *American Journal of Political Science*, 56(1): 84–97.

Stuart, Paul. 1979. *The Indian Office: Growth and Development of an American Institution, 1865–1900. Studies in American History and Culture*, Ann Arbor: UMI Research Press.

Suryanarayan, Pavithra. 2024. "Endogenous state capacity." *Annual Review of Political Science*, 27: 223–243.

Trennert, Robert A. Jr. 1988. *The Phoenix Indian School: Forced Assimilation in Arizona, 1891–1935.* Norman: University of Oklahoma Press.

Tyson, Scott A. 2018. "The Agency Problem Underlying Repression." *The Journal of Politics*, 80(4): 1297–1310.

Törnberg, Petter. 2023. "ChatGPT-4 Outperforms Experts and Crowd Workers in Annotating Political Twitter Messages with Zero-Shot Learning." *arXiv preprint arXiv:2304.06588*.

United States. 1887. "An Act to Provide for the Allotment of Lands in Severalty to Indians on the Various Reservations (General Allotment Act or Dawes Act)." *Statutes At Large 24*, https://commons.und.edu/indigenous-gov-docs/177/.

United States. 1893. *An Act Making Appropriations for Current and Contingent Expenses of the Indian Department, and for Fulfilling Treaty, Contract, and Other Obligations, for the Fiscal Year Ending June 30, 1894, and for Other Purposes.* Vol. 27, Washington, D.C.: Government Printing Office. 52nd Cong., 2nd sess., ch. 209 (Mar. 3, 1893).

Unrau, William E. 1972. "The Civilian as Indian Agent: Villain or Victim?" *The Western Historical Quarterly*, 3(4): 405–420.

U.S. Bureau of the Census and U.S. Civil Service Commission. 1867–1905. "Official Register of the United States." *https://catalog.hathitrust.org/Record/011712460*, U.S. Government Printing Office, Washington. Digitized volumes available via HathiTrust Digital Library.

U.S. Department of the Interior. 1881. "Annual Report of the Commissioner of Indian Affairs to the Secretary of the Interior." U.S. Government Printing Office Government Report, Washington, D.C.

U.S. Department of the Interior. 1883. "Annual Report of the Commissioner of Indian Affairs to the Secretary of the Interior." U.S. Government Printing Office Government Report, Washington, D.C.

U.S. Department of the Interior. 1889. "Annual Report of the Commissioner of Indian Affairs to the Secretary of the Interior." U.S. Government Printing Office Government Report, Washington, D.C.

U.S. Department of the Interior. 1890. "Annual Report of the Commissioner of Indian Affairs to the Secretary of the Interior." U.S. Government Printing Office Government Report, Washington, D.C.

U.S. House of Representatives. 1882. "Apaches at the Mescalero and Jicarilla Agencies: H.R. Exec. Doc. No. 207, 47th Cong., 1st Sess."

U.S. Office of Indian Affairs. 1837–1906. "Annual Report of the Commissioner of Indian Affairs." *https://catalog.hathitrust.org/Record/008696072*, Digitized volumes available via HathiTrust Digital Library.

Vuckovic, Myriam. 2024. *Voices from Haskell: Indian Students Between Two Worlds, 1884–1928.* Lawrence: University Press of Kansas.

White, Jules, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C Schmidt. 2023. "A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT." New York, NY, USA: ACM.

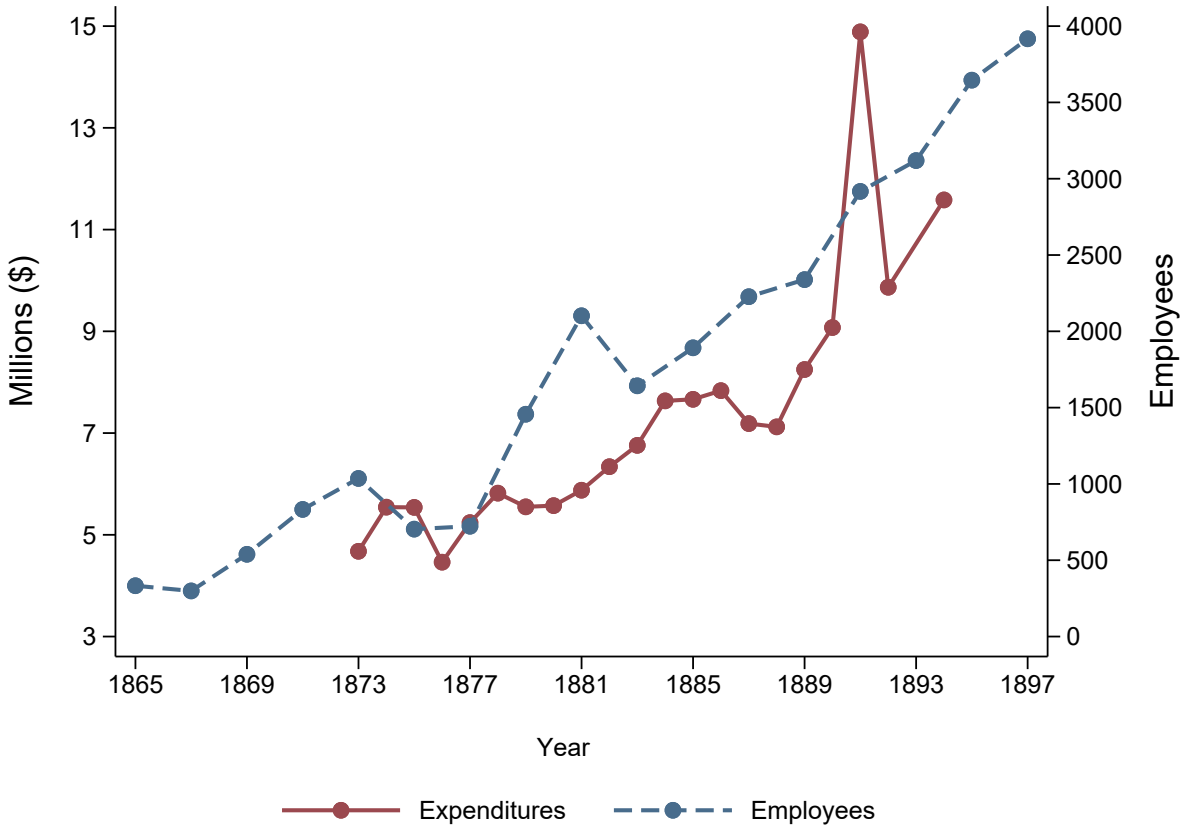
Woodward, C. Vann. 1955. *The Strange Career of Jim Crow.* . 3rd ed., New York: Oxford University Press.

Zhao, Wayne Xin, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, Peiyu Liu, Jian-Yun Nie, and Ji-Rong Wen. 2023. "A Survey of Large Language Models." Continuously updated, latest version available at https://github.com/RUCAIBox/LLMSurvey.

Zheng, Mingqian, Jiaxin Pei, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. 2024. "When "A Helpful Assistant"' Is Not Really Helpful: Personas in System Prompts Do Not Improve Performances of Large Language Models." *arXiv preprint 2311.10054*.

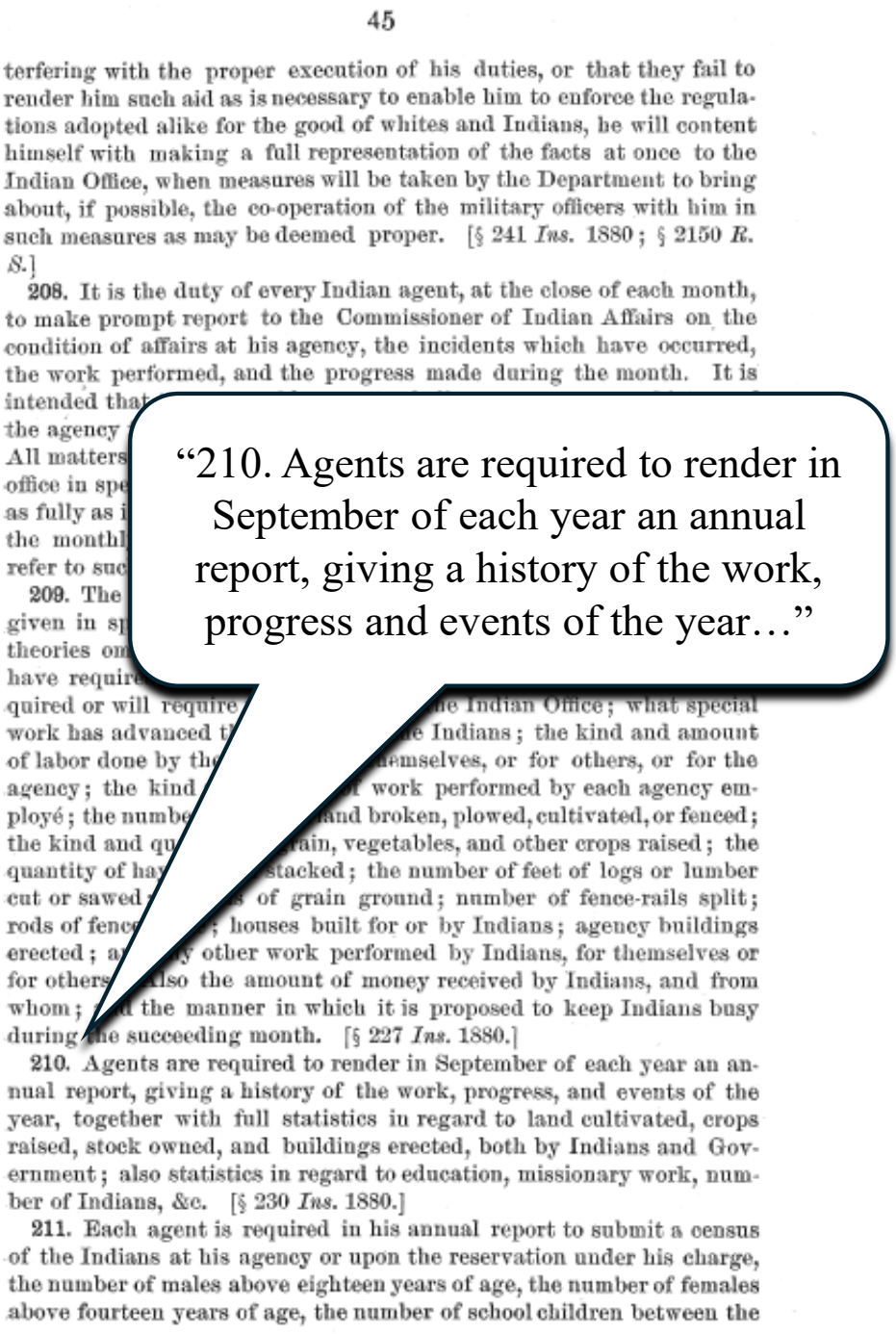
A Appendix Figures and Tables

Figure A1: Office of Indian Affairs: Expenditures and Employees



Notes: The underlying data was digitized from Tables IV and VII in Stuart (1979). The spending statistics were compiled from annual statements of appropriations and expenditures printed in the Executive Documents of the U.S. House of Representatives for the fiscal years ending in 1874 to 1893. Employment data are from the Official Register of the U.S. from 1865 to 1897.

Figure A2: Annual Reporting Requirements for Indian Agents (1884)



45

terfering with the proper execution of his duties, or that they fail to render him such aid as is necessary to enable him to enforce the regulations adopted alike for the good of whites and Indians, he will content himself with making a full representation of the facts at once to the Indian Office, when measures will be taken by the Department to bring about, if possible, the co-operation of the military officers with him in such measures as may be deemed proper. [§ 241 *Ins.* 1880; § 2150 *R. S.*]

208. It is the duty of every Indian agent, at the close of each month, to make prompt report to the Commissioner of Indian Affairs on the condition of affairs at his agency, the incidents which have occurred, the work performed, and the progress made during the month. It is intended that

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emselves, or for others, or for the

work performed by each agency em-

and broken, plowed, cultivated, or fenced;

rain, vegetables, and other crops raised; the

stacked; the number of feet of logs or lumber

of grain ground; number of fence-rails split;

; houses built for or by Indians; agency buildings

other work performed by Indians, for themselves or

lso the amount of money received by Indians, and from

the manner in which it is proposed to keep Indians busy

e succeeding month. [§ 227 *Ins.* 1880.]

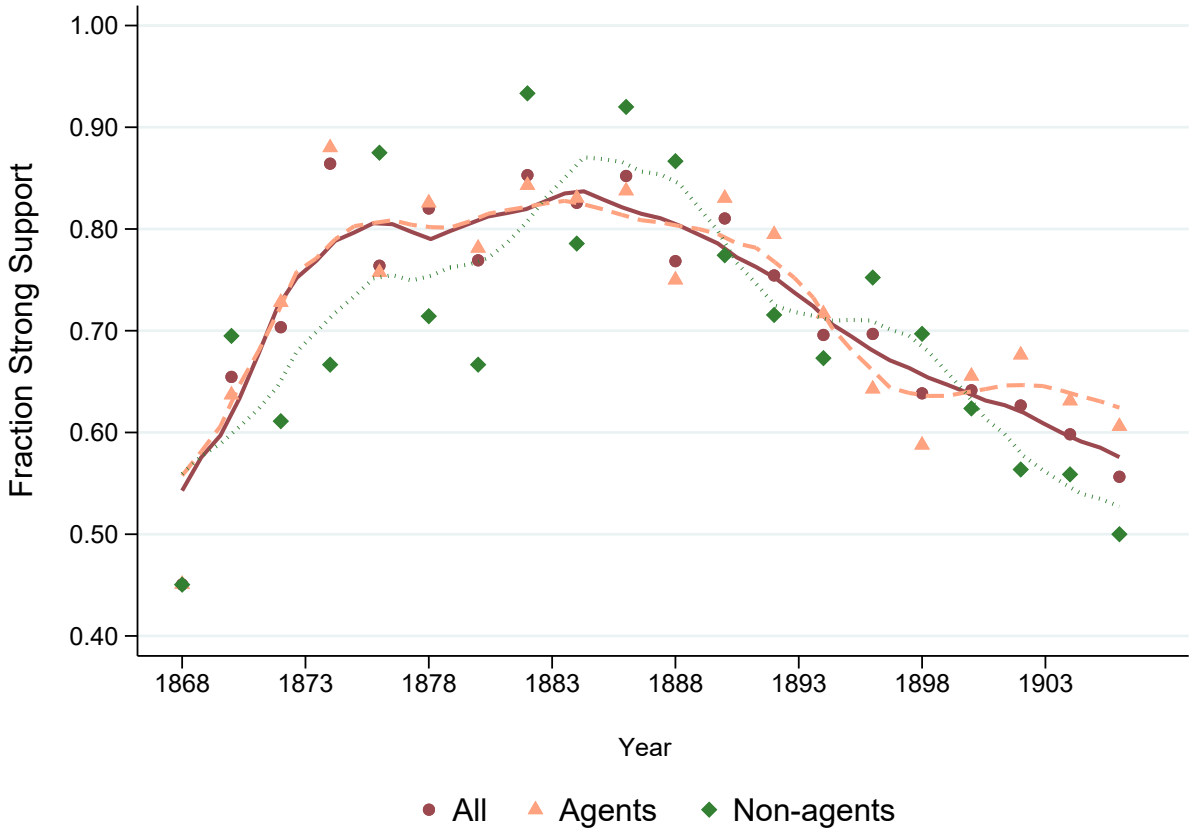
210. Agents are required to render in September of each year an annual report, giving a history of the work, progress, and events of the year, together with full statistics in regard to land cultivated, crops raised, stock owned, and buildings erected, both by Indians and Government; also statistics in regard to education, missionary work, number of Indians, &c. [§ 230 *Ins.* 1880.]

211. Each agent is required in his annual report to submit a census of the Indians at his agency or upon the reservation under his charge, the number of males above eighteen years of age, the number of females above fourteen years of age, the number of school children between the

"210. Agents are required to render in September of each year an annual report, giving a history of the work, progress and events of the year…"

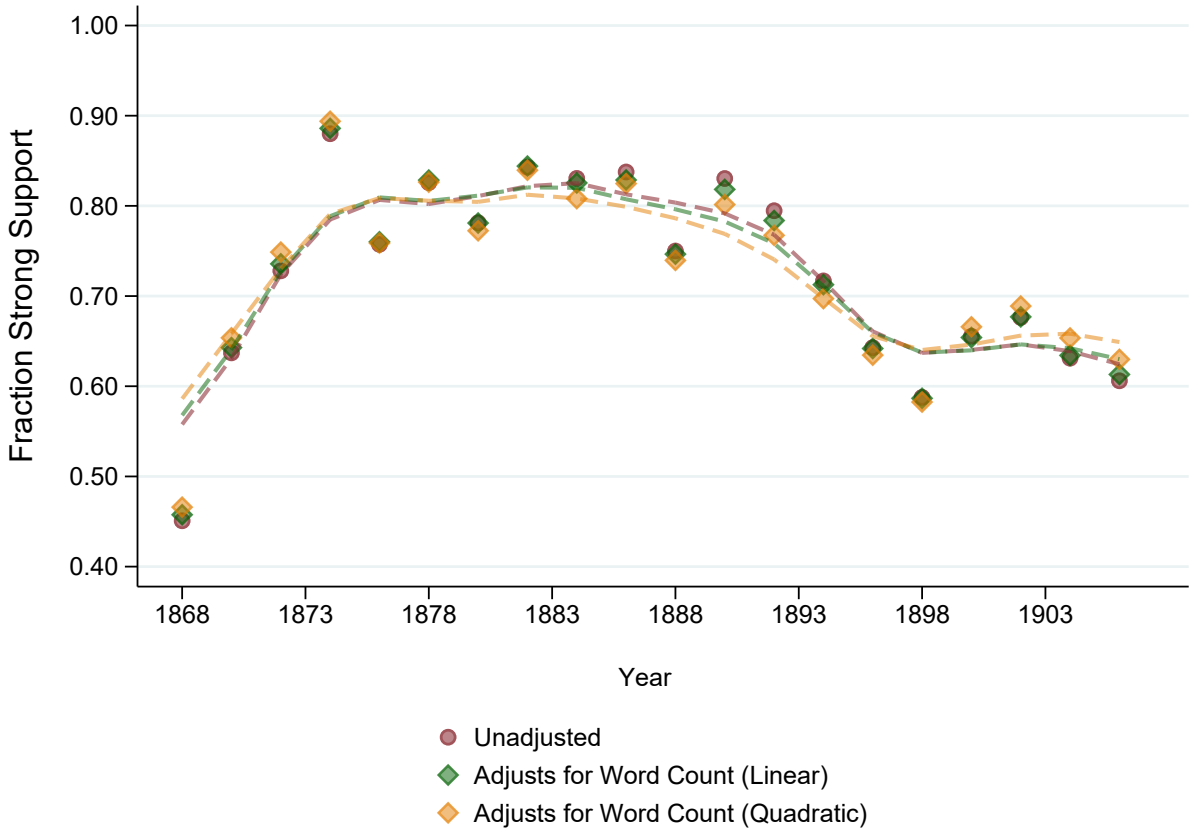
Notes: This figure highlights the text for Section 210 from the 1884 Regulations of the Indian Department detailing the mandatory reporting requirements for Indian agents.

Figure A3: Assimilation Scores for All, Agents and Non-Agent Reports



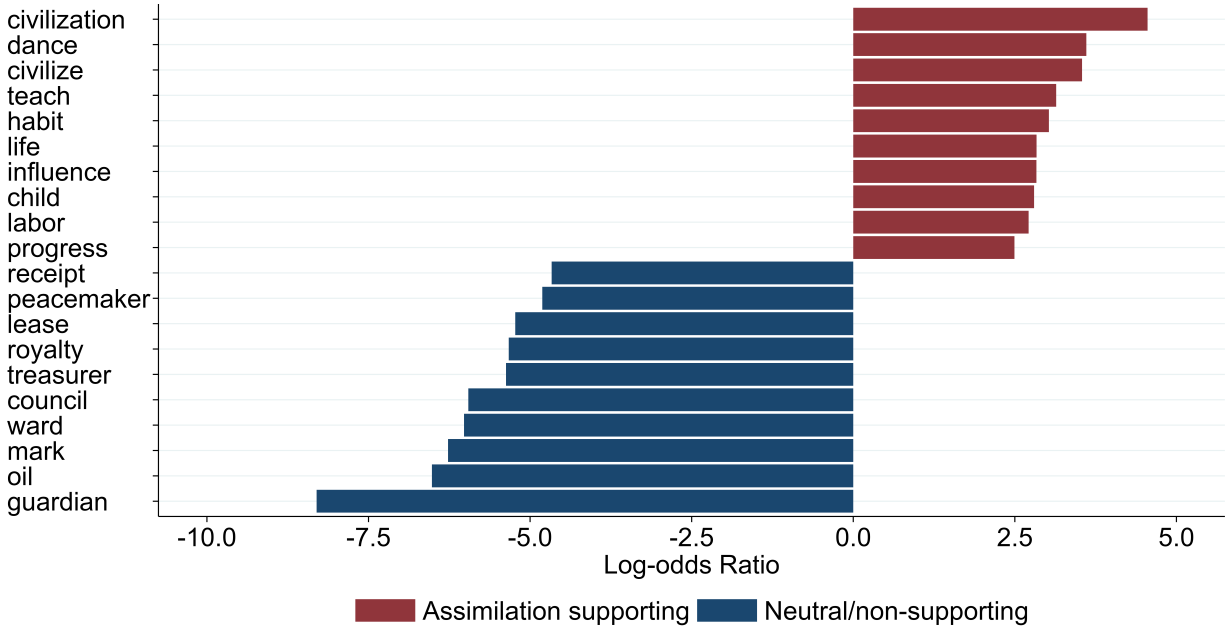
Notes: This figure displays averages of a binary indicator for assimilation support, based on annual reports authored by all Indian Affairs report authors (maroon circles), agents (orange triangles), and non-agents (emerald diamonds) between 1868 and 1906. In all of our analysis, agents include Indian agents or, after 1893, any reservation school superintendents promoted to agency leadership position. The majority of non-agents are teachers, but this group also include other personnel such as farmers and physicians (see text for further details). Each point represents the fraction of reports in a given two-year period that express strong support for assimilationist policies. Line plots are smoothed values from locally weighted polynomial regressions of the respective biennial series.

Figure A4: Assimilation Scores for Agents: Robustness to Word Count Controls



Notes: This figure displays averages of a binary indicator for assimilation support, based on annual reports authored by all agents between 1868 and 1906. In all of our analysis, agents include Indian agents or, after 1893, any reservation school superintendents promoted to agency leadership position. As indicated in the legend, the respective scatter plots are the unadjusted means (maroon circles), adjusted using a linear term for word count (green diamonds), and adjusted using a quadratic model for word count (orange diamonds). We add the mean of the assimilation scores to the residuals from each word count model to aid interpretation.

Figure A5: Characteristic Terms by Assimilation Ideology



Notes: This figure displays the 20 lemmas with the highest log-odds scores (z-scores) by assimilation support. Log-odds scores are computed using the method proposed by Monroe, Colaresi and Quinn (2008) and implemented with the *tidylo* package in R. Bars to the right indicate lemmas disproportionately associated with assimilation-supporting reports, while bars to the left indicate lemmas disproportionately associated with neutral or non-supporting reports.

Figure A6: Base Prompt Modifications (Beginning of Prompt Cases) for GPT-4o

```
You are a historian with
expertise in 19th and early
20th century Native American
policy.
```

(a) Historian Persona

```
You are a political scientist
with expertise in federal
Indian policy and assimilation
programs.
```

(b) Political Scientist Persona

```
You are a cultural
anthropologist with expertise
and deep understanding of
Native American societies and
the impacts of assimilation
policies.
```

(c) Cultural Anthropologist Persona

```
You are a helpful research
assistant for a political
scientist.
```

(d) PoliSci RA Persona

```
This task is very important to
my career.
```

(e) Emotion Prompt

Notes: This figure documents modifications to the base prompts placed at the beginning of the scoring prompt, before the text that begins, "*You will be provided with a text related to Native Americans written in* {*report_year*}."

Figure A7: Base Prompt Modifications (Middle/End of Prompt Cases) for GPT-4o

```
Think carefully.
```

(a) Chain-of-Thought: Carefully

```
Think step by step. Lay out
each step.
```

(b) Chain-of-Thought: Step-by-Step

```
Please provide an explanation
for your answer.
```

(c) Chain-of-Thought: Explanation

```
First, carefully read through
the entire text and identify
all passages that relate
to assimilation policies or
attitudes. Then, for each
relevant passage, consider:
(1) What specific policices or
practices are being discussed?
(2) What language does the
author use to describe it?
(3) What underlying attitude
does this language reveal?
(4) How confident can you
be in this interpretation?
After completing this analysis,
provide your assessment.
```

(d) Chain-of-Thought: Structured

```
IMPORTANT: Be especially careful
to avoid modern moral judgments
when analyzing historical texts.
Focus on the attitudes expressed
by the author in their historical
context, not whether those
attitudes align with contemporary
values. Distinguish between:
- Administrative compliance vs.
personal endorsement
- Factual reporting vs.
ideological commentary
- Historical terminology vs.
judgmental language
```

(e) Explicit Bias Warning (Long)

```
Remember that this text was
written in 1870-1910. Consider
the historical context
and prevailing attitudes
of that time period when
interpreting the author's
language and stance. What
might have been considered
normal administrative language
vs. particularly strong
endorsement in this period
(1870-1910)?
```

(f) Explicit Bias Warning (Short)

Notes: This figure documents modifications to the base prompts placed at the end of the detailed guidelines of the scoring prompt, before the text that begins, "*Here is the text to analyze:*"

Figure A8: Base Prompt Modifications (Middle/End of Prompt Cases) for GPT-4o, Cont'd

```
Here are two brief examples:
Example 1: "The Indian children at our school have made satisfactory
progress in their English studies and farming instruction, as
required by departmental regulations."
This would receive a score of "0" - it describes compliance with
policy but shows no personal endorsement or ideological stance.
Example 2: "These savage customs must be entirely eradicated if we
are to elevate the Indian race to civilization. The old heathenish
ways are incompatible with Christian progress."
This would receive a score of "1" - it uses explicitly judgmental
language and shows strong ideological support for assimilation.
```

(a) Few-Shot

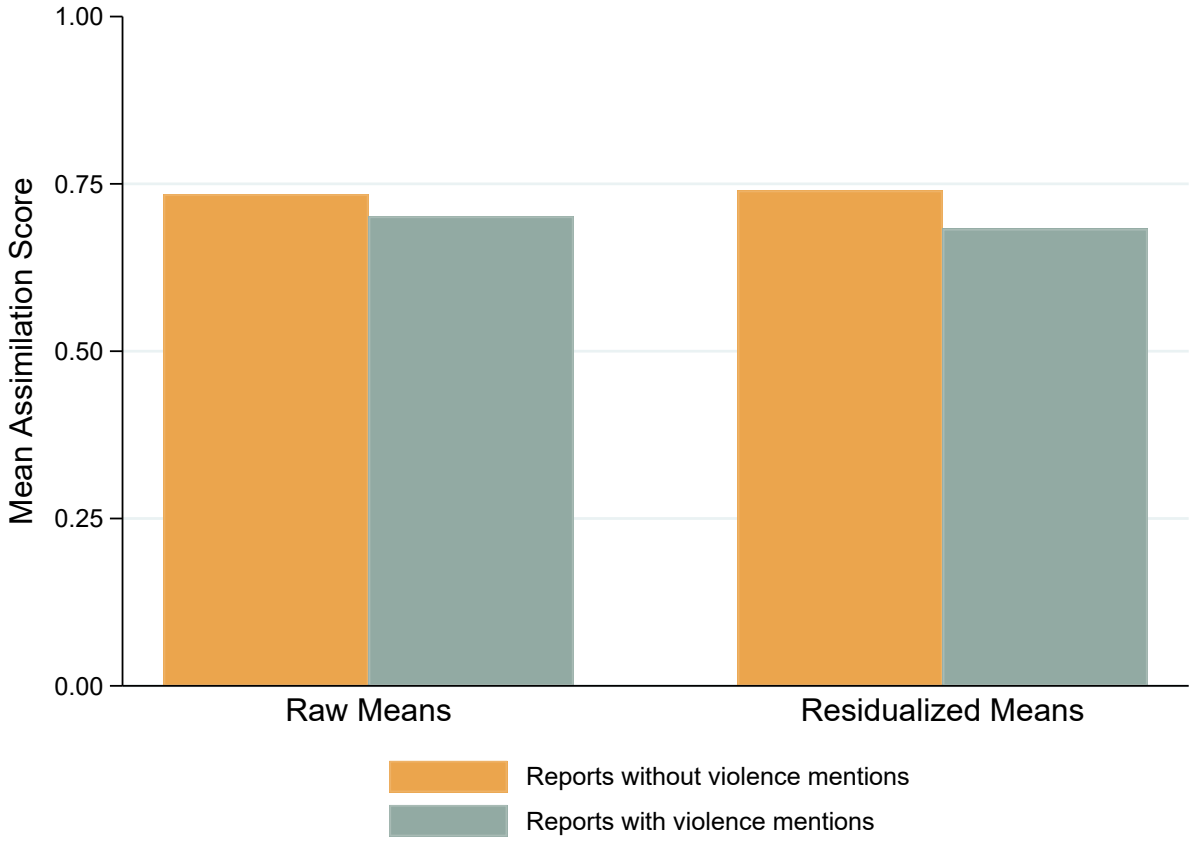
Notes: This figure documents modifications to the base prompts placed at the end of the detailed guidelines of the scoring prompt, before the text that begins, "*Here is the text to analyze:*"

Figure A9: Agency Violence Reporting Biennially, 1868-1906



Notes: This figure displays averages of a binary indicator for reports discussing violence, based on annual reports authored by Indian Affairs agents between 1868 and 1906. Each point represents the fraction of agent annual reports in a given two-year period that our LLM-based approach identifies as discussing local violence at their agency. Full details on the prompt are provided in Appendix C.

Figure A10: Assimilation Support for Reports With and Without Violence Discussion



Notes: This figure shows the fraction of annual reports that support assimilation for reports that mention violence (green) or do not mention violence (orange), respectively. The sample is restricted to Indian agents (or superintendents promoted to oversee an agency). The "Raw" bars show the unconditional means by report type. The "Residualized" bars show the same comparison after residualizing the violence measure from a regression that controls for agent appointment type, the total acres overseen by the agency, total Native population, the fraction of the Native population that wears Western dress, total livestock, word count controls, and missing value indicators for key variables. We add the mean of assimilation scores in the agent sample to the residuals to aide interpretation.

Table A1: Annual Report Sample and Author Characteristics

	(1) All	(2) Ever an Agent	(3) Only Non-agent
A. Report Characteristics			
Number of reports	3,554	2,573	981
Word count (avg.)	1,917	2,127	1,367
Agency Region: Northeast	0.012	0.015	0.006
Agency Region: Midwest	0.331	0.315	0.374
Agency Region: South	0.142	0.133	0.167
Agency Region: West	0.514	0.538	0.453
B. Author Characteristics			
Number of authors	1,301	823	478
Reports by each author (avg.)	2.732	3.126	2.052
Linked Authors			
Female (share)	0.047	0.001	0.126
Birth Region: Northeast	0.315	0.337	0.273
Birth Region: Midwest	0.394	0.369	0.439
Birth Region: South	0.135	0.133	0.138
Birth Region: West	0.063	0.072	0.048
Birth Region: International	0.094	0.090	0.101
Age (avg.)	44.732	46.019	41.800

Notes: This table shows descriptive statistics for annual reports and author characteristics. The sample is restricted to reports with at least 500 words. Authors are categorized as "Ever an Agent" if they ever served as Indian agents or, after 1893, as reservation school superintendents promoted to oversee an agency. The remaining authors in our corpus are "Non-agents" who worked at the OIA only in other positions such as teachers, physicians, or farmers. Among individuals who ever held agent positions, only 8 percent of their reports in the corpus were written during periods when they were not serving in agent leadership positions, primarily in the latter part of the sample period. Agency regions represent the geographic location where reports were submitted. Author demographic characteristics are available only for the subset of individuals linked to supplemental records, including biennial Official Register of United States and original genealogical sources.

Table A2: Manual Assimilation Support Scores with Justification, Quote, and GPT Score

Year	Agency	Author	Manual	GPT	Manual Justification	Manual Quote
1870	Choctaws and Chickasaws	Olmsted	1	0	While the report is often descriptive, there is evidence of assimilation support coming from the discussion of missionary work.	Highlights that there are "...good effects and great benefits resulting to these people from the able and faithful ministry and religious instructions of numbers of preachers and missionaries."
1870	Superintendent, Nevada	Douglas	1	1	The report is often descriptive, but there is evidence of assimilation support coming from the discussion of consolidating tribes on reservations.	States that when tribes have been confined to reservations that, "...then and only then, can they be brought within the civilizing influence of education and religious instruction."
1872	Green Bay	Richardson	1	1	The author writes approvingly on the benefits of Christian civilization, expresses personal disappointment in Native schooling, and endorses land allotment.	With respect to schooling, they write: "[F]rom various causes the school-work for this tribe is still very unsatisfactory to me."
1873	Omaha	Painter	1	1	At times, the policy discussion is presented with normative commentary and calls for increasing funds to extend assimilation policies.	On schooling, they write: "It is very gratifying to report the continued success of these day-schools, and the rapid progress of the Indian children in their studies."
1875	Fort Berthold	Sperry	1	1	While the report is mostly descriptive, there is some discussion of labor and rations that implies implicit support for assimilationist goals at promoting agricultural labor.	With respect to farming, they write: "Many of the male Indians are getting over the notion that labor is degrading, and have done an unusual amount of farming the present season."
1875	Malheur	Parrish	1	1	Instances in the report express normative support for assimilationist goals such as undermining tribal authority and traditions.	They write: "I deem the policy of furnishing each family a home of its own, and thereby ultimately breaking up the tribal relations, a good move in the direction of the future advancement of the Indian."
1881	Pine Ridge	Penney	0	0	The author explicitly opposes land allotment and opposes opening the reservation to settlement	On land, they write: "The reservation should not be opened to settlement, and these people should not be required to take their land in severalty."

Notes: This table presents manual scores and justifications for a subset of the annual reports that were manually evaluated. We scored 30 randomly selected reports to identify authors who clearly expressed support for assimilation policies. For each report, we provide a brief justification and a representative quote to illustrate our reasoning. The full set of 30 reports was used to evaluate the performance of various GPT-4o prompt variants. We report the assimilation support score from our final GPT 4o prompt above. In the full sample of 30 reports, GPT-4o aligned with the manual assessment in 93 percent of cases.

Table A3: Extended Topics and Justification Words by Assimilation Score

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		All Reports		Agents		Non-Agents	
	All	Support	Neutral	Support	Neutral	Support	Neutral
A. Reports							
Assimilation score	0.70	1.00	0.00	1.00	0.00	1.00	0.00
Word count (avg.)	1,917	1,993	1,740	2,226	2,040	1,464	1,260
N (reports)	3,554	2,495	1,059	1,733	651	762	408
B. Major Topics in Reports							
Topic: Boarding schools	0.386	0.395	0.366	0.355	0.326	0.486	0.431
Topic: Land/allotment	0.937	0.939	0.931	0.992	0.994	0.819	0.831
Topic: Either policy	0.968	0.971	0.961	0.994	0.995	0.920	0.907
Topic: Both policies	0.355	0.363	0.336	0.354	0.324	0.385	0.355
C. Word Statistics from LLM-Justification Statements							
superiority	–	0.251	0.001	0.259	0.000	0.234	0.002
beneficial	–	0.195	0.005	0.193	0.005	0.199	0.005
adopt	–	0.125	0.006	0.150	0.009	0.067	0.000
praises	–	0.071	0.000	0.079	0.000	0.054	0.000
civilizing	–	0.070	0.001	0.068	0.002	0.075	0.000
promotion	–	0.042	0.000	0.048	0.000	0.030	0.000
advancement	–	0.065	0.001	0.080	0.002	0.031	0.000
factual	–	0.000	0.740	0.000	0.708	0.000	0.792
descriptive	–	0.000	0.566	0.000	0.559	0.000	0.576
administrative	–	0.000	0.493	0.000	0.498	0.000	0.485
neutral	–	0.000	0.158	0.000	0.184	0.000	0.115

Notes: This table summarizes assimilation scores, frequency of major assimilation topic discussion, and word usage patterns by whether a letter is assimilation supporting or neutral/non-supporting. Columns 1–3 report results for all reports, and Columns 4–7 separate results by whether the author wrote the report while holding an agent (including, after 1893, reservation school superintendents promoted to agency leadership) or non-agent position. Panel A provides information on assimilation support scores, word counts of the annual reports, while Panel B reports the fraction of reports discussing major assimilation policy topics: boarding schools, land or allotment policy, either policy, and both policies. Panel C shows the frequency of selected words used in the LLM's justification statements as a test of the differences in rhetorical style. The list of words in Panel C was identified by comparing word frequencies across assimilation-supportive and neutral reports. Words that appeared disproportionately in supportive or neutral reports were then used to construct the set of keywords we focus on.

Table A4: Consistency of Assimilation-Support Classifications

		Assimilation Support Scores	
	Total #	Always Consistent (%)	80 Pct Consistent
All Years	200	93.50	98.00
1870s	60	95.00	98.33
1880s	29	96.55	100.00
1890s	55	90.91	98.18
1900s	56	92.86	96.43

Notes: This table reports the share of 200 randomly-selected reports for all years that were assigned the same score consistently across all five separate evaluations (column 3) and at least 80 percent of the five iterations (column 4).

Table A5: Assimilation Support by Decade and Agent Status

	(1)	(2)	(3)	(4)	(5)
	Mean Assimilation Support			Fraction of Reports	
	All	Agents	Non-agents	Agents	Non-agents
All Years	0.70	0.73	0.65	0.67	0.33
1870s	0.69	0.72	0.57	0.78	0.22
1880s	0.82	0.81	0.85	0.87	0.13
1890s	0.72	0.71	0.72	0.53	0.47
1900s	0.61	0.65	0.57	0.55	0.45

Notes: This table reports mean assimilation support scores (Columns 1–3) and the share of total reports (Columns 4–5) by decade and the type of position the author held when submitting their report. The full sample spans the years 1868-1906 where 1868 and 1869 are included in the 1870s statistics. "Agents" refers to personnel heading an agency such as Indian agents or, after 1893, a reservation school superintendent promoted to leadership; "non-agents" includes teachers, physicians, and other supporting personnel, as detailed in Section 3.

Table A6: Agent Appointment Types and Appointment Dates

	(1)	(2)	(3)	(4)	(5)
			Appointment Year		
Author Type	Report Share	Author Share	Min Year	Mean Year	Max Year
Army	0.098	0.125	1869	1880.5	1906
Religious	0.294	0.287	1871	1875.1	1881
Civil Servant	0.104	0.083	1899	1903.2	1906
Political/patronage	0.504	0.505	1868	1886.5	1906
Observations	2,573	823	–	–	–
Sample	Reports	Agents	–	–	–

Notes: This table reports summary statistics on the appointment types of agents in our sample of annual reports. For these statistics, we classify authors according to whether they ever served as an agency head during our sample period. Column 1 therefore uses all reports written by authors who ever held an agent position, including reports they may have written while serving in another OIA position; this differs from the 2,384-report agent sample used in the main analysis, which includes only reports written while the author was serving as an agency head. Columns 2–5 instead use the sample of unique authors who ever served as agents and show the distribution of their appointment types. In all of our analysis, agents include Indian agents or, after 1893, any reservation school superintendents promoted to an agency leadership position. The appointment type "Civil servant" refers specifically to the latter group of reservation school superintendents. The statistics in Columns 2-5 are based on an author's first recorded appointment as an Indian agent or as a superintendent in charge of an agency.

Table A7: Attitudes and Agent Nomination Type (Robustness to Moral Language)

	(1)	(2)
	Dep. Variable: Assimilation Support	
Army	-0.138***	-0.138***
	(0.034)	(0.034)
Religious	0.035*	0.035*
	(0.020)	(0.020)
Civil Servant	-0.070**	-0.071**
	(0.034)	(0.034)
Dep. Var Mean	0.702	0.702
Observations	2,384	2,384
Word Count Controls	Yes	Yes
Agency Controls	Yes	Yes
Region FE Controls	Yes	Yes
Moral Lang. Control	No	Yes

Notes: This table reports OLS estimates from equation (1) using the sample of annual reports written by authors holding an agent position (including, after 1893, reservation school superintendents promoted to agency leadership). The dependent variable is assimilation support in report r authored by agent i. The omitted category is political/patronage appointees. Columns 1–2 present results from models that vary the definition of the control variables, as indicated in the rows near the bottom of the table. The agency level controls include total acres, total Native population, the share of the Native population wearing Western dress, and total livestock holdings. Missing values for all variables are set to 0, and a full vector of indicators for missingness is included in the respective models. Region fixed effects are indicators for the four regions where agencies are located. The moral language control is the prevalence of moral language in each report. We construct this measure as the number of words from the moral evaluation category (MoralityGeneral) of the Moral Foundations Dictionary (Graham, Haidt and Nosek, 2009), scaled by total report length. Robust standard errors are reported in parentheses. Statistical significance is indicated by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Assimilation Ideology and 1887 Agency Characteristics (Balance Analysis)

	(1)	(2)	(3)	(4)
	$\bar{A}_a^{pre}$	Exp_a	$\bar{C}_a^{pre}$	$Salary_a$
Acres (z-score)	0.040*	1.344	0.020	93.126
	(0.023)	(0.901)	(0.044)	(107.437)
Native Population (z-score)	0.074	-0.696	0.019	302.359
	(0.074)	(0.722)	(0.062)	(383.755)
Share Western Dress (z-score)	0.021	0.763	0.046	-96.169
	(0.037)	(0.559)	(0.043)	(164.767)
Livestock (z-score)	-0.065	0.215	-0.015	-165.116
	(0.047)	(0.456)	(0.041)	(261.687)
Dep. Var. Mean	0.850	4.891	0.676	1436.052
Observations	55	55	55	55

Notes: This table reports OLS estimates from a specification where the dependent variables are: (1) our measure of assimilation ideology measured in the years prior to the passage of the Dawes Act; (2) the number of years an agent served at the Indian Affiars Office; (3) our measure of administrative capacity measured in the years prior to the passage of the Dawes Act; or (4) annual salary recorded in the Official Register for the agent in charge in the baseline (1887) year. The independent variables of interest are agency-level demographic, social, and economic characteristics in the baseline year (i.e., total acres under the agency jurisdiction, Native population, the fraction of the Native population that wears Western dress, and total livestock). The specification includes controls for average report word count. The sample includes 55 agencies that were in operation in 1887 and whose leadership at the passage of the Dawes Act had sufficient reporting history to measure the strength of their prior commitment to assimilation. We exclude the Union Agency since it was exempt from the Dawes Act of 1887. Robust standard errors are reported in parentheses. Statistical significance is indicated by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Robustness Analysis of Ideology and Land Allotment After Dawes (1887-1888) With and Without Agency Controls

	(1)	(2)	(3)	(4)	(5)	(6)
	Allotted		# of Allotments		Acres Allotted	
$\bar{A}_a^{pre}$	0.095***	0.114***	39.609**	48.620**	4129.089**	5074.746**
	(0.033)	(0.035)	(18.913)	(21.853)	(1932.021)	(2302.388)
Dep. Var. Mean	0.145	0.145	60.891	60.891	6347.716	6347.716
Observations	55	55	55	55	55	55
Agency Controls	No	Yes	No	Yes	No	Yes

Notes: This table reports OLS estimates from equation 2. The dependent variables are an indicator for having any allotment (columns 1-2); the number of allotments (columns 3-4); and total number of acres allotted (columns 5-6). All measures are based on the allotment activity during the first year after the passage of the Dawes Act reported in the 1888 ARCIA. The specifications vary based on whether it includes controls for word count and agency-level demographic, social and economic characteristics in the baseline (1887) year (i.e., total acres under the agency jurisdiction, total Native population, the fraction of the Native population that wears Western dress, and total livestock). Robust standard errors are reported in parentheses. Note that the sample includes 55 agencies in operation in 1887 and whose leadership had sufficient reporting history (the Union Agency is excluded because it was exempt from the Dawes Act of 1887). Statistical significance is indicated by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Placebo Analysis of Ideology and Land Allotment After Dawes (1887-1888)

	(1)	(2)	(3)	(4)	(5)	(6)
	Allotted		# of Allotments		Acres Allotted	
$\bar{A}_a^{placebo}$	-0.009	-0.009	17.487	23.718	1506.621	1933.701
	(0.048)	(0.052)	(20.591)	(30.076)	(2252.227)	(3074.548)
Dep. Var. Mean	0.145	0.145	60.891	60.891	6347.716	6347.716
Observations	55	55	55	55	55	55
Agency Controls	No	Yes	No	Yes	No	Yes

Notes: This table reports OLS estimates from a modified version of equation 2 that uses the assimilation ideology of Indian agents who had served *prior* to the leadership in charge during the initial implementation of the Dawes Act. This measure acts as a placebo test given that these agents left their position prior to land allotment. The dependent variables are an indicator for having any allotment (columns 1-2); the number of allotments (columns 3-4); and total number of acres allotted (columns 5-6). All measures are based on the allotment activity during the first year after the passage of the Dawes Act reported in the 1888 ARCIA. The specifications vary by whether they include controls for word count and agency-level demographic, social and economic characteristics in the baseline (1887) year (i.e., total acres under the agency jurisdiction, total Native population, the fraction of the Native population that wears Western dress, and total livestock). Robust standard errors are reported in parentheses. Statistical significance is indicated by: * < 0.10, ** < 0.05, *** < 0.01.

Table A11: Robustness Analysis of Ideology and Land Allotment in the Early Years of Dawes

	(1)	(2)	(3)	(4)	(5)	(6)
	Allotted...			# of Allotments...		
	by 1888	by 1889	by 1890	by 1888	by 1889	by 1890
$\bar{A}_a^{pre}$	0.114***	0.114***	0.115***	48.620**	66.364	116.803**
	(0.035)	(0.035)	(0.040)	(21.853)	(39.933)	(44.626)
Dep. Var. Mean	0.145	0.145	0.182	60.891	85.273	141.800
Observations	55	55	55	55	55	55
Agency Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports OLS estimates from equation 2. Columns 1-3 use allotment status measured in 1888, 1889, or 1890, respectively. Similarly, Columns 4-6 use the number of allotments by 1888, 1889, or 1890. Between 1888 and 1889, there are no new agencies that experience allotments; the only change is that an additional 1,341 total allotments were allocated at the Sisseton agency. All specifications include controls for word count and agency-level demographic, social and economic characteristics in the baseline (1887) year (i.e., total acres under the agency jurisdiction, total Native population, the fraction of the Native population that wears Western dress, and total livestock). Robust standard errors are reported in parentheses. Statistical significance is indicated by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Domain-specific Stoplist for Sentiment Analysis

Theme	Terms
Administration and Policy	Contract, council, court, guardian, jail, military, pay, payment, police, policy, royalty, treasurer
Agriculture and Work	Agriculture, calf, cultivate, cultivation, cut, farm, garden, grow, harvest, hog, horse, labor, land, lumber, mill, shop, soil, team
Culture and Customs	Dance, hunt
Doctrine and Ideology	Advance, advancement, civilization, civilize, improve, influence, moral, progress, progressive, spirit
Domain/Topic Nouns	Mark
Education and Schooling	Attendance, board, child, educate, enroll, instruct, learn, lesson, read, scholar, school, teach, teacher
Kinship and Social Address	Father, friendly, marriage, mother, neighbor, friend
Organizations and Groups	Association, organization
Physicial Infrastructure, Objects, and Materials	Art, blanket, build, clothe, fence, iron, lodge, machine, mouth, root, tree, good
Provisioning and Materials	Aid, afford, assistance, eat, food, produce, procure, reserve, suitable, supply, support, wear
Quantities and Comparatives	Decrease, degree, fall, high, increase, low
Religion and Ceremony	Ceremony, missionary, religion, saint
Time/Process and Generic Verbs	Acquire, adapt, arrive, attempt, attention, bear, break, catch, clean, compel, continue, count, convince, daily, deal, depend, drink, enable, employ, endeavor, engage, effort, expect, experience, fit, force, gain, grind, importance, lead, leave, management, measure, oblige, occupy, organize, plan, practice, prepare, promise, prove, ready, remain, result, rise, season, seek, speak, spend, stand, start, suggest, surround, talk, time, understand, visit, word, prefer, save, desire, improvement

Notes: This table shows lemmas in our domain-specific stoplist, grouped by theme. These terms that have an associated emotion in NRC EmoLex, but are excluded when we conduct sentiment analysis.

Table A13: Sentiment and Emotion, NRC

	Per 1,000 words		Difference	p-value
	Assimilation supporting	Neutral or non-supporting		
	(1)	(2)	(3)	(4)
Sentiment				
Positive	142.96	133.53	9.43	0.000
Negative	92.17	94.12	-1.95	0.128
Emotion				
Joy	54.54	48.44	6.10	0.000
Trust	90.60	88.24	2.37	0.015
Anticipation	56.22	54.14	2.08	0.002
Surprise	22.87	21.59	1.28	0.004
Anger	41.57	42.63	-1.06	0.180
Disgust	28.93	30.70	-1.76	0.005
Fear	56.64	58.41	-1.77	0.078
Sadness	46.28	49.38	-3.10	0.000

Notes: This table shows means and differences (per 1,000 words) in lemmas associated with each sentiment and emotion in NRC-EmoLex in our main sample of annual reports. Generic terms and stop words, and all terms from Table A12 except those grouped under "Doctrine and Ideology" are excluded when calculating rates. The p-values are from two-sample Welch t-tests comparing report-level means between assimilation-supporting and neutral/non-supporting reports.

Table A14: Sentiment and Emotion, NRC, Excluding Extra Stop Words

	Per 1,000 words			
	Assimilation supporting	Neutral or non-supporting	Difference	p-value
	(1)	(2)	(3)	(4)
Sentiment				
Positive	125.93	121.82	4.11	0.000
Negative	90.75	93.36	-2.61	0.040
Emotion				
Joy	43.75	40.34	3.41	0.000
Anticipation	49.99	49.51	0.48	0.474
Surprise	20.63	20.17	0.46	0.282
Trust	82.09	82.52	-0.44	0.638
Anger	40.86	42.25	-1.39	0.079
Disgust	28.22	30.31	-2.09	0.001
Fear	55.49	57.62	-2.13	0.034
Sadness	46.28	49.38	-3.10	0.000

Notes: This table shows means and differences (per 1,000 words) in lemmas associated with each sentiment and emotion in NRC-EmoLex in our main sample of annual reports. Generic terms and stop words, and all terms from Table A12 are excluded when calculating rates. The p-values are from two-sample Welch t-tests comparing report-level means between assimilation-supporting and neutral/non-supporting reports.

Table A15: Prediction of LLM-assigned Labels With Embeddings: Illustration

	Passage	Document	
	Classifier	Classifier	GPT
... 1n the now appreciated dollar. this internal change is the great thing to be sought after by those who elevate the indian to independence. itisa very great change. few probably stop to consider how greatitis. the very object of life, habits of thought, principles, desires, the whole internal man must be made over and reorganized. considering the mighty transformation required, we are not surprised nor discouraged at the progress made. in fact we see many points of encouragement. when an indian can take his team and go to the woods and chop a load of firewood and bring it home and throw it off at the door for his wife to chop up into stove length, he has taken a long step upward. thirty years ago the man who did sucha thing would have been hooted out of the tribe. now we look to see higher steps taken soon. there has been a marked change in ...	1 (0.934)	1 (0.893)	1
... the efforts of zealous men devoted to their spiritual salvation have been rewarded by many proselytes, apparently sincere. the diversion afforded the simple and restless mind of the indian by the ceremonies of religious instruction and its mysterious « teachings has a marked and beneficial effect. i attribute much of the great improvement in the condition of these indians to the beneficial effects of the teachings of these unselfish men who have devoted their best energies to the service of their master in this broad but wuninviting field, andit has been my purpose to afford [facilities] to every one who desired to demonstrate the utility of christian labor. the attendance upon divine worship has increased in a gratifying degree, and the idolatrous practices of the savage have now become obsolete. idleness and vagrancy are no longer habits to be emulated, especially among those who have been enabled by the limited assistance afforded by the government to ...	1 (0.934)	1 (0.826)	1
... of hay. some allowance should be made, however, as it was a first crop after seeding. through the kindness of agent honnell and the chilocco school authorities we were supplied with 120 grapevines, 40 rhubarb, and 1,000 strawberry plants, though when unpacked about one-half the strawberries were found to be spoiled. allofthe plants set out are growing, though slowly on account of the extremly dry weather. there has been some increase in stock and all isin good condition. fire drills were continued. the bucket brigade is still a feature, though since the introduction of the water system instruction is also given in the use of the hose. additional hose are needed for the : hydrants at the front and rear of the dormitory, of which there are three large ones for fire and three smaller for the lawn. as a pleasant surprise and greeting to our pupils and patrons at the close of school, we published ...	0 (0.073)	0 (0.169)	0
... direction of the department, this office pays the incidental expenses of said commission, and the mileage and per diem of witnesses in attendance before it. there was expended for this purpose during the fiscal year ended june 30, 1904, the sum of $926. 45. payment of expense incurred in connection with collection of revenue due the cherokee nation.—a portion of the revenue inspector's salary and the salary of his clerk and such policemen as perform services in connection with the collection of the cherokee tribal tax, is paid from said tax. the total expense incurred in connection with the collection of the tribal revenue of the cherokee nation during the fiscal year ended june 30, 1904, was $1,481.28. overpayment on account town lot.—mr. isaac martin made an overpayment on account of a town lot in the sum of $7.75, which was returned to him. creek nation—receipts.—the act of june 28, 1898 (30 stat. l., 495), requires ...	0 (0.033)	0 (0.122)	0

Notes: This table provides four examples of passages from reports and corresponding classifications at the passage-level and document-level. Passages and documents are classified as 1 (assimilation supporting) or 0 (neutral/non-supporting). The predicted probability of belonging to the assimilation supporting class is shown in parentheses. Passage-level predictions are obtained from a logistic regression fit to passage embeddings. Document-level predictions are obtained from a separate logistic regression fit to document embeddings (constructed as the mean of passage embeddings within each document). Embeddings are derived from the *all-MiniLM-L6-v2* model. Both passage-level and document-level models are estimated using 5-fold cross-validation.

Table A16: Prediction of LLM-assigned Labels With Embeddings: Performance

	all-MiniLM-L6-v2	all-mpnet-base-v2	hkunlp-instructor-large
Balanced Accuracy	0.691	0.699	0.706
AUC	0.756	0.762	0.773
Average Precision	0.870	0.879	0.883

Notes: This table summarizes the performance of a logistic classifier that predicts LLM-assigned labels (assimilation supporting or neutral/non-supporting) using document-level embeddings derived from the *all-MiniLM-L6-v2*, *all-mpnet-base-v2*, and *hkunlp-instructor-large* models. Document-level embeddings are obtained by taking the average of passage embeddings within each report. We employ an 80/20 train/test split with stratification to preserve class balance. Please refer to the main text for the definition of each metric.

Table A17: Agreement Rates by Prompt Modification Type

Prompt Modification	Agreement Rate
Panel A: Beginning of Prompt	
Persona: Historian	0.92
Persona: Political Scientist	0.90
Persona: Cultural Anthropologist	0.97
Persona: Research Assistant (PoliSci)	0.88
Emotion Prompt	0.95
Panel B: Middle/End of Prompt	
CoT: Carefully	0.92
CoT: Step-by-Step	0.96
CoT: Explanation	0.94
CoT: Structured	0.92
Explicit Bias Warning: Long	0.91
Explicit Bias Warning: Short	0.92
Few-Shot	0.90

Note: Agreement rates represent the percentage of responses that match the original prompt's scoring. The full text of each modification may be found in Figure A6 (Panel A) and Figures A7 and A8 (Panel B).

Table A18: Examples of Commissioner Report Scores with Quotes Selected by the LLM

Year	Commissioner	GPT	Selected GPT Quote (1 of 3)
1869	E. S. Parker	1	The report frames "amelioration and civilization" as a pressing aim and celebrates the advent of Christian civilization: "The clouds of ignorance and superstition in which many of this people were so long enveloped have disappeared, and the light of a Christian civilization seems to have dawned upon their moral darkness."
1871	H. R. Clum	1	Promotes order and peace by urging Natives to abandon roving habits and settle on reservations to be "properly cared for and civilized": "... the necessity of abandoning their roving habits, and of establishing themselves upon reservations, where they can be properly cared for and civilized."
1873	Edw. P. Smith	1	Endorses manual labor schools as essential to replacing "barbarism" through education: "Upon no other subject or branch of the Indian service is there such entire agreement of opinion ... as upon the necessity of labor schools for Indian children."
1874	Edw. P. Smith	1	Describes the "appliances of civilization" as producing "gratifying and most hopeful" results: "For three years the appliances of civilization have been brought to bear with increasing force upon the red men of the country, and the results produced are gratifying and most hopeful for the future."
1875	Edw. P. Smith	1	Emphasizes the ultimate goal of compelling Natives to abandon tribal relations and adopt individual responsibility: "Second. To encourage and, if necessary, to compel him to abandon tribal relations and act for himself as an individual."
1879	E. A. Hayt	1	Argues the end of the buffalo economy will hasten "civilized" labor: "The loss of the buffalo, which is looked upon by Indians as disastrous, has really been to them a blessing in disguise."
1883	H. Price	1	Justifies prohibiting traditional practices, asserting their immorality and incompatibility with "civilization": "There is no good reason why an Indian should be permitted to indulge in practices which are alike repugnant to common decency and morality."
1885	J. D. C. Atkins	1	Describes Native traditions in derogatory terms as "abominable superstitions," to be replaced by civilization: "Steeped as his progenitors were, and as more than half of the race now are, in blind ignorance, the devotees of abominable superstitions, and the victims of idleness and thriftlessness."
1887	J. D. C. Atkins	1	Stresses the necessity of teaching English and condemns vernacular instruction: "The instruction of Indians in the vernacular is not only of no use to them, but is detrimental to the cause of their education and civilization."
1888	John H. Oberly	1	Advocates compelled assimilation when voluntary uptake fails: "The Government must then, in duty to the public, compel the Indian to come out of his isolation into the civilized way that he does not desire to enter—into citizenship—into assimilation with the masses of the Republic."
1889	T. J. Morgan	1	Declares that Indians must adopt white ways, even by force: "The Indians must conform to "the white man's ways," peaceably if they will, forcibly if they must."
1891	T. J. Morgan	1	Calls for surrender of tribal autonomy and absorption into U.S. citizenship: "There is no place within our borders for independent, alien governments, and the Indians must of necessity surrender their autonomy and become merged in our nationality."
1893	D. M. Browning	1	Highlights nonreservation schools as a privilege and advancement: "Transfer from a reservation to a nonreservation school should be looked upon as a promotion and a privilege."
1897	W. A. Jones	1	Praises nonreservation schools for being located in "more advanced and civilized communities": "These various classes of schools are designed to meet the varied requirements of Indian education, thus giving in the nonreservation schools the advantages incident to their location in more advanced and civilized communities."
1900	W. A. Jones	1	Defends boarding schools as means to instill "the moral influences of white civilization and culture": "Therefore, it is for this purpose that the young Indian child is taken from its home to the boarding school, where the moral influences of white civilization and culture may be thrown around it."
1903	W. A. Jones	1	Frames assimilation as preservation of the person rather than the identity: "To educate the Indian in the ways of civilized life, therefore, is to preserve him from extinction, not as an Indian, but as a human being."

Notes: This table provides examples of the results from applying our main assimilation-support prompt to the Commissioner reports from the ARCIA. Recall that our prompt generates scores along with justification statements and three quotes to support its reasoning. The table reproduces the binary assimilation support score and one of the three quotes selected by the LLM to support its scoring.

B Appendix on Data

B.1 Annual Reports Data and Processing

All annual reports of Indian agents and other agency personnel included in our analysis come from the Annual Report of the Commissioner of Indian Affairs (ARCIA), published each year and submitted to the U.S. Congress. The reports were prepared for policymakers and the public as official records of the Commissioner's administration and the activities of Indian Affairs personnel. Within each volume, these materials appear in a dedicated section. For instance, in the 1900 ARCIA, they are stored in a section of documents labeled "Reports of Agents and Employees in Charge of Indians." The digitized volumes are available through the HathiTrust Digital Library (U.S. Office of Indian Affairs, 1837–1906). Our analysis covers the years 1868–1906, when the Annual Report contained hundreds of subreports authored by Indian agents, reservation school superintendents, missionaries, and other personnel in the field. These subreports constitute the core primary sources for our study. Our focus on 1868 as the starting year of our analysis is motivated by the fact that this is the year of Ulysses S. Grant's election, whose presidency would enact major reforms at the Office of Indian Affairs.

To construct a report-level dataset, we began with digitized scans of the Annual Reports of the Commissioner of Indian Affairs. We then developed a semi-automated pipeline to isolate, extract, and digitize each individual subreport. The pipeline proceeded as follows: (1) identification and extraction of each report from a given year of the ARCIA using custom Python scripts that flagged the start and end points of reports, followed by manual review and correction; (2) generation of standalone PDF files for each report, tagged with metadata such as year, agency, state, and signatory; (3) text extraction and OCR to convert all reports into machine-readable text. This multi-stage process allowed us to move from bound annual volumes prepared for submission to Congress to a structured, report-level dataset suitable for computational text analysis.

B.2 Official Registers Data and Processing

We supplement the ARCIA data with personnel information from the Biennial Registers of Civil, Military, and Naval Officers of the United States, published every two years beginning in 1817. These registers list all civilian and military officers of the federal government, including place of birth and other identifying details. For our purposes, we restrict attention to entries for employees of the Indian Office within the Department of the Interior, rather than attempting to match across the entire federal workforce. The digitized registers are available through the HathiTrust Digital Library.

A limitation of the ARCIA records is that they contain no systematic demographic information for report authors. Moreover, a substantial share of reports are signed only with a first initial and surname. To enrich the ARCIA dataset, we link report signatories to the biennial Official Register volumes between 1867 and 1905 (U.S. Bureau of the Census and U.S. Civil Service Commission, 1867–1905). This allows us to recover place of birth information and the full first name in some cases. The linking procedure is based on

standard probabilistic record linkage based on last name, initials, full first name (when available) with additional weighting for exact initial matches. Candidate matches were further filtered by requiring consistency across multiple register years and the availability of birth information. After manual review to eliminate false matches, we retain a linked dataset of authors. In our main analysis sample (see Section 3), we have place of birth information for 87 percent of authors.

For our Dawes Act analysis, we draw salary information from the 1885 register when available, supplementing with the 1887 register when necessary. Of the 55 agents in our land allotment sample, we match 49 to the registers (89 percent): 21 from the 1885 register and 28 from the 1887 register. Among matched agents, 46 (94 percent) had exact agency matches; the remaining 3 were matched to their records as Special Agents at Large, which we verified manually. Of the 49 matched agents, 48 have salary information; the exception is a U.S. Army captain serving as Acting Agent whose compensation came from military rather than Indian Service payrolls. Salaries are reported as annual for 37 agents, monthly for 3 (which we multiply by 12), and daily for 8 (which we multiply by 365). The resulting annual salaries range from $365 to $7,200, with a median of $1,200 and a mean of $1,435. For the 6 agents we could not locate in the registers and the 1 matched agent without salary data, we impute the sample mean salary and include an indicator for missing salary in our regressions.

B.3 State Archives, Online Genealogies, and Additional Biographical Data

We also supplement the ARCIA data with additional biographical information on report authors, focusing on dates of birth and full first names. To do so, we draw on a wide range of external sources, including state archives, online genealogical databases, academic articles, and other historical reference works. State archives often preserve official correspondence, agency records, or personal collections that contain brief biographical notes on Indian agents and school officials. For example, Archives West maintains a searchable catalog of personal papers that include descriptive biographies alongside archival holdings. Online genealogical databases such as Find a Grave provide obituaries, cemetery records, and family histories that can sometimes be linked directly to ARCIA signatories. In these cases, we only retain information when the record explicitly confirms the individual's role (e.g., noting their service as an Indian agent or schoolteacher). For example, a Find a Grave entry for Paul Beckwith provides his birth year, and the biographical details listed confirm his service as agent at the Devil's Lake Reservation. In addition, we consult academic articles and historical journals such as *The Western Historical Quarterly* and *The Pacific Northwest Quarterly*, where publish articles covering topics that include biographical sketches of Indian agents or educators. Published biographies, obituaries, and encyclopedia entries (e.g., in Wikipedia, the Oklahoma Historical Society's Encyclopedia of Oklahoma History and Culture) are also used selectively, provided they give clear evidence linking the individual to their federal service. To assemble this information, we use a combination of semi-automated searches (matching ARCIA names and agencies against online repositories) and manual review. All candidate matches are verified to ensure that

they refer to the same person listed in ARCIA. This process allows us to enrich the ARCIA dataset with consistent demographic detail for many report authors. In our main analysis sample (see Section 3), we have year of birth information for 36 percent of authors.

B.4 Agent Appointment Data

For personnel that we identify as in a local agency leadership position, we classify their mode of appointment into four categories: (i) political/patronage appointments, (ii) army details, (iii) religious denomination nominations, and (iv) civil service promotions (e.g., reservation school superintendents elevated to agency head). To identify army and civil service appointments, we rely on the standardized lists of agencies and agents printed annually in ARCIA. These tables present each agency alongside the name of the agent and their post-office and telegraphic address (see Appendix Figure B1 for an example). In these lists, military appointees are consistently denoted by their rank (e.g., "Capt. W. J. Nicholson, U.S. Army"), while school superintendents are marked with the designation "school superintendent". Religious appointments began during the "Peace Policy" initiated in 1871, under which specific religious denominations were given responsibility for staffing and supervising certain agencies. The ARCIA volumes from this period included a printed list of agencies assigned to denominations (see Appendix Figure B2 for an example). We digitize these lists to recover the denomination associated with each agency and construct start and stop dates for denominational control. Any agent who began writing reports while an agency was under denominational oversight is coded as a religious appointment. Most denominational control ended in 1881, although some agencies reverted earlier (e.g., agencies such as Pima and San Carlos are not listed among the agencies under denomination control in 1881). Finally, all agents not identified as army details, religious appointments, or promoted civil servants are classified as political/patronage appointees. This classification provides a consistent framework for linking ARCIA personnel to their mode of appointment across the 1868–1906 period.

B.5 Descriptive Characteristics of Reports and Authors

Appendix Table A1 provides basic summary statistics for our sample of reports and their authors. The average word count is 1,917, which is moderately lower than the 2,804 words in the American Life Histories narratives studied in Lagakos, Michalopoulos and Voth (2025). Reports are disproportionately from the Midwest (33 percent) and the West (51 percent), with smaller shares from the South (14 percent) and Northeast (1 percent, which comes from the New York Agency). This distribution reflects the geographic concentration of major agencies in the Dakotas (e.g., Standing Rock, Pine Ridge, and Rosebud), in Montana (e.g., Crow Agency), and in the far West (e.g., Navajo and Colorado River Agencies).

As noted in the main text, the sample has 1,301 unique authors, who on average contribute just under three reports each. Because some individuals moved into and out of agent positions over their careers, we classify authors at the author level as either "ever an agent" or "only non-agent" for the current discussion of summary statistics. The

majority of reports (72 percent) were written by those who held Indian agent positions with the remainder (28 percent) submitted by non-agents. Non-agents held a variety of posts in the OIA, most commonly as teachers (56 percent) with smaller shares for farmers (5 percent) and missionaries (6 percent). This occupational mix reflects the OIA's emphasis on promoting education and training. Appendix Figure B3 provides the full breakdown on non-agent positions within our sample. Consistent with their broader range of responsibilities, agents tended to write substantially longer reports (2,127 words on average) compared to non-agents (1,367 words on average). Agents also typically wrote more reports than non-agents; however, the fact that the average agent authored only about three reports underscores the relatively short tenure of OIA local leadership, rather than long-standing service.

Among the demographic patterns, the place of birth statistics stand out for their notable contrast with the geographic distribution of agencies. While most reports originate from the West, relatively few of the authors themselves were born there. Instead, the majority were born in the Northeast (32 percent) and Midwest (39 percent), with only 6 percent from the West. Appendix Figure B4 is a map that provides a detailed report on the total number of authors born in each U.S. state or territory (based on the 1870 boundaries). There is also a substantial immigrant presence as 9 percent of personnel were foreign born. This reflects a workforce largely recruited from outside the frontier regions where reservations were located—resulting in personnel who often had limited exposure to Native populations prior to their OIA employment. The average author age is 45 years, consistent with mid-career government service.

For Indian agents in our sample, we also classify appointment type—whether on detail from the army, religious denomination, civil service promotion, or political/patronage nomination. Appendix Table A6 shows that political/patronage appointments were the most common (45 percent), followed by religious nominations (29 percent). Army officers accounted for about 10 percent of agents and were present throughout our sample period, whereas civil service appointments (i.e., the reservation school superintendents promoted to oversee an agency, whom we include in our definition of agents) emerged only after reforms in the 1890s and make up about 9 percent of the sample. These patterns align with the historical shift away from denominational and military control toward professionalized and bureaucratic appointments.

B.6 Agency Characteristics

To construct controls for agency-level conditions, we digitize statistics from the ARCIA for three benchmark years (1875, 1885, and 1895) and, for the Dawes Act analysis, for 1887. These benchmarks were chosen to maximize coverage and comparability across the study period. Each annual report in our corpus is linked to the closest benchmark year within its decade (e.g., reports from 1868–1879 are linked to 1875), ensuring that all reports are associated with a consistent set of agency-level controls. The agency characteristics are as follows. Total population by agency is drawn from the ARCIA tables on salaries and incidental expenses, which report the number of Indians attached to each agency—these

figures are consistently reported at the agency level. To proxy for culture, we digitize the total number of Native persons that are recorded as wearing Western Dress, as documented in the ARCIA records (e.g., in 1895 it is under the table titled, "Table relating to population, dress, intelligence, dwellings, and subsistence of Indians, together with religious, vital, and criminal statistics"). We aggregate the underlying tribal level statistics to the agency level. If a given tribe is missing information on reported Western Dress, we use all non-missing statistics to compute the agency-level total. For the number of acres overseen by an agency, we use the land statistics from the schedule showing the names of Indian reservations in the United States, agencies, denomination nominating agents, tribes occupying or belonging to the reservation reported in the ARCIA. We aggregate the underlying reservation level statistics to the agency level. Total livestock is digitized from the ARCIA Table of statistics relating to area and cultivation of Indian lands, crops raised, stock owned by Indians, and miscellaneous products of Indian labor. As with the Western Dress statistics, we aggregate the underlying tribal level statistics to the agency level using all non-missing statistics.

B.7 Agency Crosswalks

We digitize agency-level characteristics from a small set of benchmark years as a practical choice that balances measurement with feasibility. Because agency structures were fluid—agencies frequently opened, merged, or were renamed—we construct harmonized crosswalks to link agency names from the annual reports to their benchmark-year counterparts. This allows us to address inconsistencies across time. For example, the Mescalero and Jicarilla agencies operated separately in 1881 but were merged in 1882 into a single agency (U.S. House of Representatives, 1882). Accordingly, all reports from these agencies in 1881 are mapped to the combined agency's statistics reported in the 1885 ARCIA.

The process to create the crosswalks was as follows. We start from the standardized lists of agencies and agents printed annually in ARCIA in our benchmark years (1875, 1885, 1895). As noted above, these tables present each agency alongside the name of the agent and their post-office and telegraphic address (see Appendix Figure B1). We identify links between the benchmark year agencies and the agencies associated with the corpus of annual reports based on detailed manual inspection. Potential links are initially identified via probabilistic string matching and then further reviewed in consultation with historical records on the history of agencies (e.g., the dates of their creation or possible consolidation). In our final sample of reports, we link 90 percent of annual reports to the crosswalk created for linking reports with agency characteristics. The annual reports that are not linked to a crosswalk are primarily from schools or military outposts that were not agencies (e.g., Fort Apache).

B.8 Dawes Land Allotment Data

We digitize records on land allotments during the first year after the passage of the Dawes Act in 1887. The statistics are taken from Table 16 of the 1888 ARCIA which reports allotments of land made since July 1, 1887. The statistics are provided at the reservation level, and we aggregate these to the agency level to align with our measure of assimilation

support for the Indian agents assigned to agencies. For our robustness analysis, we create additional measures of allotment status and the number of allotments as of 1889 and 1890. These measures are created based on similar reports on allotment activity in their respective ARCIAs.

B.9 Board of Indian Commissioners Data

We draw on correspondence collected by the Board of Indian Commissioners (BIC) to validate our main measure of assimilation ideology. Established by Congress in 1869, the BIC served as a quasi-official advisory body tasked with overseeing federal Indian policy and recommending reforms (Prucha, 1984). In 1875, the BIC circulated letters to Indian agents across the reservation system, asking whether they would support the creation of federally funded Indian police forces—a reform explicitly intended to undermine tribal authority and advance assimilation. We digitized these responses, published in the Board's Seventh Annual Report (Board of Indian Commissioners, 1875), and matched them to the same set of agents for whom we observe assimilation scores from their annual reports in that year. Because the letters were produced outside of the standard reporting process and before the widespread establishment of police forces, they provide an external benchmark to test whether assimilation support expressed in annual reports predicted contemporaneous policy preferences.

B.10 Annual Reports of Commissioners and the Carlisle Off-Reservation Boarding School

In addition to the agency and personnel reports described above, we rely on data from two other types of documents included in the ARCIA. First, we use the annual reports written by the Commissioner of Indian Affairs. These documents are printed as the first document in each volume of the ARCIA and provide a comprehensive account of federal policy, administrative actions, and conditions related to Indian Affairs policies. Second, for the years 1880 to 1900, we use the annual reports of the Carlisle Indian Industrial School, founded in 1879 in Carlisle, Pennsylvania. These reports first appear in the 1880 ARCIA and were authored by the school's founder and longtime superintendent, Richard Henry Pratt. Unlike the agency-level reports, the Carlisle reports are stored in a separate section of the ARCIA (e.g., in 1900, they were printed in the section "Reports of Superintendents of Bonded Schools"). To prepare these reports for analysis, we followed the same digitization and processing steps described above: isolating the relevant sections of the ARCIA, generating standalone PDF files tagged with metadata such as year and author, and applying OCR to render the text machine-readable. This procedure ensured that the Commissioner's reports and the Carlisle reports were integrated into the same structured, report-level dataset as the agency and personnel reports, allowing us to analyze them using parallel computational methods.

Figure B1: List of Indian Agencies and Agents (1901 ARCIA)

List of Indian agencies and independent schools, with post-office and telegraph addresses of agents and superintendents.

Agency.	Agent.	Post-office address.	Telegraphic address.
ARIZONA.			
Colorado River	Chas. S. McNichols	Parker, Yuma County, Ariz	Needles, Cal.
Fort Apache	A. A. Armstrong	Whiteriver, Ariz	Whiteriver, via Holbrook, Ariz.
Navaho	Geo. W. Hayzlett	Fort Defiance, Ariz	Gallup, N. Mex.
Pima	Elwood Hadley	Sacaton, Pinal County, Ariz	Casa Grande, Ariz.
San Carlos	Capt. W. J. Nicholson, U. S. Army	San Carlos, Ariz	San Carlos, Ariz.
Walapai	Henry P. Ewing *a*	Hackberry, Ariz	Hackberry, Ariz.
CALIFORNIA.			
Hupa Valley	Wm. B. Freer, school supt	Hoopa, Humboldt County, Cal	Eureka., Cal.
Mission Tule River (consolidated)	Lucius A. Wright	San Jacinto, Riverside County, Cal	San Jacinto, Cal.
Round Valley	Harry F. Liston, school supt	Covelo, Mendocino County, Cal	Covelo, via Cahto, Cal.
COLORADO.			
Southern Ute	Jos. O. Smith	Ignacio, La Plata County, Colo	Ignacio, Colo.
IDAHO.			
Fort Hall	A. F. Caldwell	Rossfork, Bingham County, Idaho	Pocatello, Idaho.
Lemhi	Edw. M. Yearian	Lemhi Agency, Lemhi County, Idaho	Red Rock, Mont.
Nez Percé	C. T. Stranahan	Spalding, Nez Perces County, Idaho	North Lapwai, Idaho.
INDIAN TERRITORY.			
Quapaw	Edgar A. Allen	Seneca, Newton County, Mo	Seneca, Mo.
Union	J. B. Shoenfelt	Muscogee, Ind. T	Muscogee, Ind. T.
IOWA			
Sauk and Fox	Wm. G. Malin	Toledo, Iowa	Toledo, Iowa.
KANSAS.			
Potawatomi and Great Nemaha.	W. R. Honnell	Nadeau, Jackson County, Kans	Hoyt, Kans.
MINNESOTA.			
Leech Lake	Capt. W. A. Mercer, U. S. Army	Odanah, Minn	Walker, Minn.
White Earth	Jno. H. Sutherland	White Earth, Becker County, Minn	Detroit, Minn.

a Industrial teacher in charge.

Notes: This figure is an excerpt from the 1901 ARCIA which is a list of Indian agencies, showing agents, post-office addresses, and designations such as military rank or "school superintendent."

Figure B2: List of Agencies Assigned to Religious Denominations (1880 ARCIA)

LIST OF INDIAN AGENCIES ASSIGNED TO THE SEVERAL RELIGIOUS DENOMINATIONS.

FRIENDS.—Great Nemaha, Omaha and Winnebago, Otoe, and Santee, in Nebraska, and Pawnee, in the Indian Territory. *B. Rush Roberts, Sandy Spring, Md.*

FRIENDS (ORTHODOX).—Pottawatomie and Kickapoo, in Kansas; Quapaw, Osage, Sac and Fox, Kiowa, Comanche, and Wichita, and Cheyenne and Arapaho, in the Indian Territory. *Dr. James E. Rhoades, Germantown, Philadelphia, Pa.*

METHODIST.—Hoopa Valley, Round Valley, and Tule River, in California; Yakama, Neah Bay, and Quinaielt, in Washington Territory; Klamath and Siletz, in Oregon; Blackfeet, Crow, and Fort Peck, in Montana; Fort Hall and Lemhi, in Idaho; and Mackinac, in Michigan. *Rev. Dr. J. M. Reid, secre tary Missionary Society, Methodist Episcopal Church, 805 Broadway, New York City.*

CATHOLIC.—Tulalip and Colville, in Washington Territory; Grand Ronde, and Umatilla, in Oregon; Flathead, in Montana; and Standing Rock and Devil's Lake, in Dakota. *General Charles Ewing, Catholic commissioner, Washington, D. C.*

BAPTIST.—Union (Cherokees, Creeks, Choctaws, Chickasaws, and Seminoles), in the Indian Territory; and Nevada, in Nevada. *Rev. Dr. H. L. Morehouse, secretary American Baptist Home Missionary Society, No. 28 Astor House offices, New York City.*

PRESBYTERIAN.—Navajo, Mescalero Apache, and Pueblo, in New Mexico; Nez Percés, in Idaho; and Uintah Valley, in Utah. *Rev. Dr. J. C. Lowrie, secretary Board of Foreign Missions of the Presbyterian Church, 23 Centre street, New York City.*

CONGREGATIONAL.—Green Bay and La Pointe, in Wisconsin; Sisseton and Fort Berthold, in Dakota; and S'Kokomish, in Washington Territory. *Rev. Dr. M. E. Strieby, secretary American Missionary Association, 56 Reade street, New York City.*

REFORMED.—Colorado River, Pima and Maricopa, and San Carlos, in Arizona. *Rev. Dr. J. M. Ferris, secretary Board of Missions of Reformed Church, 34 Vesey street, New York City.*

PROTESTANT EPISCOPAL.—White Earth, in Minnesota; Crow Creek, Lower Brulé, Cheyenne River, Yankton, Rosebud, and Pine Ridge, in Dakota; Ponca, in Indian Territory; and Shoshone, in Wyoming. *Rev. A. H. Twing, secretary Board of Missions of the Protestant Episcopal Church, 30 Bible House, New York City.*

UNITARIAN.—Los Pinos and White River, in Colorado. *Rev. Rush R. Shippen, secretary American Unitarian Association, 7 Tremont Place, Boston.*

UNITED PRESBYTERIAN.—Warm Springs, in Oregon. *Rev. John G. Brown, D. D., secretary Home Mission Board United Presbyterian Church, Pittsburgh, Pa.*

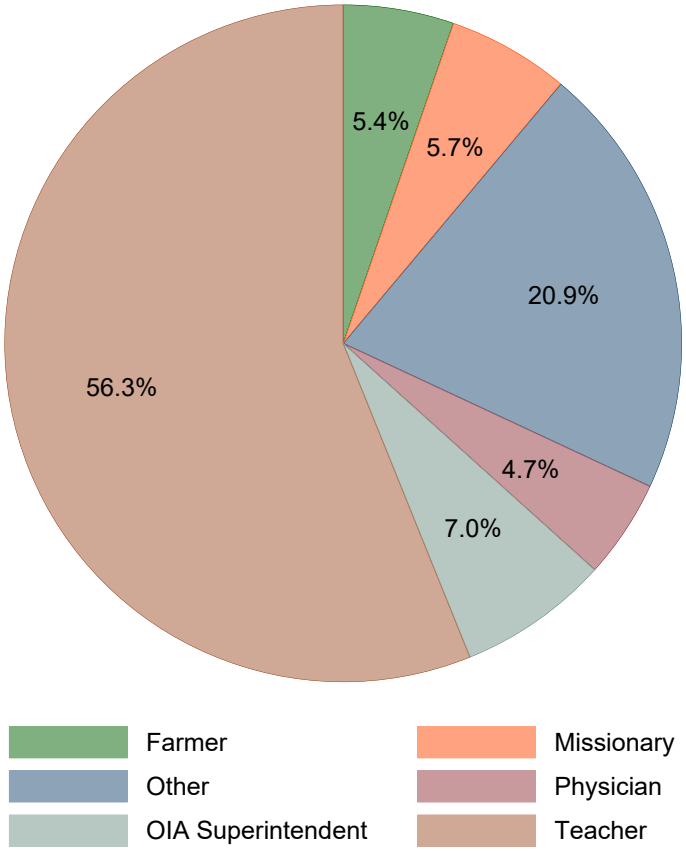
CHRISTAIN UNION.—Malheur, in Oregon. *Rev. J. S. Rowland, Salem, Oreg.*

EVANGELICAL LUTHERAN.—Southern Ute, in Colorado. *Rev. J. G. Butler, Washington, D. C.*

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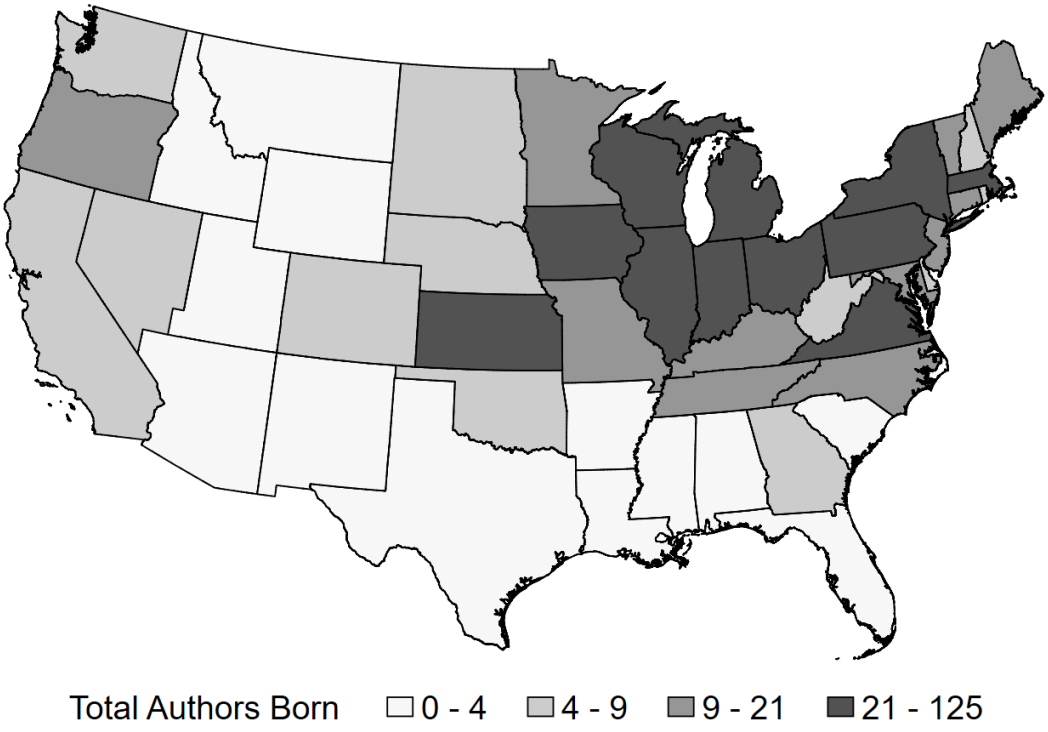
Notes: This figure is an excerpt from the 1880 ARCIA schedule of agencies assigned to religious denominations under the federal "Peace Policy."

Figure B3: Non-Agent Position Distribution



Notes: This figure shows the distribution of positions for the non-agents in the main sample of annual reports. We highlight the largest categories of positions: farmers, teachers, physicians, missionaries, and Office of Indian Affairs superintendents. Note that prior to 1878, the OIA operated a superintendency system, in which superintendents oversaw multiple agencies within large geographic jurisdictions. These high-level administrative roles are distinct from the superintendents of reservation schools that appear elsewhere in our sample. The remaining category labeled "other" combines a wide range of roles—including field matrons, interpreters, inspectors, and special commission members—that supported the OIA's assimilation agenda in more specialized ways.

Figure B4: State Birthplaces for Authors in the Annual Reports Sample



Notes: This figure shows a map of the United States based on 1870 state and territory boundaries. The shading indicates the number of authors in our main sample of annual reports born in each state. We have birthplace location data for 78 percent of the total authors ($N = 1,302$) in our main analysis sample of annual reports. Of the authors with birthplace information, 91 percent were born in the United States and are included in the statistics reported in this map.

C Appendix on LLM Prompts

C.1 Assimilation Prompt Details

Our development of a classification prompt using GPT-4o began with background reading of foundational historical works on the assimilation era (e.g., Prucha, 1984; Stuart, 1979) along with close readings of individual annual reports. This exercise highlighted variation in report length, topical scope, and author voice. Most importantly, this review confirmed that normative commentary on assimilation policy was prevalent enough to make text-based inference of beliefs and ideology plausible.

To calibrate our prompt quantitatively, we drew a random sample of 30 reports from the full corpus and manually scored each one. We jointly read and discussed each report, assigning a binary score of "1" if the text contained sufficient evidence that the author personally supported assimilation policies or implicitly supported them through criticism of Native cultural or social practices. Reports that lacked such evidence or expressed opposition to assimilation policies were coded as "0". Our focus on binary classification follows the approach in Lagakos, Michalopoulos and Voth (2025). In addition to the binary scores, we provided a brief justification statement and identified a representative quote from each report to justify our decision. In this calibration sample, 22 reports (73 percent) were scored as supportive of assimilation.

Prompt development proceeded in multiple rounds. Our initial attempts used short prompts instructing the model to classify reports on the binary scale and to provide both a short justification and three direct quotes from the report as supporting evidence. In these first iterations, we observed that the model tended to assign relatively more assimilation supportive classifications compared to our manual benchmark. To address the high Type I error (false positives) rate, we refined the prompt by incorporating a detailed set of decision rules, including explicit examples of what did and did not constitute normative endorsement of assimilation. This additional structure provided clearer guidance and reduced ambiguity in borderline cases.

The final version of our prompt achieved a 93 percent match rate, correctly classifying 28 of the 30 benchmark reports. The two disagreements were evenly split where the model identified assimilationist support in a report we judged neutral/opposed, and one where the model failed to detect assimilationist support that we judged present. The full text of our final prompt is provided in Appendix Figure C1. As representative illustrations, Appendix Table A2 presents the benchmark reports alongside our manual scores, GPT scores, as well as the manual justifications and quotes that we wrote and identified.

Figure C1: GPT-4o Instructions: Assessing Support for Assimilation

Your task is to read through the text and assess the writer's attitudes toward the social, cultural, religious, linguistic, or economic assimilation of Native Americans, or toward assimilation policies targeting Native Americans at the time (e.g., boarding schools, land allotment).

Return only a valid JSON object containing the following fields:

- "attitude_score": a score of "1" (support for assimilation or related policies) or "0" (a neutral stance, or opposition to assimilation or related policies).

- "attitude_score_justification": a brief justification (2 to 4 sentences) explaining your answer to "support_score".

- "attitude_score_reasoning_1": reasoning supported by a direct quote from the text.

- "attitude_score_reasoning_2": additional reasoning supported by a direct quote from the text.

- "attitude_score_reasoning_3": further reasoning supported by a direct quote from the text.

Use the scale and decision rules below to determine your answer to "attitude_score":

0: Opposition, or a neutral, descriptive, or ambiguous stance, toward assimilation or related policies. Assign a score of "0" if:

- The text discusses assimilation or related policies in a factual, descriptive, or administrative tone, without any clear positive or negative judgment, personal endorsement, or ideological stance.

- The text expresses moderate or qualified opposition to assimilation or related policies, using language that conveys skepticism or concern, or expresses respect for Native American culture.

- The text includes strong or ideological criticism of assimilation or related policies, using language that portrays assimilation as harmful, coercive, or unjust, and affirms the value, rights, or sovereignty of Native American culture and traditions.

1: Support for assimilation or related policies. Assign a score of "1" if:

- The text includes a clear, positive assessment of assimilation or related policies, using language that expresses personal approval or endorsement.

- The text includes a strong or ideological endorsement of assimilation or related policies, using language that frames assimilation as morally right and necessary for civilization, or strongly criticizes or denigrates Native American culture and traditions.

Please adhere to the following general guidelines:

- Treat "0" as the default score. Do not assign a score of "1" unless the text includes either personal endorsement or ideological or judgmental language in support of assimilation or related policies.

- Do not assign a score of "1" if the writer merely speaks approvingly of schooling or farming, or indicates compliance with government policy or instructions, but does not express a personal endorsement of or ideological commentary on assimilation or related policies.

- When reading the text, always assume that the word "Indian", tribal names (e.g., "Crow", "Apache"), and outdated or offensive terms (e.g., "squaw") refer to Native Americans. Do not assign a score of "1" merely because the text uses historical or outdated referential terms for Native Americans. However, do assign a score of "1" if the text uses explicitly judgmental or derogatory language (e.g., "savage", "heathenish", or "uncivilized") to describe Native Americans, their culture, or their traditions.

- Each reasoning field should include a short explanation (2 to 4 sentences) interpreting a direct quote from the text that supports the assigned score. Include the quote after the explanation. Do not include formulaic or procedural phrases (e.g., report openings like "in compliance with the regulations...") as evidence.

- The "attitude_score" and "attitude_score_confidence" fields must be returned as strings (e.g., "attitude_score": "1", not "attitude_score": 1).

C.2 Violence Prompt Details

To examine whether violence might be linked to bureaucratic attitudes, we developed a second classification prompt aimed at detecting local reports of violent incidents in the annual reports. As described in the main text and detailed in Appendix Section E, discussions of violence in these records were often indirect or embedded in narrative detail, making keyword-based searches less reliable. Our LLM-based classification therefore provides a more systematic measure of whether a given report references local violence, regardless of the author's framing. Appendix Figure A9 shows that such references are most common in the 1870s and 1880s and decline after 1890, consistent with the broader historical context of conflict in the assimilation era.

The design of this violence prompt differs from the assimilation prompt in two ways. First, it uses a simpler binary classification task, returning "yes" if a report clearly describes a specific, recent, and local episode of violence involving Native Americans, and "no" otherwise. Second, the prompt includes explicit temporal and geographic restrictions to ensure that only contemporaneous, site-specific incidents were captured, excluding generic commentary on violence or accounts of distant historical events. The full text of the violence prompt is provided in Appendix Figure C2.

Figure C2: GPT-4o Instructions: Violence Mentions Classification

```
You will be provided with a text related to Native Americans written in
{report_year}.

Your task is to determine whether the text describes one or more specific,
recent, and local episodes of violence committed by or against Native
Americans.


Return only a valid JSON object containing the following fields:

- "mentions_violence": "yes" or "no".

- "mentions_violence_reasoning": provide reasoning for your answer in no more
than two sentences.


Please adhere to the following general guidelines:

- Answer only "yes" or "no" in the field "mentions_violence".

- For an answer of "yes", the text must refer to a specific violent event. Do
not answer "yes" if the text only contains a general description of violence
committed by or against Native Americans.

- For an answer of "yes", the text must refer to a local violent event (i.e.,
an event that occurred at or near the location of the writer). If the location
of the event is unclear, but plausibly near the location of the writer, you may
still answer "yes".

- Only answer "yes" if the text refers to a violent event that occurred within
3 years of the year the text was written. Do not answer "yes" if the text
describes historical violence or conflict.

- When reading the text, always assume that the word "Indian", tribal names
(e.g., "Crow", "Apache"), and outdated or offensive terms (e.g., "squaw") refer
to Native Americans.
```

C.3 Administrative Capacity Prompt Details

A potential concern in interpreting the relationship between agents' assimilation attitudes and early allotment activity is that the same agents who expressed stronger pro-assimilation views may also have had greater bureaucratic skill or administrative effectiveness. In that case, higher allotment rates could reflect differential capacity to implement new federal directives, rather than ideology per se. To address this possibility, we developed a separate LLM-based classification prompt to measure agents' administrative capacity using the text of their pre-Dawes annual reports. As with the assimilation and violence measures, references to administrative competence in these reports are often qualitative and embedded in narrative detail (e.g., discussions of recordkeeping, staffing,

enforcement, procurement, and the organization of agency operations), making simple keyword-based approaches unreliable.

The administrative capacity prompt is designed as a conservative binary classification task. It returns "1" only when the text contains clear evidence that the author demonstrates strong administrative performance—for example, careful recordkeeping and accounting, systematic supervision of subordinates, timely execution of directives, organized management of agency resources, or the production of detailed operational statistics. In the absence of such evidence, the prompt defaults to a score of "0." Importantly, the instructions explicitly direct the model to ignore the author's policy views (including assimilationist rhetoric) and to focus solely on signals of administrative competence. The full text of the administrative capacity prompt is provided in Appendix Figure C3.

Figure C3: GPT-4o Instructions: Administrative Capacity Classification

```
You are scoring **administrative capacity** in historical Indian Affairs
reports. Administrative capacity means the presence of concrete,
procedural, and repeatable office practices for running an agency (e.g.,
planning/scheduling, record-keeping with metrics, resource/logistics
management, coordination/compliance with orders, problem diagnosis followed
by remedial action and verification).

Your task is **not** to evaluate ideology, moral judgments, or writing quality.
Ignore policy preferences (e.g., for or against assimilation) except when
they explicitly describe administrative actions or procedures. Reward only
**explicit evidence** of administration.

Define five evidence buckets:
1) planning_procedures - plans, schedules, protocols, targets, standard forms,
written rules;
2) recordkeeping_measurement - ledgers/rolls/inventories, attendance/production
figures, budget tallies, internal checks;
3) resource_management - procurement, ration issuance workflow, stock control,
maintenance, staffing rosters;
4) coordination_compliance - references to Commissioner circulars,
inspections/audits, inter-office coordination, following instructions;
5) problem_solving_followthrough - identifies bottlenecks, takes a remedy, then
verifies results.


**Binary rule:** Assign capacity_binary = 1 **only if** the report contains
concrete evidence from **at least one distinct bucket** above. Otherwise assign
capacity_binary = 0. Do not infer beyond the text. Do not reward eloquence or
normative claims.
```

D Appendix on Standard Textual Analysis Exercises

This appendix provides methodological details and full results for the three complementary textual analyses summarized in Section 4 of the main text. We examine whether assimilation-supporting and neutral/non-supporting reports differ systematically in (i) sentiment and emotive language, (ii) characteristic vocabulary, and (iii) broader semantic content captured by sentence embeddings.

D.1 Sentiment and Emotion

We first use the NRC Word-Emotion Association Lexicon (NRC EmoLex) (Mohammad and Turney, 2013) to explore sentiment and emotive language. This lexicon contains around 14,000 associations between English words, sentiment (positive and negative), and eight emotions (anger, fear, anticipation, trust, surprise, sadness, joy, and disgust).[40]

In line with best practices when using emotion lexicons (e.g., Mohammad, 2023), we first develop a domain-specific "stoplist" (i.e., terms that we exclude when conducting sentiment analysis) to account for words that are assigned a positive/negative sentiment or emotion in NRC EmoLex, but are used in an administrative or neutral fashion in our corpus. For example, our stoplist includes the word "teacher", which is associated with "trust" and "positive" sentiment in the lexicon, but arguably does not carry these connotations in 19th-century reports by Indian Office agents. Our final stoplist contains around 160 words, covering themes such as education and school (e.g., "pupil", "teacher"), agriculture and work (e.g., "hog", "labor"), and physical infrastructure, objects, and materials (e.g., "building", "iron").[41] We then preprocess reports in our main sample, removing generic terms (e.g., the names of tribes, standard stop words) and terms on our domain-specific stoplist, and lemmatize (normalize) remaining words.[42] Finally, we require that lemmas appear at least 10 times in the corpus, ensuring robustness against idiosyncratic terms and OCR noise.

We then examine differences in the number of lemmas per 1,000 words associated with each NRC EmoLex sentiment or emotion in assimilation supporting and neutral/non-supporting reports. Appendix Table A13 shows the results of this exercise. Differences are small in magnitude, but assimilation supporting reports do appear to have a slight "positive" slant, with roughly 9 more positive-sentiment terms per 1,000 words (after removing standard and domain-specific stop words). Consistent with this, assimilation supporting reports are more likely to use terms associated with the emotions of "joy", "trust", and "anticipation" (e.g., "civilize", "encourage", and "hope"), and are less likely to use terms

[40] The terms in EmoLex contain English nouns, verbs, adjectives, and adverbs. Term-emotion associations were identified using manual annotations obtained through Amazon MTurk. A term may be associated with multiple emotions.

[41] Appendix Table A12 contains the full list of themes and terms.

[42] We use the *stop_words* function from the R package *tidytext* to generate our list of standard stop words (Silge and Robinson, 2016). Lemmatization converts variations of a word into their dictionary form (e.g., the words "civilizes," "civilized," and "civilizing" all become "civilize"), allowing us to treat them as the same word. This reduces sparsity and noise in the text.

associated with the emotions of "fear", "disgust", and "sadness" (e.g., "evil", "offense", and "punish"). For robustness, we repeat this exercise excluding plausibly ideological or value-laden terms that are associated with positive sentiment or emotions in NRC EmoLex (e.g., "civilize", "influence", and "progress").[43] Appendix Table A14 presents these supplemental results. While class-level differences with respect to "trust" and "anticipation" disappear after applying these exclusions, there is still a detectable positive slant in assimilation supporting reports. Taken together, these patterns provide complementary evidence that assimilation-supporting and neutral/non-supporting reports differ in evaluative tone even after explicitly ideology-laden terms are excluded.

D.2 Characteristic Word Analysis

While count-based methods are straightforward and transparent, the lexicons that underlie them involve a degree of subjective judgment in their construction, and require researcher discretion in their application (e.g., the stoplist described above). Therefore, we complement our analysis of sentiment and emotion with a data-driven investigation of differences in vocabulary. Here, we apply the log-odds method from Monroe, Colaresi and Quinn (2008), which identifies words that are statistically overrepresented in one category of texts versus another—in our setting, the categories being the LLM-labeled assimilation supporting reports versus neutral/non-supporting ones. This method—which uses the overall corpus to form a Dirichlet prior—addresses the tendency of the standard log-odds ratio to overemphasize differences in very rare words (Jurafsky and Martin, 2025).[44] We calculate standardized log-odds scores (equivalent to z-scores) using a version of our main sample of annual reports filtered using the same preprocessing procedure as in our sentiment analysis, except that here we retain words from the domain-specific stoplist.

Appendix Figure A5 shows the 20 lemmas with the largest absolute log-odds scores; positive values indicate that the lemma is characteristic of assimilation supporting reports, whereas negative values indicate that the lemma is characteristic of neutral/non-supporting reports. There are marked differences across classes. The lemmas most strongly associated with assimilation supporting reports—such as "civilize", "dance", and "progress"—reflect clear concerns with assimilation and cultural practices. In contrast, the terms most characteristic of neutral/non-supporting reports (e.g., "guardian", "council", and "lease") are largely administrative in nature.

D.3 Semantic Analysis Using Sentence Embeddings

To assess whether LLM-assigned labels correspond to broader semantic differences beyond count-based features, we examine semantic differences between assimilation supporting and neutral/non-supporting reports using sentence embeddings. Sentence embeddings are vector representations of sentences (or passages) of text; if two passages have similar meanings (e.g., promoting the adoption of Christianity), their corresponding

[43] These terms are grouped under "Doctrine and Ideology" in Table A12.

[44] The Dirichlet prior (i.e., parameters based on a word's frequency in the entire corpus) shrinks the influence of very infrequent words when conducting comparisons between categories.

embeddings will be more similar to each other than to the embedding of an unrelated passage (e.g., describing financial statements).

We use embeddings from the Sentence-BERT family of models (Reimers and Gurevych, 2019). Broadly, these embeddings are generated by fine-tuning transformer-based models (i.e., based on a neural network architecture that captures dependencies between distant words or phrases) on combinations of sentences (e.g., pairs and triplets) such that sentences with similar meanings have similar numerical representations. In our application, each report is divided into smaller textual passages of roughly 256 words, which are encoded into vector representations using one of three pre-trained models: *all-MiniLM-L6-v2*, *all-mpnet-base-v2*, and *instructor-large*.[45] We then average passage-level embeddings to obtain a single embedding per report, and train a logistic regression classifier on these document-level embeddings. The evaluation of classification performance uses an 80/20 train/test split, so that the reported performance is calculated on reports not used to estimate the classifier.

Appendix Table A15 presents illustrative output from the classification procedure. Each row displays: (i) a passage of text, (ii) an indicator for whether the passage is classified as assimilation supporting by the logistic classifier (with predicted probability in parentheses), (iii) an indicator for whether the document from which the passage is drawn is classified as assimilation supporting by the logistic classifier (with predicted probability in parentheses), and (iv) the label assigned by GPT-4o. The first two passages express approval of cultural assimilation (e.g., "the whole internal man must be made over and reorganized... when an [I]ndian can take his team and go to the woods to chop a load of firewood... he has taken a long step upward"), whereas the latter two are purely administrative. A logistic regression classifier trained on passage-level embeddings assigns the first two passages to the assimilation supporting class and the latter two to the neutral/non-supporting class.[46] Aggregating passage-level embeddings and training the classifier on document-level embeddings yields consistent classifications, which align with the (document-level) LLM-assigned labels.

To evaluate and interpret our classification exercise more generally, we report standard performance metrics (e.g., Hausladen, Fochmann and Mohr, 2024). These are: balanced accuracy (the average of the correct prediction rate for each class, giving equal weight to both classes), AUC (area under the receiver-operating characteristic curve, which is the probability that the model ranks a randomly chosen positive example higher than a randomly chosen negative one), and AP (average precision, which is a summary of the precision–recall trade-off).[47] We compare these with two relevant benchmarks. The first is

[45]The *all-mpnet-base-v2* model produces higher-dimensional embeddings that capture more fine-grained nuance than *all-MiniLM-L6-v2*, at the cost of speed. Neither model is fine-tuned for the analysis of historical documents. The *instructor-large* model generates task-specific embeddings using natural language instructions (e.g., "represent the passage to capture stances and attitudes in historical reports by United States government officials"), which allows for domain adaptation without additional fine-tuning.

[46]Note that, since we do not have LLM-assigned labels at the passage level, we assign document-level labels to all passages within the report for the purposes of this illustrative exercise.

[47]In our setting, "positive" refers to the assimilation supporting label, and a "true positive" is a report

a majority-class classifier that always predicts the assimilation supporting label, and the second is random guessing in proportion to class prevalence. Both benchmarks would achieve balanced accuracy of 0.50, AUC of 0.50, and AP of 0.70.

Appendix Table A16 reports the performance of a logistic regression classifier trained to distinguish between assimilation supporting and neutral/non-supporting reports with document-level embeddings. When using embeddings generated by the "baseline" *all-MiniLM-L6-v2* model, the classifier achieves balanced accuracy of 0.691, AUC of 0.756, and AP of 0.870. These results exceed both benchmarks across all metrics, and performance improves when using embeddings from the higher-dimensional *all-mpnet-base-v2* and *instructor-large* models. Taken together, these results support the view that there are meaningful semantic differences between assimilation supporting and neutral/non-supporting reports, as classified by the LLM.

D.4 Sensitivity to Prompt Modifications

We assess sensitivity to prompt specification by rescoring a randomly selected sample of 100 reports under a range of modifications to our baseline prompt. For each alternative specification, we calculate the share of reports receiving the same classification as under the baseline prompt. The modifications cover several approaches emphasized in the prompt-engineering literature, including personas, reasoning instructions, and few-shot prompting (Lu et al., 2022; Liu et al., 2023; Sahoo et al., 2024; White et al., 2023). For context, our separate stability exercise finds that 93.5 percent of reports receive the same classification in all five applications of the unchanged prompt.

In Panel A of Appendix Table A17, we compare the baseline prompt with variants that add instructions at the very beginning of the prompt, as shown in Appendix Figure A6. The first set of modifications request the LLM to adopt one of four possible personas: "historian with expertise in 19th and early 20th century Native American policy," "political scientist with expertise in federal Indian policy and assimilation programs," "cultural anthropologist with expertise and deep understanding of Native American societies and the impacts of assimilation policies," or a "helpful research assistant for a political scientist." We varied not only the role ("political scientist"), but also the description of their expertise, as the goal is to see how much variation we can produce with these adjustments. This type of prompt augmentation has been shown in other domains to affect performance by anchoring the model's perspective or tone, with some finding significant performance improvements (Salewski et al., 2023; Hu and Collier, 2024), though others have found modest and unpredictable improvements (Zheng et al., 2024). Reassuringly, the impact is relatively minimal as the associated agreement rates of 92, 90, 97, and 88 percent are all very close to the sampling variation benchmark of 93 percent. Finally, motivated by Li et al. (2023), we use an "emotion prompt" that starts the prompt with "This task is very important to my career," a technique that produced large improvements in their context. This adjustment produces 95 percent agreement wiht the baseline classifications.

In Panel B of Appendix Table A17, we turn to the prompt modifications added to the

classified as assimilation supporting by GPT-4o.

middle or end of the original prompt (but before the text of the letter being evaluated), as written out completely in Appendix Figure A7. The first four cases prompt "chain-of-thought" reasoning via simple (zero-shot) modifications shown to affect performance in previous research (e.g., Kojima et al., 2022). These simply ask the LLM to "Think carefully," to "Think step by step. Lay out each step," or to "Please provide an explanation for your answer." The most extensive of these (the structured chain-of-thought prompt) writes out a set of steps, though one could argue this is somewhat redundant with the text earlier in the original prompt. We find that these prompts induce even less variation from our original prompt than the persona modifications (with agreement rates ranging from 92 to 96 percent). The next two prompt modifications are not discussed in the literature but aimed at reminding the LLM of the historical context of the text and to avoid modern moral judgments. One achieved 91 percent agreement while the other achieved 92 percent agreement, but in the latter case all deviations were in the direction of assigning more cases to neutral rather than strong support, whereas the other warning led to roughly an equal number of deviations in either direction. Finally, we consider few-shot prompting, a technique that can improve performance significantly by providing labeled examples, though prior research has shown this class of prompting can be quite sensitive to the ordering (Lu et al., 2022). To avoid pushing the limits of the LLM context window, we provide condensed hypothetical sentences that are then scored and justified. We find that this modified prompt agrees with the original prompt in 90 percent of cases, with all deviations being in the direction of assigning what would be neutral/non-supporting letters to instead be supportive. Overall, agreement with the baseline classification remains high across a diverse set of prompt modifications, ranging from 88 to 97 percent across specifications.

E Additional Evidence on the Evolution of Assimilation Support

This appendix provides additional evidence on two potential explanations for the evolution of assimilation support documented in Section 5. We first examine whether the moderation among Indian agents coincided with a similar change in assimilation support among higher-level OIA leadership and other prominent federal assimilation institutions. We then examine whether the changing prevalence of local violence is associated with the evolution of agent assimilation support. Across both exercises, we find little evidence that either changing signals from senior OIA leadership or declining local violence provides the main explanation for the moderation in assimilation support among agents.

E.1 Assimilation Support Among Senior OIA Leadership

One potential explanation for the moderation in assimilation support after 1890 is that agents were responding to changing views or directives from higher-level OIA leadership, or to a broader change in how assimilationist ideology was expressed within the federal government. To assess this, we apply our classification prompt to the 39 Commissioner reports in our sample and find that all are scored as assimilation-supporting except for 1868, which is coded as non-supporting in this first run.[48] Notably, these findings are consistent with the historical literature, which emphasizes the unwavering assimilationist commitments of Commissioners. Writing about Commissioner Thomas Jefferson Morgan (1889–1893), Prucha described him as having an "absolute and unwavering conviction that the Indians must all be completely Americanized" (1984, p. 723).

An analysis of reports from the Carlisle Indian Industrial School, covering 1880–1900, provides an additional comparison. Richard H. Pratt, the school's founder and longtime superintendent, consistently expressed strong support for assimilation throughout this period. Using the digitized annual reports from Carlisle and applying our classification prompt, we likewise find unbroken support for assimilationist ideology across the two decades. This contrasts with the moderation observed in the agent-level reports, underscoring that the shift among agents reflected a particular evolution within the OIA bureaucracy rather than a broader weakening of federal assimilation policy. Pratt's role as superintendent of Carlisle is also distinct from the reservation school superintendents in our sample, who appear more moderate in tone. In line with this, historical accounts emphasize that off-reservation schools such as Carlisle were intended as uncompromising instruments of assimilation, whereas reservation schools were often perceived as being

[48] As with the agent reports, we repeated the GPT-4o classification script five times as a consistency check (here for the full Commissioners sample rather than a subset). We find complete stability across 1869-1904 (and 1906); however 1868 and 1905 were borderline cases, as 1868 was coded as supporting in four out of five of the re-runs, while 1905 was coded as non-supporting in all five of the re-runs. One caveat for this analysis is that the Commissioner's reports are much longer and more wide-ranging than the agent letters. A concern is that such length might approach the limits at which GPT-4o's performance deteriorates, although Appendix Table A18 shows no clear evidence of this: the LLM's justifications for the Commissioner reports are consistent with our analysis of the annual reports, and manual inspection of the support quotes confirms their accuracy.

less forceful and effective in "civilizing" Indian children (Adams, 1995).[49]

E.2 Local Violence and Assimilation Support

Another potential explanation for the moderation we observe is that support for assimilation may have waned as the marked social tension and violence that characterized the 1870s and 1880s subsided. The early years in the sample of annual records covered a period of conflict between the federal government and tribal communities. For instance, in 1879, Nathan Meeker, the Indian Agent leading the White River Indian Agency, and several members of his staff were killed by Ute tribal members following escalating disputes over land use and cultural practices. More broadly, historians emphasize the frequency of local confrontations where land encroachment by white settlers set the stage for broader conflict, including high-profile conflicts such as the Nez Perce War in 1877 and the Apache Wars that culminated in Geronimo's 1886 surrender (Brown, 1970).

To examine whether violence might be linked to bureaucratic attitudes, we leverage a second large language model (LLM) approach to create an original collection of reports of violent incidents mentioned in the annual reports. Discussions of violence are often indirect or embedded in narrative detail, making keyword searches potentially less reliable. Our LLM-based classification provides a more systematic measure of whether a given report references local violence, regardless of how it is framed. Appendix Figure A9 shows the fraction of agent-authored reports that mention violence in each year. Consistent with the broader historical context, references to violence are most common in the 1870s and 1880s and decline after 1890.

Despite the similar timing of a moderation in ideology and violence, we find little evidence of a systematic relationship between reported local violence and assimilation attitudes. Reports that mention violence do not exhibit meaningfully different assimilation attitudes compared to those that do not mention violence. Summary statistics reported in Appendix Figure A10 show that the average assimilation support score is very similar across reports with and without violence mentions.[50] Thus, while violence was an important contextual feature of the assimilation era, these comparisons provide little evidence that declining local violence accounts for the moderation in assimilation support among agents.

[49]This view was shared within the OIA. In 1890, for example, Commissioner Thomas Jefferson Morgan opined that reservation schools faced "influences which necessarily hamper them very seriously in their work." Among these were the fact that these schools were "far removed from civilization," and Indian parents had "ready access" (U.S. Department of the Interior, 1890, pp. XII–XIII).

[50]The figure also demonstrates that there is a similar pattern in assimilation support scores between reports with and without violence after residualizing out the influence of a large set of observables.